

Protocol

Levosimendan to Facilitate Weaning From ECMO in Patients With Severe Cardiogenic Shock

The LEVOECMO Randomized Clinical Trial

Principal Investigator–Coordinator and Corresponding Author:

Alain Combes, MD, PhD

ORCID: 0000-0002-6030-3957

Service de Médecine Intensive–Réanimation

Assistance Publique–Hôpitaux de Paris, Hôpital Pitié–Salpêtrière

Sorbonne Université

INSERM, UMRS_1166-ICAN, Institute of Cardiometabolism and Nutrition

47, boulevard de l'Hôpital, F-75013 Paris, France

Phone: 33.1.42.16.38.18

e-mail: alain.combes@aphp.fr

This appendix has been provided by the authors to give readers additional information about their work.

LEVOECMO trial Supplementary Appendix

This supplement contains the following items:

1. Original protocol (Version 1.1, 12/04/2020)
2. Second version of the protocol (Version 2.0, 05/31/2023)
3. Final version of the protocol (Version 3.0, 06/26/2024)
4. Summary of changes

**" LEVOSIMENDAN to facilitate weaning
from ECMO in severe cardiogenic shock patients "
LEVOECMO**

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

Version N° 1.1 of 04/12/2020

Project code number: P170914J - EUDRACT No.: 2019-004319-29

Coordinating Investigator:

Alain COMBES

Intensive Care Unit, Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Tel. : + 33 1 42 16 38 18

Email : alain.combes@aphp.fr

Methodologist:

Pr Florence TUBACH

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Email: florence.tubach@aphp.fr

Sponsor:

Assistance publique – Hôpitaux de Paris and by delegation: Clinical Research and Innovation
Delegation (DRCI) - Hôpital Saint-Louis - 1, avenue Claude Vellefaux

DRCI head office project advisor: Carla VANDENABELE

Tel : 01 40 27 57 47

Fax : 01 44 84 17 01

Email : carla.vandenabele@aphp.fr

Entity responsible for monitoring the trial:

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

Head: Pr Florence TUBACH

Project advisor: Jessica PALMYRE

Email: jessica.palmyre@aphp.fr

Tel: 01 42 16 19 35

Fax : 01 42 16 24 40

Agence générale des équipements et produits de santé, (APHP)

AGEPS, Département Essais Cliniques - 7, rue du Fer à Moulin, 75005 Paris

Céline DUPRE : Email : celine.dupre@aphp.fr

Nancy TAMIR-GEDDIS: Email : nancy.tamirgeddis@aphp.fr

Clinical Research and Innovation Delegation (DRCI)
Hôpital Saint Louis 75010 PARIS

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

PROTOCOL SIGNATURE PAGE

Research code number: P170914J

Title: LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients –
"LEVOECMO"

Version N° of: 1.1 of 04/12/2020

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:

Pr. Alain COMBES
Intensive Care Unit
Hospital Pitié Salpêtrière
47 - 83 Boulevard de l'Hôpital
75651 Paris Cedex 13

Date: 2, 7, 2021

Signature:

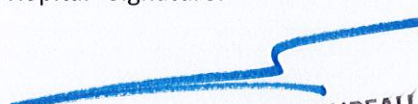


Sponsor

Assistance Publique-Hôpitaux de Paris
Clinical Research and Innovation Delegation Hôpital Saint Louis
1 avenue Claude Vellefaux
75010 PARIS

Date: 19 JUL. 2021

Signature:



Serge BUREAU
DRCI - AP-HP
Pôle Promotion

The study was approved by the Ethic committee (CPP) of on and authorised by the ANSM
on

TABLE OF CONTENTS

1	SUMMARY	5
2	SCIENTIFIC JUSTIFICATION FOR THE TRIAL	9
2.1	SHORT-TERM MECHANICAL CIRCULATORY SUPPORT FOR REFRACTORY CARDIOGENIC SHOCK	9
2.2	LEVOSIMENDAN, AN ORIGINAL INODILATOR	9
2.3	WHY RECENT LARGE LARGE RANDOMIZED CONTROLLED TRIALS OF LEVOSIMENDAN WERE ALL NEGATIVE?.....	10
2.4	RATIONALE FOR A TRIAL TESTING LEVOSIMENDAN IN VA-ECMO PATIENTS AND JUSTIFICATION FOR THE CHOICE OF SUBJECTS	11
2.5	NAME AND DESCRIPTION OF THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	11
2.6	DESCRIPTION AND JUSTIFICATION OF THE DOSAGE, ROUTE OF ADMINISTRATION, ADMINISTRATION SCHEDULE AND TREATMENT DURATION	11
2.7	SUMMARY OF THE KNOWN AND FORESEEABLE BENEFITS AND RISKS FOR THE STUDY PARTICIPANTS	12
3	OBJECTIVES	12
3.1	PRIMARY OBJECTIVE	12
3.2	SECONDARY OBJECTIVES	12
4	DESCRIPTION OF THE TRIAL	12
4.1	CONCISE DESCRIPTION OF THE PRIMARY AND SECONDARY ENDPOINTS	13
4.2	RESEARCH METHODOLOGY	13
5	PROCEDURE FOR THE TRIAL	16
5.1	BASELINE VISIT = J1 (INCLUSION AND RANDOMIZATION)	16
5.2	FOLLOW-UP VISITS	17
5.3	EXPECTED LENGTH OF PARTICIPATION, CHRONOLOGY AND DURATION OF THE STUDY.....	18
5.4	TABLE OR DIAGRAM SUMMARISING THE CHRONOLOGY OF THE STUDY	18
5.5	DISTINCTION BETWEEN STANDARD CARE AND RESEARCH	18
5.6	TERMINATION AND EXIT RULES.....	19
6	SELECTION CRITERIA.....	21
6.1	IDENTIFICATION OF ELIGIBLE PATIENTS: INCLUSION AND EXCLUSION CRITERIA.....	21
6.2	CONSENT AND RANDOMIZATION PROCEDURE.....	22
6.3	RECRUITMENT METHODS	22
7	TREATMENT ADMINISTERED TO STUDY PARTICIPANTS	23
7.1	THE INVESTIGATIONAL MEDICINAL PRODUCT(S).....	23
7.2	TRACEABILITY INFORMATION FOR THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	24
7.3	COMMON MANAGEMENT REGARDING OTHER INOTROPIC DRUGS, HEART FAILURE MEDICATIONS AND ANTIAGREGANTS AND ANTICOAGULANTS.....	24
7.4	COMMON MANAGEMENT REGARDING THE ECMO CIRCUIT	24
7.5	AUTHORISED AND PROHIBITED TREATMENTS (MEDICINAL, NON-MEDICINAL, SURGICAL), INCLUDING EMERGENCY MEDICATIONS	25
7.6	METHODS FOR MONITORING COMPLIANCE WITH THE TREATMENT	25
8	EFFICACY ASSESSMENT.....	25
8.1	ANTICIPATED METHODS AND TIMETABLE FOR MEASURING, COLLECTING AND ANALYSING THE EFFICACY DATA	25
9	SPECIFIC COMMITTEES FOR THE TRIAL	25
9.1	STEERING COMMITTEE (SC)	25
10	SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY	25
10.1	RECORDING AND REPORTING ADVERSE EVENTS.....	26
11	DATA MANAGEMENT	32
11.1	DATA COLLECTION PROCEDURE	32
11.2	IDENTIFICATION OF DATA RECORDED DIRECTLY IN THE CRFs WHICH WILL BE CONSIDERED AS SOURCE DATA	32
11.3	RIGHT TO ACCESS SOURCE DATA AND DOCUMENTS.....	32
11.4	DATA PROCESSING AND STORAGE OF DOCUMENTS AND DATA	33

11.5	OWNERSHIP OF THE DATA.....	33
12	STATISTICAL ASPECTS	33
12.1	PLANNED STATISTICAL METHODS, INCLUDING THE TIMETABLE FOR ANY PLANNED INTERIM ANALYSES	33
12.2	SAMPLE SIZE CALCULATION	35
12.3	STATE WHETHER SUBJECTS WHO EXIT THE STUDY PREMATURELY WILL BE REPLACED AND IN WHAT PROPORTION.	35
12.4	ANTICIPATED LEVEL OF STATISTICAL SIGNIFICANCE.....	35
12.5	MANAGEMENT OF MODIFICATIONS MADE TO THE ANALYSIS PLAN FOR THE INITIAL STRATEGY.	36
12.6	SELECTION OF POPULATIONS	36
13	QUALITY CONTROL AND ASSURANCE	36
13.1	GENERAL ORGANISATION	36
13.2	QUALITY CONTROL	36
13.3	CASE REPORT FORM.....	37
13.4	MANAGEMENT OF NON-COMPLIANCES.....	37
13.5	AUDITS/INSPECTIONS.....	37
13.6	PRINCIPAL INVESTIGATOR'S DECLARATION OF RESPONSIBILITY	37
14	ETHICAL AND LEGAL CONSIDERATIONS	38
14.1	METHODS FOR INFORMING AND OBTAINING CONSENT FROM THE RESEARCH PARTICIPANTS.....	38
14.2	<i>PROHIBITION OF CONCOMITANT CLINICAL STUDIES PARTICIPATION AND EXCLUSION PERIOD AFTER THE TRIAL, IF APPLICABLE</i>	38
14.3	COMPENSATION FOR SUBJECTS	38
14.4	REGISTRATION ON THE NATIONAL REGISTER OF SUBJECTS PARTICIPATING IN HUMAN RESEARCH TRIALS ON THE PRODUCTS LISTED IN ARTICLE L. 5311-1 OF THE FRENCH PUBLIC HEALTH CODE.....	38
14.5	AUTHORISATION FOR THE RESEARCH LOCATION.....	39
14.6	LEGAL OBLIGATIONS.....	39
15	FUNDING AND INSURANCE	40
15.1	SOURCES OF FUNDING FOR THE TRIAL	40
15.2	INSURANCE	40
16	PUBLICATION	40
16.1	MENTION OF AP-HP AFFILIATION FOR PROJECTS SPONSORED OR MANAGED BY AP-HP	40
16.2	MENTION OF THE AP-HP MANAGER (DRCI) IN THE ACKNOWLEDGEMENTS OF THE TEXT	41
16.3	MENTION OF THE FUNDER IN THE ACKNOWLEDGEMENTS OF THE TEXT	41
17	BIBLIOGRAPHY	42
18	LIST OF ADDENDA	45
18.1	LIST OF INVESTIGATORS	45
18.2	SERIOUS ADVERSE EVENTS REPORT FORM	45
18.3	PREGNANCY REPORT FORM	45
18.4	QUESTIONNAIRE/SCALE.....	45
18.5	SCP'S LEVOSIMENDAN.....	45

1 SUMMARY

Full title	LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients
Acronym	LEVOECMO
Coordinating Investigator	COMBES ALAIN, GROUPE HOSP. PITIE-SALPETRIERE (AP-HP), PARIS, France
Sponsor	Assistance Publique-Hôpitaux de Paris
Scientific justification	In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with refractory cardiogenic shock. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery. In patients with potentially reversible cardiac failure (e.g. myocarditis, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery. Although considered as the ultimate life-saving technology for refractory cardiac failure, veno-arterial ECMO is still associated with severe complications. Specifically, excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. ECMO support also exposes to many complications such as infections, hemorrhage or peripheral vascular embolism. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs. Levosimendan, which acts to sensitize myocardial contractile proteins to calcium, improves cardiac contractility without increasing the intracellular calcium concentration. Unlike traditional inotropes such as dobutamine, levosimendan neither increases myocardial oxygen consumption nor impairs diastolic function or possess proarrhythmic effects. It also influences the opening of ATP-dependent potassium channels, including those in vascular smooth muscle cells, leading to coronary, pulmonary, and peripheral vasodilation and antiinflammatory, antioxidative, antiapoptotic, anti-stunning and cardioprotective effects. Additionally, Levosimendan which has a long lasting action (up to 7–9 d), resulting from the formation of active metabolite, may be used as a single 24h perfusion. In recent preliminary studies, the drug was associated with accelerated weaning from VA-ECMO and even improved survival. Therefore, a multicenter randomized trial with sufficient statistical power is needed in refractory cardiogenic shock patients supported by VA-ECMO to test if the early administration of Levosimendan can facilitate and accelerate VA-ECMO weaning, and ultimately translate in significantly less morbidity, reduced ICU and hospital length of stays and associated costs.
Main objective and primary endpoint	Main objective: To assess the efficacy of early administration of Levosimendan on the time to successful ECMO weaning (defined below) within the 30 days following randomization, in patients with VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Primary endpoint: Time to successful ECMO weaning within the 30 days following randomization.

	- ECMO weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal. - Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization). - Patients still under ECMO 30 days after randomization will be considered as censored. - The number of days alive without ECMO was not an appropriate primary outcome due to the high proportion of patients dying with ECMO and thus ex-aequo (0 days). The analysis will therefore model the risk of successful ECMO ablation in the presence of two competing risks (death and weaning failure).
Secondary objectives and endpoints	Secondary objectives/endpoints: To assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function. - mortality on D30 and D60 - duration of ECMO support between inclusion and D30, and between inclusion and D60; number of ECMO-free days between inclusion and D30/D60 - duration of ICU stay and duration of hospitalization; - major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, rehospitalization for heart failure on days 30 and 60 - time to improvement into hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization - number of days with organ failure, defined by SOFA score, and number of days alive without organ failure between inclusion and D30; - duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30; - duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60; - LV function assessed with echocardiography at Day 30 following randomization - Incidence of adverse drug reactions (atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia). The primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.
Design of the trial	Multicenter, Double Blind Randomized clinical trial
Target Population	Adults patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support and for whom LV recovery is likely.
Inclusion criteria	1. Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h. 2. Obtain informed consent from a close relative or surrogate. According to the specifications of emergency consent, randomization without the close relative or surrogate consent

	could be performed. Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.
Exclusion criteria	1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes before ECMO 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) Prior history of torsades de pointes 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study
Investigational medicinal product(s)	A continuous infusion of Levosimendan administered over 24h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate limiting side effects.
Comparator treatment	The placebo will be made of riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used as a dietary supplement, especially in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing levosimendan (Gordon, NEJM 2016) The route of administration, administration procedures and duration of treatment will be similar to those of the experimental arm.
Risks added by the trial	Treatment by levosimendan or placebo
Procedure of the trial	<u>Inclusion visit (= Randomisation) Day 1</u> – Informed consent form. – Randomisation – Administration of a continuous infusion of Levosimendan or Placebo – Comorbidities – Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses – Record of AE/SAE Day 2 to Day 30 or date of death or date of discharge from the ICU – Record of AE/SAE – Clinical data – ECMO parameters

	– Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses Day 30, Day 60 – Vital status – ECMO: yes/no – On other mechanical cardiac support device – Heart transplantation – Mechanical ventilation – ICU discharge and date – Hospital discharge and date
Number of subjects included	206 (103 per arm)
Number of sites	National trial involving 13 centers
Duration of the trial	- inclusion period: 24 months - participation period (treatment+follow-up):2 months - total duration: 26 months
Number of enrolments expected per site and per month	0.7
Statistical analysis	The analysis will model the risk of successful VA-ECMO ablation in the presence of two competing risks (death and weaning failure).
Trial will have a Data Monitoring Committee	YES
Funding source	PHRC 2017 / French Ministry of Health

2 SCIENTIFIC JUSTIFICATION FOR THE TRIAL

2.1 Short-term mechanical circulatory support for refractory cardiogenic shock

In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with acute cardiogenic shock in the setting of acute myocardial infarction, myocarditis, intoxication with cardiotoxic drugs, end-stage dilated cardiomyopathy or post-cardiotomy [1-6]. Most of these “crash and burn” patients receive the device as salvage therapy after having already developed signs of multiple organ failure. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery [1, 2]. Several considerations must be taken into account before instituting ECMO. First, the device should be inserted before the patient has developed multiple organ failure or myocardial failure has led to refractory cardiac arrest, since these conditions are associated with significantly poorer outcomes [2, 3, 6]. Second, highly unstable patients may benefit from urgent on-site ECMO initiation by a rapid resuscitation team able to operate a portable and quick-to-prime ECMO circuit before transportation to the ECMO referral center [4]. Third, cardiac failure and other organ injuries should be deemed reversible and the patient’s underlying condition should not contraindicate a bridge to a more permanent device or to transplantation. Fourth, management of patients on ECMO for refractory cardiogenic shock is complex and should be conducted in experienced medical-surgical centers [7].

VA-ECMO might be used as a bridge to decision-making if the patient survives the first days to reach the “decision-making” point. In patients with potentially reversible cardiac failure (e.g. myocarditis, drug intoxication, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery [1]. Alternatively, it might be used as a bridge to transplantation or VAD implantation because of absence of LV functional recovery [8]. ECMO can also be simply withdrawn in cases of therapeutic futility (severe brain lesions, end-stage multiple organ failure or absence of myocardial recovery in the context of a definitive contraindication to transplantation or VAD implantation). Long-term survival after VA-ECMO is 70-80% after myocarditis or cardiotoxic drug poisoning, 40-50% after myocardial infarction and 15-25% when the device is used to rescue refractory cardiac arrest [2, 3, 6, 8-10]. Survivors reported a preserved quality of life, despite some limitations in physical activities and social functioning in previous series [2, 3, 6].

Although considered as the ultimate life-saving technology for refractory cardiac failure, venoarterial ECMO is still associated with severe complications. It increases left ventricular (LV) afterload that may delay myocardial recovery. Excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. Prolonged ECMO support also exposes to many complications such as local hemorrhage (10-20%), cannulation site infection and bacteremia (10-15%), limb ischemia (5-10%), ischemic or hemorrhagic stroke (5%) [2, 3, 6, 11, 12]. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs [2, 11-13].

2.2 Levosimendan, an original inodilator

Levosimendan is a calcium sensitizer with inotropic, vasodilatory and cardioprotective properties. It facilitates contact between actin and myosin and prolongs the time of contact between these two proteins, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect that unlike other positive inotropes, does not modify intracellular calcium concentration (which can cause arrhythmia and cell death) or myocardial diastolic relaxation. It also increases the contractile force of the myocardium without modifying its O₂ consumption. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on coronary arteries). The drug was also reported to exert cardioprotective, anti-stunning and anti-ischemic effect through this K⁺-ATP-dependent channel pathway.

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four

to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with prolonged half-life of about three days [16].

2.3 Why recent large randomized controlled trials of levosimendan were all negative?

Levosimendan has been tested in large RCTs against dobutamine or placebo in patients with decompensated heart failure, septic shock and the low cardiac output syndrome after cardiac surgery.

In patients with acute decompensated heart failure, while a first proof of concept trial suggested that levosimendan was associated with improved haemodynamic performance and lower mortality than dobutamine [15], the large (n=1,327) randomized double-blind SURVIVE clinical trial failed to confirm these findings [17]. In this trial, patients received a loading dose of levosimendan (12 µg/kg) followed by an infusion of 0.1 up-to 0.2 µg/kg/min for an additional 23 hours, as tolerated. Although a greater decrease in B-type natriuretic peptide level was observed in the first 5 days of treatment in the levosimendan group, all-cause mortality at 180 days was similar between groups, respectively occurring in 26% in the levosimendan group and 28% in the dobutamine group (HR 0.91; [0.74-1.13]; p=0.40). Marked hypotension, probably related to the levosimendan loading dose, atrial fibrillation, and hypokalemia were more frequent in the levosimendan group, possibly explaining the negative results of this trial.

The Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis (LeoPARDS) trial tested whether the addition of levosimendan to standard care would reduce the severity of organ dysfunction among patients with septic shock [18]. The administration of levosimendan or placebo was started at a rate of 0.1 µg/kg/min, increased to 0.2 µg/kg/min for a further 20 to 22 hours in the absence of rate-limiting side effects. The addition of levosimendan to standard treatment was not associated with less severe organ dysfunction or lower mortality. The mean (±SD) SOFA score over the ICU stay was 6.68±3.96 in the levosimendan group and 6.06±3.89 in the placebo group (P = 0.053) and mortality at 28 days was 34.5% in the levosimendan group and 30.9% in the placebo group (P = 0.43). More patients in the levosimendan group than in the placebo group had supraventricular tachyarrhythmia, tachycardia and hypotension. The negative results of this trial may relate to the inclusion of very few patients with septic shock and confirmed myocardial dysfunction and to the administration of relatively high dose of levosimendan (up to 0.2 µg/kg/min), which induced marked tachycardia and hypotension, the vasodilating properties of drug possibly aggravating sepsis-induced vasoplegia in this setting.

Meta-analyses of small trials [19] suggested that levosimendan may result in a higher rate of survival among patients at risk of developing or presenting the low cardiac output syndrome after cardiac surgery and set the stage for 3 large RCTs testing the drug in this setting. The LICORN [20] (n=336 patients) and the LEVO-CTS [21] (n=882 patients) trials assessed the ability of preoperative levosimendan to prevent postoperative low cardiac output syndrome in patients with pre-operative left ventricular dysfunction. The continuous prophylactic infusion of 0.1 µg/kg/min was started just after anesthetic induction and was maintained over 24 hours in the absence of serious adverse reactions. The occurrence of the composite end-points of prolonged catecholamine infusion, use of left ventricular mechanical assist device, or renal replacement therapy (LICORN) and of death, renal-replacement therapy, perioperative myocardial infarction, or use of a mechanical cardiac assist device (LEVO-CTS) and the rates of adverse events were not different between levosimendan and placebo treated patients in these studies. It should be mentioned that only 22% of the patients developed low cardiac output syndrome in the LEVO-CTS trial and that in both the LICORN and the LEVO-CTS trials, only half of the patients needed catecholamine support for >24 hours. Thus, the preventive use of levosimendan does not seem to confer clinical benefits in this setting. Alternatively, the CHEETAH trial [22] included only patients who developed perioperative or early post-operative cardiovascular dysfunction and low cardiac output syndrome, defined as the need for support with an intraaortic balloon pump or high-dose inotropic support in patients with preoperative left ventricular ejection fraction of less than 25%. Patients were randomly assigned to receive levosimendan at a dose of 0.025 to 0.2 µg/kg/min or placebo, with no loading dose, for up to 48 hours or until discharge from the intensive care unit. The trial was stopped early for futility after 506 patients were enrolled and there

was no significant difference in the primary end-point of 30-day mortality between the levosimendan group and the placebo group (32 patients [12.9%] and 33 patients [12.8%], respectively; absolute risk difference, 0.1 percentage points; 95% confidence interval [CI], -5.7 to 5.9; $P = 0.97$). Rates of hypotension or cardiac arrhythmias were also not different between the levosimendan and placebo groups. It should however be noted that the mean infusion dose of levosimendan ($0.07 \mu\text{g/kg/min}$) was lower in this trial than in the 2 other RCTs of cardiac surgery patients.

2.4 Rationale for a trial testing levosimendan in VA-ECMO patients and justification for the choice of subjects

Preliminary retrospective cohort studies suggested a benefit of the early administration of levosimendan in refractory cardiogenic shock patients on VA-ECMO. In a small “before-after” study of medical cardiogenic shock patients, Affronti et al. [23] showed that the drug was associated with a higher rate of VA-ECMO weaning compared to historical controls, respectively 83.3% vs 27.3% ($p=0.498$). Similarly, Tafer et al [24] reported more ECMO-free days (22 vs 9) and lower 30-day mortality (24% vs 56%, $p<0.05$) in a cohort of 80 patients of whom 37 had received levosimendan early after ECMO initiation. In a series of 240 patients undergoing VA-ECMO after cardio-vascular surgery, of whom 75% received levosimendan immediately after ECMO initiation [25], the treatment was also associated with higher rates of successful ECMO weaning [HR; 0.41; 95% CI, 0.22–0.80; $P=0.008$]. Additionally, lower 30-day (adjusted HR 0.52; 95% CI 0.30–0.89; $P=0.016$), and long-term survival (adjusted HR 0.64; 95% CI 0.42–0.98; $P=0.04$) were reported, although levosimendan-treated patients had higher baseline EuroSCORE ($P<0.001$) and SAPS-3 ($P=0.02$) and lower LVEF ($p=0.04$).

Indeed, the rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support. Furthermore, it might be possible to administer higher doses of levosimendan to this specific group of patients, since VA-ECMO also provides robust peripheral vascular support that might protect against vasoplegia and hypotension that were suspected to hamper the benefit of the drug in previous trials.

In conclusion, the probability of demonstrating lower mortality with levosimendan in cardiogenic shock patients on ECMO appears unlikely based on the results of recent large RCTs. However, any medical therapy that may be able to hasten myocardial recovery and shorten the duration of VA-ECMO support might translate in significantly less morbidity and ultimately reduced ICU and hospital length of stays and associated costs. Highly relevant pathophysiological and clinical reasons indicate that levosimendan might fulfill these objectives. A multicenter randomized trial with sufficient statistical power is therefore urgently needed to test the early administration of Levosimendan to facilitate and accelerate VA-ECMO weaning in refractory cardiogenic shock patients supported by VA-ECMO.

2.5 Name and description of the investigational medicinal product(s)

- Levosimendan (ZIMINO)
- Placebo of Levosimendan

See section 7 for more details

2.6 Description and justification of the dosage, route of administration, administration schedule and treatment duration

See section 7 for further information about the administration of the product

2.7 Summary of the known and foreseeable benefits and risks for the study participants

Reducing the duration of ECMO support with levosimendan may be associated with less ECMO-related complications such as hemorrhages, stroke or infections. It may also be accompanied by a reduction in the duration of ICU and hospital stays, therefore reducing the costs associated with the care of these critically ill patients. Better long-term outcomes might also be expected with earlier ECMO weaning in these severe cardiogenic shock patients.

Known risk associated with levosimendan infusion include hypotension and supraventricular and ventricular arrhythmias. However, this side effects may be significantly mitigated in patient on VA-ECMO.

3 OBJECTIVES

3.1 Primary objective

The primary objective will be to assess the efficacy of the early administration of levosimendan on the time to successful ECMO weaning in patients on VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory device or heart transplantation 30 days after ECMO removal [8, 36]. Patients still under ECMO 30 days after randomization will be considered as censored at 30 days.

Patients with unsuccessful ECMO weaning will be considered as weaning failure (competing risk) at the date of the weaning attempt.

3.2 Secondary objectives

Secondary objectives of the trial are to assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function.

- Mortality on D30 and D60
- Duration of ECMO support between inclusion and D30, and between inclusion and D60; and number of ECMO-free days between inclusion and D30/D60
- Duration of ICU stay; of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60.
- Time to improvement in hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization
- Number of days with organ failure, defined by SOFA score, and number of days without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of drug adverse effects (atrial fibrillation, ventricular arrhythmias, ventricular fibrillation, “torsades de pointes”, hypokalemia)
- Primary and secondary endpoints in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4 DESCRIPTION OF THE TRIAL

4.1 Concise description of the primary and secondary endpoints

4.1.1 Primary endpoint

- Time to successful ECMO weaning within the 30 days following randomization

- The event of interest is the successful weaning of ECMO (within 30 days after randomization) defined as: the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal [8, 36]. Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization).
- ECMO weaning failure (i.e. unsuccessful weaning), within 30 days after randomization, is a competing event.
- Death within 30 days after randomization, without an attempt to wean, is a competing event.
- No event after 30 days after randomization will be considered and all patients without any of the three events defined above are censored at 30 days post randomization.
- A protocolized management regarding weaning of VA-ECMO will be applied to both groups (see paragraph 5.2.1 for details) [8, 36].
- The number of days alive without ECMO was not an appropriate primary outcome criterion due to the high proportion of patients dying on ECMO and thus ex-aequo (0 days). The analysis will therefore **model the risk of successful ECMO weaning in the presence of competing risks (death and weaning failure)**.

4.1.2 Secondary endpoints

- Mortality on Day 30 and Day 60
- Total duration of ECMO support between inclusion and D30 and between inclusion and D60;
- Number of ECMO-free days between inclusion and D30/D60.
- Duration of ICU stay and of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60
- Time to improvement in hemodynamic parameters (systolic, diastolic, mean blood pressure, heart rate ...) and time to hemodynamic stabilization
- Days with organ failure, defined by the SOFA score, and days alive without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of adverse drug reactions (including atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia).

The Primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4.2 Research methodology

4.2.1 Design of the trial

Characteristics of the trial are as follows:

- Multicenter clinical trial
- Phase III

- Randomized
- Double blind
- Comparative, Placebo controlled
- Superiority design
- Two treatment arms
 - Early administration of Levosimendan vs. placebo
 - 1:1 ratio

All consecutive eligible patients will be proposed to participate.

Allocation concealment will be guaranteed by use of a centralized randomization process with a web server (see below) and use of the minimization method of randomization. Investigator masking to group assignment after randomization will be guaranteed by use of a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

Because acute cardiogenic shock is a complex and multifactorial process, one key issue in every trial evaluating a new therapeutic strategy is how patients in both groups are managed and how standard care for these patients recommended by current guidelines are implemented in the participating ICUs. In the LEVOECMO trial, common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

4.2.2 Number of participating sites

The study will involve 13 sites in France. Subjects will be recruited in ICU departments.

4.2.3 Identification of the subjects

All consecutive eligible patients will be proposed to participate.

The subjects participating in this study will be identified as follows:

Site number (3 digits) - Sequential selection number for the site (4 digits) - surname initial - first name initial

This reference number is unique and will be used for the entire duration of the trial.

A randomisation number will also be assigned when the participant is randomised. This number will have the following format: XXX.

A treatment number corresponding to the treatment arm will be also assigned to the patient when the participant is allocated to a treatment arm. This number will have the following format: TXXXXX.

A registry of eligible patients not included will be completed, with reason for non-inclusion.

4.2.4 Randomization

After baseline screening, patients will be randomly assigned in a 1:1 ratio to receive levosimendan or placebo. The computer-generated randomization process will be implemented using the minimization method with a probability of assignment of 0.8 after the 20 first included patients (being assigned by balanced random) by a statistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence TUBACH. Stochastic minimization will be according to the primary etiology of cardiogenic shock (acute myocardial infarction, myocarditis, post-heart surgery and other causes) and by center. The assignments of participants will be concealed by use of a centralized 24-hour Internet service (CleanWeb, Télémédecine Technologies, Boulogne, France) and thus without knowledge of the next assignment in the sequence.

4.2.5 Blinding methods and measures put in place to protect the blinding

Investigator masking to group assignment after randomization will be guaranteed by use a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

The study will be double blinded. Treatments will be prepared at ICU by a nurse.

Levosimendan and the placebo will be administered as non-identifiable preparation. The placebo will be made of riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used daily as a dietary supplement in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing Levosimendan [18]. The appearance (color and presentation) of placebo of levosimendan will be strictly identical to levosimendan to maintain double blind.

4.2.6 Unblinding procedures, if applicable

The investigating doctor may request unblinding for any reason he considers essential, by contacting:

- the DRCI (sponsor) **in non-urgent situations**, on working days and during working hours; the request should be addressed to the DRCI's project advisor, tel.: 01 40 27 57 47
- the poison centre of Fernand Widal Hospital, in the case of an emergency (see emergency situations requiring unblinding), on weekends, bank holidays, when the DRCI is closed and when unblinding cannot be carried out by the DRCI. Telephone: **+33 (0)1 40 05 48 48**
- the hospital pharmacy

5 **PROCEDURE FOR THE TRIAL**

5.1 **Baseline visit = J1 (inclusion and randomization)**

5.1.1 *Verification of inclusion and exclusion criteria (see paragraph 6)*

5.1.2 *Consent (see paragraph 14.1 for details)*

Whose consent must be obtained	Who informs the individual and collects their consent	When is the individual informed	When is the individual's consent collected
• Patient's legal representative, family member or a close relative	• Local site investigator	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow

5.1.3 **Treatment**

- Reconstitution/Preparation of the treatment before administration
- Administration by IV perfusion during 24h
- Administration as soon as possible after randomization; no later than 24 hours after randomization.

5.1.4 **Clinical data**

- Date, time of hospital and ICU admission
- Date, time of ECMO initiation
- Reason for ECMO initiation
- Date of birth
- Sex
- McCabe and Jackson score
- Charlson comorbidity index score
- history of systemic hypertension, myocardial infarction, diabetes mellitus, preexisting cardiovascular diseases (stroke, carotid surgery), severe left ventricle dysfunction (NYHA stage and ejection fraction determined by echocardiography or angiography), chronic renal insufficiency (serum creatinine, chronic dialysis), and hepatic insufficiency, defined as serum bilirubin >100 µmol/L or albuminemia <35 g/L with no other cause found
- immunocompromised: yes/no,
- SAPS II (table to check-off)
- Sequential organ-failure assessment (SOFA) score (table to check-off)
- Hemodynamic parameters (including arterial pressure and heart rate)
- Catecholamine doses

5.1.5 **Biological and echocardiographic data**

- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers

- Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,)

5.1.6 ECMO parameters

- FiO₂ and membrane ventilation
- Pump turns per minute and flow
- Fibrin/clots on the membrane: yes/no
- Clogged circuit requiring change: yes/no
- Intravascular hemolysis caused by the circuit: yes/no
- Hemorrhage at the cannula insertion site: yes/no
- Infection at the cannula insertion site: yes/no
- Circuit changed: yes/no
- Leg ischemia: yes/no

5.2 Follow-up visits

5.2.1 The following Information will be collected on days 1, 2, 3... 30 or date of death or date of ICU discharge

- Hemodynamic parameters (arterial pressure, heart rate).
- Catecholamine doses
- Mechanical ventilation: yes/no
- Renal replacement therapy: yes/no
- Urine output
- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers
- SOFA score
- ECG
- Echocardiographic parameters (LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus).
- Weanability test [8, 36]: yes/no
 - A protocolized management regarding weaning of VA-ECMO will be applied to both groups.
 - An ECMO-weaning trial will be undertaken every day as soon as the patient will be considered hemodynamically stable, i.e. baseline MBP >60 mm Hg while receiving no or low-dose vasopressors and a pulsatile arterial waveform maintained for at least 24 h.
 - The ECMO flow will be decreased gradually decreased to 66%, and to 33% of its baseline value and then to a minimum of 1–1.5 L/min.
 - If MBP drops significantly and constantly <60 mm Hg during the trial, ECMO flow will be returned to 100% of the initial flow and the trial will be stopped.
 - Patients achieving aortic VTI ≥12 cm, LVEF >20-25% and TDSa ≥6 cm/s at minimal ECMO flow support, will be considered weanable from ECMO [8, 36].
- ECMO parameters until device removal
 - FiO₂ and membrane ventilation
 - Pump turns per minute and flow
 - Fibrin/clots on the membrane: yes/no
 - Clogged circuit requiring change: yes/no
 - Intravascular hemolysis caused by the circuit: yes/no
 - Hemorrhage at the cannula insertion site: yes/no
 - Infection at the cannula insertion site: yes/no
 - Circuit changed: yes/no

- Leg ischemia: yes/no

5.2.2 Outcomes on days 30 and 60

- Living/dead; died: date and hour
- On ECMO: yes/no
- In case of ECMO weaning, need for a second ECMO run yes/no
- On other mechanical cardiac support device: yes/no
- Heart transplantation: yes/no
- Mechanical ventilation: yes/no
- In the ICU: yes/no; discharged from the ICU: date
- Hospitalized; yes/no; discharged from the hospital: date
- Patient's contact information will be collected at ICU discharge to permit 60 day outcome data collection.

5.3 Expected length of participation, chronology and duration of the study.

- Length of enrolment period: 24 months
- Duration of participation for each subject: 60 days
- Treatment period: 24 hours
- Follow-up period: 60 days
- Total study duration: (start = trial start date; end = trial end date) = 26 months
- When randomization and blinding will take place: immediately after patients are deemed eligible.

5.4 Table or diagram summarising the chronology of the study

Actions	D1	D2 to D30 or date of death or date of ICU discharge	D60
Information	X		
Signature of the consent form	X		
Randomization	X		
Past medical history	X		
Clinical exam	X	X	
Medical procedures (ECG, Echo.)	X	X	
Dispensation of LEVOSIMENDAN or Placebo	X		
Adverse events	X	X	X
Vital status	X	X	X
ECMO status	X	X	X
Pursuit consent		X	X (if not obtained before)

5.5 Distinction between standard care and research

Procedures and treatments carried out as part of the research	Procedures and treatments associated with <u>care</u>	Procedures and treatments added because of <u>the research</u>
Treatments	None	Levosimendan or Placebo at D1
Consultations	D1 to D30 : During patient's hospitalisation's	Follow up by phone :Day 60

Blood samples	Hematology - Biochemistry	None
Imaging, etc.	ECG Echocardiography	None
Act	ECMO	None

In the LEVOECMO trial, only the initial treatment undergoing randomization (levosimendan vs. placebo) will differ between groups.

Common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

5.6 Termination and exit rules

5.6.1 Criteria and procedures for prematurely terminating the study treatment

- Temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the subject's source file and the case report form (CRF). The subject remains enrolled in the study
- Premature termination of treatment, but the subject remains enrolled in the study until the end of the subject's participation: the investigator must document the reason
- Premature termination of treatment and exit from the study. The only reasons for premature exit from the study are: death, lost to follow-up, or patient's consent withdrawal. Premature termination of treatment or adverse reaction should not imply premature exit from the study.

The investigator must:

- Document the reason(s)
- Collect all endpoints at the moment the subject exits from the study, if the subject agrees
- Schedule further follow-up visits, especially in case of a serious adverse event.

Study treatment must be discontinued if they experience any of the following:

- Anaphylactic shock
- Ventricular fibrillation
- "Torsades de pointes"
- Ventricular tachycardia
- Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for more than 60 minutes despite:
 - vascular filling with two 500 mL successive crystalloid boluses AND
 - norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.

Criteria and procedure for premature withdrawals and exits from the study

- Subject may exit the study at any time and for any reason.
- The investigator can temporarily or permanently withdraw a subject from the study for any safety reason or if it is in the subject's best interests
- Subject lost to follow-up: the subject cannot be located. The investigator must make every effort to reconnect with the subject and record his attempts in the source file (at least to determine whether the subject is alive or dead).

If a participant leaves the research prematurely, data relating to the participant can be used unless an objection was recorded when the participant signed the consent form.

If a subject exits the trial prematurely or withdraws consent, any data collected prior to the date of premature exit may still be used.

The case report form must list the various reasons why the subject exited or was withdrawn from the study:

- ☐ Lack of efficacy
- ☐ Adverse reaction
- ☐ Other medical problem
- ☐ Subject's personal reasons
- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up
- ☐ Death

5.6.2 Monitoring subjects after the premature termination of treatment

If a subject exits the trial this will in no way affect the standard care received for his/her condition. In case of severe adverse events, the investigator must notify the sponsor and monitor the subject for 30 days following the premature termination of treatment. If the treatment is stopped prematurely due to a serious adverse event, a serious adverse event report will be sent by mail (vigilance.drc@aphp.fr) to the sponsor. The serious adverse reaction will be monitored until it is resolved.

5.6.3 Full or partial cancellation of the study

AH-HP (the sponsor) or the Competent Authority (ANSM) may prematurely discontinue all or part of the trial, temporarily or permanently, upon the recommendation of the Data Monitoring Committee in the following situations:

- first, if suspected unexpected serious adverse reactions (SUSARs) are observed in one of the treatment arms or if there is a discrepancy in the serious adverse reactions between the two treatment arms, requiring a reassessment of the benefit-risk ratio for the trial.
- similarly, AH-HP, as the sponsor, or the Competent Authority (ANSM) may prematurely cancel the trial due to the discovery of unexpected facts or new information about the product, in light of which the objectives of the study or clinical programme are unlikely to be achieved .
- AP-HP, as the sponsor, reserves the right to permanently suspend enrolment at any time if the enrolment targets have not been met.

If the study is cancelled prematurely, AP-HP will inform the Competent Authority (ANSM) and the Institutional Review Board of its decision within 15 days, together with justification for the decision and any recommendations from the Data Monitoring Committee (if applicable).

6 SELECTION CRITERIA

6.1 Identification of Eligible Patients: Inclusion and Exclusion Criteria

Adult patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support will be screened for eligibility.

6.1.1 Inclusion criteria

1. **Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h.**

(The rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support).

2. **Obtain informed consent from a close relative or surrogate.** According to the specifications of emergency consent, randomization without the close relative or surrogate consent could be performed.

Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

6.1.2 Exclusion criteria

1. Age <18
2. Pregnant or lactating women
3. Initiation of ECMO >48 h
4. Resuscitation >30 minutes before ECMO
5. Irreversible neurological pathology
6. End-stage cardiomyopathy with no hope of LV function recovery
7. Mechanical complication of myocardial infarction
8. Aortic regurgitation > II
9. VA-ECMO for pulmonary embolism
10. VA-ECMO for cardiotoxic drug intoxication
11. VA-ECMO after left-ventricle assist device implantation
12. VA-ECMO in heart transplant patients
13. Patient moribund on the day of randomization, SAPS II >90
14. Liver cirrhosis (Child B or C) and other severe hepatic insufficiency
15. Chronic renal failure requiring hemodialysis
16. Known hypersensitivity to levosimendan
17. Prior history of "torsades de pointes"
18. History of epilepsy
19. Individuals under guardianship, or permanently legally incompetent adults
20. Participation to another interventional study

6.2 Consent and randomization procedure

Patients satisfying all of the Inclusion and Exclusion Criteria are classified as 'Eligible'.

To randomize an eligible patient, consent must be obtained **from a close relative or surrogate**.

Should such a person be absent, the patient will be randomized according to the specifications of **emergency enrollment** and the patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

Reasons for not enrolling an eligible patient will be recorded (lack of consent from substitute decision maker; refusal from attending physician; study team not available; study team missed the patient, patient enrolled in another trial).

6.3 Recruitment methods

Patients will be recruited in medical and surgical intensive care units participating to the study.

Number of subjects to be included during the enrolment period, number of subjects to be enrolled per site per month, based on the number of sites and the duration of the enrolment period are provided in the following tables.

Study characteristics	Value
Total number of subjects	206
Number of sites	13
Enrolment period (months)	24
Number of subjects/site	15,8
Number of subjects/site/month	0.66

7 TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

7.1 The investigational medicinal product(s) : levosimendan and placebo

7.1.1 Levosimendan characteristics

Levosimendan facilitates contact between actin and myosin and prolongs the time of contact between these two molecules, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on the coronary arteries) and a cardio protective anti-ischemic effect [29-35].

Metabolism and Pharmacokinetics

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with a half-life of about three days. The metabolite OR-1896 has been shown to have hemodynamic and pharmacologic properties similar to those of the parent drug in preclinical models [16].

Pharmacodynamics

Levosimendan produces significant, dose-dependent increases in cardiac output, stroke volume and heart rate, and decreases in pulmonary capillary wedge pressure, mean blood pressure, mean pulmonary artery pressure, mean right atrial pressure and total peripheral resistance.

There is no sign of development of tolerance even with a prolonged infusion up to 48 hours. Due to the formation of the active metabolite, the hemodynamic effects are maintained several days after stopping levosimendan infusion. Compared with dobutamine, levosimendan produces a slightly greater increase in cardiac output and a profoundly greater decrease in pulmonary capillary wedge pressure. In contrast to dobutamine, the hemodynamic effects are not attenuated with concomitant beta-blocker use. It has also been shown that at 48 hours after the start of infusion, a 24-hour infusion achieves superior hemodynamic effects over a 48-hour dobutamine infusion in patients with severe acute decompensated heart failure on beta-blockers. decrease in natriuretic peptides. The beneficial effects of levosimendan on hemodynamics and neurohormones are not associated with any significant increase in myocardial energy consumption, as evidenced using dynamic positron emission tomography (PET) in hospitalized patients with NYHA III-IV heart failure.

7.1.2 Description of investigational products

Presentation

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, containing 5 mL of solution for infusion will be provided free of charge by the Orion Corporation.

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, will be packaged in double-blinded boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP).

Each boxes contains 3 vials of 5 ml of levosimendan or placebo.

Each box will be assigned a unique treatment number according to the treatment numbers list.

Supply :

The numbered boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, will be sent on investigational sites by CTD of AGEPS.

Storage

Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, must be kept at +2 to +8 °C in the outside packaging, shielded from the light.

A continuous infusion of Levosimendan will be administered over 24 h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate-limiting side effects.

7.1.3 Treatment preparation and dispensation

The pharmacy of each center will be responsible dispensing treatment's boxes in the ICU after initiation visit. The products will be packaged and labelled as packs of three vials of levosimendan or placebo labelled and numbered according to the list of treatment numbers provided by the sponsor.

The treatment will be prepared by a nurse in the ICU (according to the following table)

Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h
40 to 46 kg	1	250 mL	8	11
47 to 89 kg	2	500 mL	16	22
90 to 132 kg	3	1000 mL	32	44

7.2 Traceability information for the investigational medicinal product(s)

Sponsor will provide traceability documents to enable the regulatory traceability of the investigational medicinal products on each investigational site.

7.3 Common management regarding other inotropic drugs, heart failure medications and antiaggregants and anticoagulants

Additional inotropic agents may be used at any time in either treatment group as clinically indicated and as recommended by the latest European Society of Cardiology [26]. Dobutamine will be the inotropic agent of choice.

Other heart failure medications (such as diuretics, ACE inhibitors, steroidal antimineralocorticoids), antiaggregants and anticoagulants will be prescribed according to the latest European Society of Cardiology [26].

Open-label Levosimendan (ZIMINO) will not be possible in either group during the entire course of the study.

7.4 Common management regarding the ECMO circuit

– Parameter settings for ECMO

The pump flow will be set to 3.5-5 l/min, to provide adequate systemic perfusion. Pump flow might be reduced in case of cessation of LV ejection or major pulmonary edema. Percentage of oxygen contained in the ventilating gaseous air–oxygen mixture will be adjusted to obtain PaO₂ between 65 and 90 mmHg and/or arterial oxygen saturation >90%. The membrane ventilation will be adjusted to maintain PaCO₂ between 40 and 45 mmHg. Extracorporeal circulation will be run at normal body temperature.

– ECMO monitoring (extracorporeal circuit, anticoagulation, possible complications)

The ECMO circuit will be monitored several times daily by the medical and nursing team caring for the patient and at least once every 48 hours by a perfusionist. Circuit and cannula surveillance is intended to verify the correct functioning of the device and early screening for complications (leg ischemia, fibrin deposits or clots on the ECMO membrane, clots in the cannulae or in the pump, bleeding or signs of

inflammation or cutaneous infection at the cannula insertion sites, unexpected drop of the ECMO outflow, appearance of clinical or biological signs of intravascular hemolysis). Should any of these complications occur, a medical–surgical consultation will be held to discuss the best therapeutic approach to take.

Anticoagulation will be obtained with non-fractionated heparin to a target aPTT of 55-70 sec or heparinemia (antiXa activity) between 0.4 and 0.6 IU/ml. Should severe bleeding occur that is not immediately controllable by specific treatment, heparin will be discontinued.

Intravascular hemolysis will be sought should unexpected dark urine be excreted or in the case of obvious circuit dysfunction. It is recommended that plasma free hemoglobin be measured every 48 hours and immediately if hemolysis linked to the circuit is suspected.

The membrane and ECMO circuit will be changed in the following situations: massive intravascular hemolysis linked to the device, severe thrombopenia linked to the circuit, clots preventing the pump and/or lines from functioning properly or systemic defibrination.

The hemoglobin threshold for red cells transfusion will be 9-10 g/dl (may be decreased to 7-8 g/dl if the patient is stabilized and does not have residual myocardial ischemia). Platelet transfusion will be discouraged except when severe thrombopenia is associated by bleeding complications or when platelet count will be <20 G/L.

Connecting an extrarenal dialysis circuit to the ECMO circuit will be permitted under strict supervision of perfusionists.

7.5 Authorised and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications

No treatment is prohibited during the study, except open-label Levosimendan (ZIMINO). Additional inotropic agents (except digoxine) may be used at any time in either treatment group as clinically indicated. Dobutamine will be the inotropic agent of choice.

7.6 Methods for monitoring compliance with the treatment

Compliance to study drug will be assured by the physician in charge of the patient during the 24 hour infusion of the treatment.

8 EFFICACY ASSESSMENT

8.1 Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

The measures of all efficacy parameters rely on the clinical evaluation of the patients from the day of randomization (Day 1) up-to day 60 post-randomization. All information will be recorded by the investigator, nurses or study coordinator.

9 SPECIFIC COMMITTEES FOR THE TRIAL

9.1 Steering Committee (SC)

- Members of the committee: All primary investigators at study sites.
- Missions: The SC will define the overall structure of the study, coordinate information, determine the initial methodology and oversee the trial. Also determine the objective, write the protocol, recommend changes to the protocol during the trial

10 SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY

10.1 Recording and reporting adverse events

10.1.1 Definitions

According to Article R1123-46 of the French Public Health Code:

- **Adverse event**

Any untoward medical occurrence in a trial subject, which does not necessarily have a causal relationship with the clinical trial or with the investigational product.

- **Adverse reaction to an investigational medicinal product**

Any adverse event occurred in a trial subject, which has a causal relationship with the clinical trial or with the investigational medicinal product

- **Serious adverse event or reaction**

Any adverse event or reaction that at any dose of medication, results in death, threatens the life of the research subject, requires hospitalisation or prolongs hospitalisation, causes a severe or long-term disability or handicap, or results in a congenital abnormality or deformity.

- **Unexpected adverse reaction to an investigational medicinal product**

Any adverse reaction to the product, whose nature, severity, frequency or outcome is inconsistent with the safety information described in the Reference Safety Information (summary of product characteristics, or the investigator's brochure if the product is not authorised).

According to Article R.1123-46 of the Code de la Santé Publique and the guidelines for clinical trial sponsors (ANSM):

- **Emerging safety issue**

Any new safety information that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials..

For the clinical trials involving the first administration or use of an investigational medicinal product in healthy volunteers, any serious adverse reaction.

Examples:

- a) any clinically significant increase in the frequency of an expected serious adverse reaction
- b) suspected unexpected serious adverse reactions in patients who have terminated their participation in the clinical trial that are notified by the investigator to the sponsor together with follow-up reports
- c) any new safety issue relating to the conduct of the clinical trial or the development of the investigational medicinal product, that may impact the safety of the trial subjects.

Examples:

- a serious adverse event likely to be related to the interventions and the trial's diagnostic procedures and which may impact the conduct of the clinical trial,
 - a significant risk on the trial subjects such as ineffectiveness of the investigational medicinal product in treating a life-threatening illness under investigation,
 - significant safety results from a recently completed non-clinical study (such as a carcinogenicity study),
 - the premature termination, or temporary suspension, of a trial conducted on the same investigational medicinal product in another country, for safety reasons,
 - an unexpected serious adverse reaction associated with a non-experimental medication required for the conduct of the clinical trial, (e.g. challenge agents, rescue treatment)
- d) recommendations from the Data Safety Monitoring Board (DSMB), if applicable, that may affect the safety of the trial subjects
- e) any suspected unexpected serious adverse reaction (SUSAR) reported to the sponsor by another sponsor of a trial carried out in a different country but relating to the same medication.

10.1.2 The role of the investigator

The investigator must **assess the seriousness criteria of each adverse event** and record all serious and non-serious adverse events in the case report form eCRF.

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the severity** of the adverse events by using:

- general terms:

- *Mild: tolerated by the patient, does not interfere with daily activities*
- *Moderate: sufficiently uncomfortable to affect daily activities*
- *Serious: preventing daily activities*

- or a severity grading scale for adverse events, attached to the protocol

The investigator must assess the **causal relationship** between the serious adverse events and the investigational medicinal products.

The method used by the investigator is based on the WHO Uppsala Monitoring Centre method and uses the following causality terms:

- Certain
- Probable/likely
- Possible
- Unlikely (not ruled out).

These terms are defined as follows (extracted from the WHO-UMC causality categories, version dated 17/04/2012).

Table : WHO-UMC causality categories (extract)

Causality term	Assessment criteria*
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake ** • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognized pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable / Likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake** • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with reasonable time relationship to drug intake ** • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake ** • that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations

*All points should be reasonably complied with

** Or study procedures

10.1.2.1 Serious adverse events that require a notification without delay by the investigator to the sponsor

According to the article R.1123-49 of the French Public Health Code (CSP), the investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any serious adverse event which occurs during a trial as described in Article L.1121-1(1) CSP, except those which are listed in the protocol (see corresponding section) and, if applicable, in the investigator's brochure as not requiring a notification without delay. These latter should be notified by the investigator to the sponsor in an appropriate delay taking into consideration the specific features of the trial, the serious adverse events and the modalities specified in the protocol or the investigator's brochure.

A serious adverse event is any untoward medical occurrence that:

- 1- results in death
- 2- is life-threatening
- 3- requires inpatient hospitalisation or prolongation of existing hospitalisation
- 4- results in persistent or significant disability/incapacity
- 5- is a congenital anomaly/birth defect

10.1.2.2 Specific features of the protocol

10.1.2.2.1 Other events that require the investigator to notify the sponsor without delay

- Adverse events judged as **being directly responsible for patient death**:

– **Under VA-ECMO**

The mortality rate of patients with refractory cardiogenic shock supported by VA-ECMO is very high (40-60%) [2, 11-13].

Therefore, **only deaths considered as directly related the experimental treatment** will be notified to the sponsor without delay in the LEVOECMO trial.

Specifically, ventricular rhythms disturbances (ventricular fibrillation, « Torsades de pointes », ventricular tachycardia) do not incur a risk of immediate death in patients under VA-ECMO support and should not be notified without delay when occurring in patients under VA-ECMO.

– **After weaning ECMO**

After ECMO weaning because of cardiac recovery, the following adverse events will also be notified without delay:

- Ventricular fibrillation
- « Torsades de pointes »
- Ventricular tachycardia
- Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for > 60 minutes despite vascular filling with two 500 mL successive crystalloid boluses AND norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of these adverse events, according to the same modalities and within the same timeline as for serious adverse events (see above).

- **In utero exposure**

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any pregnancy that occurs during the trial, even if not associated with an adverse event.

10.1.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are simply recorded in the case report form. A CRF extraction of these serious adverse events will be done according Data Safety Monitoring Board (DSMB) or AP-HP's Safety department's request.

- **Normal and natural course of the condition**:
 - Death considered as related to the natural evolution of cardiogenic shock or multiple organ failure and not to the infusion of the study drug
 - Prolongation of hospitalization due to surgery
 - Rehospitalization considered as not related to cardiogenic shock or for disease follow-up

- Events likely to be exclusively due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion...)

Special rules for trials with a high mortality rate:

The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13].

Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form.

A CRF extraction of all deaths whatever the cause will be realized every month by clinical research unit. These data will be sent to the DSMB members and the AP-HP's safety department's .

If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay.

NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor.

- *Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care*

The investigator must report these events to his *Centre Régional de Pharmacovigilance* (CRPV).

- *Adverse events specifically due to the treatment with Levosimendan*
 - Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation)
 - Moderate tachycardia (HR <110 bpm)
 - Premature ventricular contraction
 - Premature atrial contraction, atrial fibrillation
 - Nausea
 - Vomiting
 - Diarrhea – Constipation - Dizziness
 - Headache
 - Insomnia
 - Mild to moderate hypokalemia (3 to 3.5 mmol/L)

10.1.2.3 Period during which the investigator must send notification of SAEs to the sponsor without delay

The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section:

- starting from the date on which the subject signs the consent form/begins treatment with the investigational medicinal product/undergoes the first intervention specific to the study
- throughout the whole follow-up period intended by the trial
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear at long term after exposure to the medication, such as cancers or congenital abnormalities)

10.1.2.4 Procedures and deadlines for notifying the sponsor

The investigator should initially complete a SAE reporting form (contained in the case report form). This report must be signed by the investigator.

The investigator must complete every section of the SAE form so that the sponsor can carry out the appropriate assessment.

The initial report sent to the sponsor must be rapidly followed up by one or more additional written reports describing the course of the event and any complementary information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful for medical assessment of the case (medical reports, laboratory test results, results of additional exams, etc.). These documents must be anonymized. In addition, the investigator must state the study acronym and the number and initials of the study participant on each paper.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the subject has terminated his participation in the trial.

The initial report, the SAE follow-up reports, and all other documents must be sent to AP-HP by e-mail (eig-vigilance.drc@aphp.fr). It is possible to send the SAE to the AP-HP's Safety department by fax No. **+33 (0)1 44 84 17 99** only in case of unsuccessful attempt to send the SAE by e-mail. This is to avoid duplicated reports.

For trials which use e-CRF

- the investigator completes the SAE report form in the e-CRF, then validates, prints and signs the form before sending it by mail;
- In case of failure to connect to the e-CRF, the investigator should complete, sign and send the SAE report form to the safety Department. As soon as the connection is restored, the investigator must complete the SAE report form in the e-CRF.

The investigator must comply with all requests for additional information from the sponsor.

For all questions relating to an adverse event report, the safety Department can be contacted via email at vigilance.drc@aphp.fr.

10.1.3 Role of the sponsor

The sponsor, represented by its safety Department, shall continuously assess the safety of each investigational medicinal product throughout the trial.

10.1.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all reported adverse events,
- the **causal relationship** between these adverse events and investigational medicinal product and any other treatments,

All serious adverse events for which the investigator and/or the sponsor suspect a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.

- the **expectedness assessment** of the serious adverse reactions

Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.

The sponsor, acting through its safety Department, assesses the expectedness of the serious adverse reaction based on the information described below.

- ❖ For serious adverse events likely to be related to the investigational medicinal product(s):
 - refer to the SmPC for Levosimendan (Zimino®) enclosed in appendix N°18.4.

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs), within the regulatory time frame, to the ANSM (French Health Products Safety Agency).

- The sponsor must send the initial report without delay upon receipt of the unexpected serious adverse reaction if it is fatal or life-threatening, or otherwise within 15 days from receipt of any other type of unexpected serious adverse reaction;
- The sponsor must provide all relevant additional information by sending follow-up reports, within 8 calendar days following receipt.

Any suspected unexpected serious adverse reaction must also be declared electronically using the Eudragilance European adverse drug reactions database managed by the European Medicines Agency (EMA).

The sponsor must notify all the investigators about any information that could adversely affect the safety of the trial subjects.

Special rules for double blind trials

After unblinding, if the administered product is the investigational medicinal product, the case will be immediately reported as suspected serious unexpected adverse reaction (SUSAR); however, if the administered product is the comparator, the unexpected nature of the adverse reaction will be re-assessed according to the reference document of the comparator found in the protocol. In the exceptional case of a clinical study about a disease with high mortality or morbidity, the sponsor may request the ANSM a readjustment of the conditions for unblinding and reporting suspected adverse reaction. These conditions should be thoroughly defined in the study protocol.

10.1.3.2 Analysis and declaration of other safety data

This relates to any new safety data that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials.

The sponsor will inform the competent authority and the Ethics committee without delay upon knowledge any emerging safety issues and, if applicable, describe what urgent safety measures have been taken by the sponsor.

Following the initial declaration of emerging safety issue, the sponsor will declare to ANSM any additional relevant information about the new safety issues in the form of a follow-up report, which must be sent no later than 8 days after becoming aware of the information.

If the suspected unexpected serious adverse reaction meets the definition of an emerging safety issue, the sponsor will report both the SUSAR and the emerging safety issue to the ANSM according to the appropriate modalities and within the regulatory timelines as previously described.

10.1.3.3 Annual safety report

The sponsor must prepare once yearly throughout the trial duration an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- an analysis of safety data concerning trial subjects
- a description of the patients included in the trial (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- cumulative summary tabulation of all the serious adverse events that have occurred since the beginning of the clinical trial,

The report must be transmitted to ANSM and the Ethics Committee no later than 60 days after the anniversary date corresponding to the date of authorization of the clinical trial by ANSM.

10.1.4 Independent Data Safety Monitoring Board (DSMB)

An independent Data Safety Monitoring Board (DSMB) will be set up by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).

The sponsor is responsible for justifying the creation or absence of a DSMB to the Competent Authority (ANSM) and to the Ethics committee.

A DSMB will be set up for this trial. Members will include experts in the field of ECMO and severe cardiogenic shock and a biostatistician/methodologist.

The DSMB must hold its first meeting before the first subject is enrolled. The DSMB's preliminary meeting should take place before the protocol submission to competent health authority (ANSM) and Ethics committee.

The members of the DSMB are:

Pr Alain VUYLSTEKE a.vuylsteke@nhs.net

Pr Christian RICHARD christian.richard@aphp.fr

Dr Caroline ELIE, caroline.elie@aphp.fr

11 DATA MANAGEMENT

11.1 Data collection procedure

During and after the clinical study, all data will be collected by the investigator and others specialised collaborators.

11.2 Identification of data recorded directly in the CRFs which will be considered as source data

None

11.3 Right to access source data and documents

11.3.1.1 Access to data

In accordance with GCP:

- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures.
- the investigators will ensure the persons in charge of monitoring and auditing the clinical trial and of quality control have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force (Articles L.1121-3 and R.5121-13 of the French Public Health Code)

11.3.1.2 Source documents

The source documents are any original document or item that proves the existence or accuracy of a data-point or fact recorded during the trial. Source documents will be kept by the investigator, or by the hospital in the case of hospital medical records, for the statutory period.

11.3.1.3 Data confidentiality

The persons responsible for the quality control of clinical studies (Article L.1121-3 of the French Public Health Code) will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular the identity of the participants and the results obtained.

These persons, as well as the investigators themselves, are bound by professional secrecy (in accordance with the conditions set out in Articles 226-13 and 226-14 of the French Criminal Code).

During and after the clinical study, all data collected about the study participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be anonymised. Under no circumstances shall the names and addresses of the participants involved be shown. The sponsor will ensure that each subject has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4 Data processing and storage of documents and data

11.4.1 Identification of the data processing manager and the location(s)

Data management of the study will be the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix. The data will be collected in the eCRFs by clinical study technicians and investigators. The data file will be declared to the Commission Nationale de l'Informatique et des Libertés (CNIL).

The statistical analysis of the data will be performed under the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix.

11.4.2 Data entry

Non-identifying data will be entered electronically via a web browser.

11.5 Ownership of the data

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12 STATISTICAL ASPECTS

12.1 Planned statistical methods, including the timetable for any planned interim analyses

The statistical analysis will be performed by a biostatistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence Tubach.

All comparisons will be done with a two-sided type I error of 5%.

Descriptive analysis will always be overall and by treatment group.

Comparative analysis between groups will be systematically performed:

- Without adjustment;
- With adjustment on baseline prognostic factors whose distribution could be, despite the randomization, unbalanced between treatment groups. These adjustments will potentially require using the appropriate models, whose relevance will be discussed according to the distribution and to the type of variables.

12.1.1 Descriptive methods

Qualitative variables will be described by numbers and percentages.

Quantitative variables will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range.

Censored data will be described by Kaplan-Meier estimation or cumulative incidence function, with rate at 15, 30 and 60 days if appropriate with 95% confidence interval.

12.1.2 Software used

Statistical analyses will be performed with the SAS® or R software with the most recent version at the time of the analysis.

12.1.3 Analysis plan

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.1.4 Description of the inclusions and follow-up

- Number of included participants
- Number of participants included in the analyses and reasons for exclusion if any
- Protocol deviations
- Inclusions curve (evolution of participants frequency between first and last inclusion)

A flow chart according to the CONSORT recommendations will be drawn.

12.1.5 Patients included in the analysis

Analyses will be conducted on the intention-to-treat population (all randomized patients), with the exception of patients who withdraw consent and asked for their data not being analysed.

12.1.6 Baseline characteristics

The following variables will be described:

- Respect of the eligibility criteria;
- Epidemiological characteristics;
- Clinical characteristics;
- Biological characteristics;
- Treatment characteristics.

12.1.7 Main outcome analysis

The main outcome is the time to successful ECMO weaning within the 30 days following randomization in the presence of competing risks.

5 types of follow-up termination can appear within the 30 days following randomization:

- Subject not weaned from ECMO at 30 days: censorship
- Death before D30 after randomization, without ECMO weaning: competitive event
- Subject with successful ECMO weaning: event of interest
- Subjects with unsuccessful ECMO weaning (i.e. death, need for repeat ECMO, need for another mechanical circulatory device or heart transplantation within 30 days after ECMO removal): competitive event
- Lost to follow-up within the 30 days following randomization: censorship.

Time to successful ECMO weaning will be defined as the difference between the date of randomization and the weaning date.

Time to unsuccessful ECMO weaning will be defined as the difference between the date of randomization and the ECMO removal date, in patients for whom repeat ECMO, need for another mechanical circulatory device or heart transplantation or death will occur within 30 days after ECMO removal.

Time to death without ECMO weaning will be defined as the difference between the date of randomization and the date of death.

Time to lost to follow-up will be defined as the difference between the date of randomization and the date of the latest news.

All other subjects will be followed up to 30 days: censorship. The time will be 30 days.

This outcome will be summarized by the cumulative incidence function, with rate at 15 and 30 days with 95% confidence intervals.

The treatments groups will be compared using the Fine and Gray model to account for competing risks. We will also use this model to account for prognostic factors.

Proportional hazard assumption will be studied with Schoenfeld residuals and the loglinearity hypothesis for quantitative covariate will be studied.

With the Fine and Gray model, subdistribution hazard ratio will be reported with 95% confidence interval.

Cumulative incidence function will be draw from the non-parametric estimation or from the model.

12.1.8 Secondary outcomes analysis

Qualitative outcomes:

- will be described by numbers and percentages;
- will be tested with χ^2 test, corrected χ^2 test or Fisher test, as appropriate.
- To account for prognostic factors or for exploratory analyses we will use a logistic model.

Quantitative outcomes:

- will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range;
- will be tested with the Student t-test, the Student t-test or the Mann-Whitney-Wilcoxon test, as appropriate
- To adjust on prognostic factors, a linear regression model will be used.

Censored outcomes:

- Mortality will be described by survival analysis with Kaplan-Meier estimation and 95% confidence interval. Only survival analysis within 60 days will be performed. In this analysis, time to death will be defined as the difference between the date of randomization and the date of death (whatever ECMO weaning had occurred before or not) or the date of the last news or the end of the follow-up (60 days). Difference between groups will be tested with the logrank test. To adjust for prognostic factors, a cox model will be used.

12.2 Sample size calculation

We estimated the cumulative incidence of successful ECMO weaning in the control arm at 50% in the presence of competing causes (death or weaning failure) from a 30-day cohort study of similar patients. According to Latouche et al. [38], with HR=1.75 (subdistribution hazard ratio), an alpha risk of 5% and a power of 80%, we need 101 events (= successful ECMO weaning) and 206 subjects in total (103 per group).

12.3 State whether subjects who exit the study prematurely will be replaced and in what proportion.

No replacement is foreseen.

12.4 Anticipated level of statistical significance

All comparisons will be done with a two-sided type I error of 5%.

12.5 Management of modifications made to the analysis plan for the initial strategy.

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.6 Selection of populations

An intention-to-treat (ITT) analysis will be performed as the principal analysis, i.e. all randomized patients will be included in the analysis in the group in which they first were randomized and all their data will be used, regardless of any changes of treatment over the study duration.

13 QUALITY CONTROL AND ASSURANCE

Every research project involving human participants managed by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored interventional research involving human participants. Every clinical study managed by AP-HP is ranked according to the projected risk incurred by the study participants using a classification system specific to AP-HP-sponsored clinical trials.

13.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the trial. The sponsor must have a quality assurance system for monitoring the implementation of the study at the research centers.

For this purpose, the sponsor shall appoint Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study sites, after completing their initial visits.

The purpose of monitoring the study, as defined in the Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1 Strategy for site opening

The strategy for opening the sites established for this study is determined using the appropriate monitoring plan.

13.1.2 Scope of site monitoring

In the case of this C risk study, the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact resulting in a study monitoring level to be implemented: level: **high level**.

13.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practice as well as with the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits by the Clinical Research Associate. During these visits, the following elements will be reviewed:

- written consent
- compliance with the study protocol and its procedures
- quality of the data collected in the case report forms: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

13.3 Case Report Form

All information required by the protocol must be entered in the case report forms. The data must be collected as and when it is obtained, and clearly recorded in these case report forms. Any missing data must be coded.

Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given instructions for using this tool.

Using on-line case report forms means the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, there are consistency checks to ensure the data are verified immediately upon being entered. The investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment. A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the study. The investigator must archive a copy of the authenticated document that was delivered to the sponsor.

13.4 Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study - with the protocol, standard operating procedures, good clinical practice or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

13.5 Audits/inspections

The investigators agree to accept the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the ground of medical secrecy.

An audit can be carried out at any time by individuals appointed by the [sponsor](#). The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the trial agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results and the storage of the data used or produced as part of the study.

13.6 Principal Investigator's declaration of responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal curriculum vitae, signed and dated less than one year, with his/her and RPPS number (Répertoire Partagé des Professionnels de Santé, Collective Database of Health Professionals). The CV must include any previous involvement in clinical research and related training.

Each investigator will commit to comply with legislation and to conduct the trial in line with GCP, in accordance with the Declaration of Helsinki.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.

The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

14 ETHICAL AND LEGAL CONSIDERATIONS

14.1 Methods for informing and obtaining consent from the research participants

In accordance with Article L.1122-1-1 of the French Public Health Code, no research can be carried out on a person without his/her free and informed consent, obtained in writing after the person has been given the information specified in Article L.1122-1 of said Code.

However, if the person is unable to give his or her written consent, consent may be obtained, in descending order of priority, from a legal representative, family member or a close relative. These persons must have no connection whatsoever to the investigator or the sponsor.

Therefore, the site investigator will obtain consent for inclusion in LEVOECMO (as approved by local IRBs) from a legal representative, family member or a close relative prior to randomization. The investigator will retain the original signed and dated consent form. The research subject relative will keep a signed and dated copy of the consent form.

Should a close relative be absent, eligible patients will be randomized according to the specifications of emergency consent (Article L.1122-1-3 CSP) after IRB approval. The patient's legal representative will be informed as soon as possible of the inclusion and asked to give his or her written consent. The patient will be informed and asked to give his/her consent for the continuation of the trial when his/her condition will allow during the subject's study participation. Investigators should indicate in the medical record the inclusion of a patient unable to consent in emergency situations in the research protocol.

Two formulations of consent form will be prepared: one for legal representative/family member/close relative of patient when patient is unable to sign consent; one for patient for authorization for pursuit of research.

When eligible patients will not be randomized, the reason for nonrandomization (lack of consent from substitute decision maker; refusal from attending physician; prior or current enrolment in a confounding RCT, patient missed by the research team) will be recorded in the screening log file.

14.2 *Prohibition of concomitant clinical studies participation and exclusion period after the trial, if applicable*

Whilst participating in this trial, subjects may not take part in any other interventional clinical study. For non-interventional or observational study, subjects may not take part without first speaking to the doctor in charge of this trial.

14.3 Compensation for subjects

No compensation will be paid to patients included in this trial.

14.4 Registration on the National Register of subjects participating in human research trials on the products listed in Article L. 5311-1 of the French Public Health Code

Not applicable

14.5 Authorisation for the research location

The study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely performed at those units. Therefore, the research location does not require specific authorisation.

14.6 Legal obligations

14.6.1 The sponsor's role

Assistance Publique Hôpitaux de Paris (AP-HP) is the sponsor of this study and by delegation, the DRCI (Clinical Research and Innovation Delegation) carries out the study's missions in accordance with Article L.1121-1 of the French Public Health Code. Assistance Publique - Hôpitaux de Paris (AP-HP) reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.6.2 Request for approval from the CPP (Research Ethics Committee)

Prior to starting the study, AP-HP, as sponsor, must obtain for this interventional study involving human participants concerning a medicinal product for human use, approval from the appropriate CPP (Research Ethics Committee), within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.6.3 Request for approval from the ANSM

Prior to starting the study, AP-HP, as sponsor, must obtain authorisation from the ANSM (French Health Products Safety Agency) for the interventional study involving human participants concerning a medicinal products for human use, within the scope of the ANSM's authority and in accordance with in force legislation and regulatory requirements.

14.6.4 Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended "Informatique et Libertés" law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

- Request for authorisation by the CNIL (French Data Protection Agency)

This research is not governed by the CNIL "Reference Method" (MR-001) because emergency consent and inability to obtain a written consent at inclusion.

The sponsor must obtain the authorisation of the CNIL (French Data Protection Agency) before implementing any data processing involving the data required to conduct the research.

14.6.5 Modifications to the trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, approval from the CPP (Research Ethics Committee) and authorisation from the ANSM within the scope of their respective authorities, before the amendment can be implemented.

The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.6.6 Final study report

The final report for the research involving human participants referred to in Article R1123-67 of the *Code de la Santé Publique* (French Public Health Code) is written and signed by the sponsor and the investigator. A report summary drafted according to the reference plan of the competent authority

must be sent to the competent authority within a period of one year following the end of the study, i.e., the end of the participation of the last participant in the study.

14.6.7 Archiving

Specific documents for an interventional study involving human participants concerning a medicinal product for human use will be archived by the investigator and the sponsor for 15 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list) :
 - the successive versions of the protocol (identified by the version number and its date), and any appendices
 - the ANSM authorisations and CPP (Research Ethics Committee) decisions
 - any correspondence
 - the enrolment list or register
 - the appendices specific to the research
 - final study report
- The data collection documents

15 FUNDING AND INSURANCE

15.1 Sources of funding for the trial

Grant from french Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministère de la Santé).

15.2 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

16 PUBLICATION

16.1 Mention of AP-HP affiliation for projects sponsored or managed by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is not important

- However, if the trial is funded by an internal call for tenders at the AP-HP, the first affiliation must be "AP-HP"
- Each of these affiliations must be identified by an address and separated by a semicolon
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France

16.2 Mention of the AP-HP manager (DRCI) in the acknowledgements of the text

- "The sponsor was *Assistance Publique – Hôpitaux de Paris* (Clinical Research and Development Department)"

16.3 Mention of the funder in the acknowledgements of the text

- "*The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministry of Health)*"

Mention that the study has been registered on the <http://clinicaltrials.gov/> website under registration number NCTXXXXX

17 BIBLIOGRAPHY

1. Werdan K, Gielen S, Ebelt H, Hochman JS, (2014) Mechanical circulatory support in cardiogenic shock. *Eur Heart J* 35: 156-167
2. Combes A, Leprince P, Luyt CE, Bonnet N, Trouillet JL, Leger P, Pavie A, Chastre J, (2008) Outcomes and long-term quality-of-life of patients supported by extracorporeal membrane oxygenation for refractory cardiogenic shock. *Crit Care Med* 36: 1404-1411
3. Mirabel M, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Chastre J, Combes A, (2011) Outcomes, long-term quality of life, and psychologic assessment of fulminant myocarditis patients rescued by mechanical circulatory support. *Crit Care Med* 39: 1029-1035
4. Beurtheret S, Mordant P, Paoletti X, Marijon E, Celermajer DS, Leger P, Pavie A, Combes A, Leprince P, (2013) Emergency circulatory support in refractory cardiogenic shock patients in remote institutions: a pilot study (the cardiac-RESCUE program). *Eur Heart J* 34: 112-120
5. Abrams D, Combes A, Brodie D, (2014) Extracorporeal Membrane Oxygenation in Cardiopulmonary Disease in Adults. *J Am Coll Cardiol* 63: 2769-2778
6. Muller G, Flecher E, Lebreton G, Luyt CE, Trouillet JL, Brechot N, Schmidt M, Mastroianni C, Chastre J, Leprince P, Anselmi A, Combes A, (2016) The ENCOURAGE mortality risk score and analysis of long-term outcomes after VA-ECMO for acute myocardial infarction with cardiogenic shock. *Intensive Care Med* 42: 370-378
7. Barbaro RP, Odetola FO, Kidwell KM, Paden ML, Bartlett RH, Davis MM, Annich GM, (2015) Association of hospital-level volume of extracorporeal membrane oxygenation cases and mortality. Analysis of the extracorporeal life support organization registry. *Am J Respir Crit Care Med* 191: 894-901
8. Aissaoui N, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Diebold B, Chastre J, Combes A, (2011) Predictors of successful extracorporeal membrane oxygenation (ECMO) weaning after assistance for refractory cardiogenic shock. *Intensive Care Med* 37: 1738-1745
9. Chen YS, Lin JW, Yu HY, Ko WJ, Jerng JS, Chang WT, Chen WJ, Huang SC, Chi NH, Wang CH, Chen LC, Tsai PR, Wang SS, Hwang JJ, Lin FY, (2008) Cardiopulmonary resuscitation with assisted extracorporeal life-support versus conventional cardiopulmonary resuscitation in adults with in-hospital cardiac arrest: an observational study and propensity analysis. *Lancet* 372: 554-561
10. Sheu JJ, Tsai TH, Lee FY, Fang HY, Sun CK, Leu S, Yang CH, Chen SM, Hang CL, Hsieh YK, Chen CJ, Wu CJ, Yip HK, (2010) Early extracorporeal membrane oxygenator-assisted primary percutaneous coronary intervention improved 30-day clinical outcomes in patients with ST-segment elevation myocardial infarction complicated with profound cardiogenic shock. *Crit Care Med* 38: 1810-1817
11. Schmidt M, Brechot N, Hariri S, Guiguet M, Luyt CE, Makri R, Leprince P, Trouillet JL, Pavie A, Chastre J, Combes A, (2012) Nosocomial infections in adult cardiogenic shock patients supported by venoarterial extracorporeal membrane oxygenation. *Clin Infect Dis* 55: 1633-1641
12. Cheng R, Hachamovitch R, Kittleson M, Patel J, Arabia F, Moriguchi J, Esmailian F, Azarbal B, (2014) Complications of extracorporeal membrane oxygenation for treatment of cardiogenic shock and cardiac arrest: a meta-analysis of 1,866 adult patients. *Ann Thorac Surg* 97: 610-616
13. Bisdas T, Beutel G, Warnecke G, Hoepfer MM, Kuehn C, Haverich A, Teebken OE, (2011) Vascular complications in patients undergoing femoral cannulation for extracorporeal membrane oxygenation support. *Ann Thorac Surg* 92: 626-631
14. Sundberg S, Lilleberg J, Nieminen MS, Lehtonen L, (1995) Hemodynamic and neurohumoral effects of levosimendan, a new calcium sensitizer, at rest and during exercise in healthy men. *Am J Cardiol* 75: 1061-1066
15. Follath F, Cleland JG, Just H, Papp JG, Scholz H, Peuhkurinen K, Harjola VP, Mitrovic V, Abdalla M, Sandell EP, Lehtonen L, (2002) Efficacy and safety of intravenous levosimendan compared with dobutamine in severe low-output heart failure (the LIDO study): a randomised double-blind trial. *Lancet* 360: 196-202
16. Kivikko M, Antila S, Eha J, Lehtonen L, Pentikainen PJ, (2002) Pharmacokinetics of levosimendan and its metabolites during and after a 24-hour continuous infusion in patients with severe heart failure. *International journal of clinical pharmacology and therapeutics* 40: 465-471

17. Mebazaa A, Nieminen MS, Packer M, Cohen-Solal A, Kleber FX, Pocock SJ, Thakkar R, Padley RJ, Poder P, Kivikko M, (2007) Levosimendan vs dobutamine for patients with acute decompensated heart failure: the SURVIVE Randomized Trial. *Jama* 297: 1883-1891
18. Gordon AC, Perkins GD, Singer M, McAuley DF, Orme RM, Santhakumaran S, Mason AJ, Cross M, Al-Beidh F, Best-Lane J, Brealey D, Nutt CL, McNamee JJ, Reschreiter H, Breen A, Liu KD, Ashby D, (2016) Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis. *N Engl J Med* 375: 1638-1648
19. Gong B, Li Z, Yat Wong PC, (2015) Levosimendan Treatment for Heart Failure: A Systematic Review and Meta-Analysis. *J Cardiothorac Vasc Anesth* 29: 1415-1425
20. Cholley B, Caruba T, Grosjean S, Amour J, Ouattara A, Villacorta J, Miguet B, Guinet P, Levy F, Squara P, Ait Hamou N, Carillon A, Boyer J, Boughenou MF, Rosier S, Robin E, Radutoiu M, Durand M, Guidon C, Desebbe O, Charles-Nelson A, Menasche P, Rozec B, Girard C, Fellahi JL, Pirracchio R, Chatellier G, (2017) Effect of Levosimendan on Low Cardiac Output Syndrome in Patients With Low Ejection Fraction Undergoing Coronary Artery Bypass Grafting With Cardiopulmonary Bypass: The LICORN Randomized Clinical Trial. *Jama* 318: 548-556
21. Mehta RH, Leimberger JD, van Diepen S, Meza J, Wang A, Jankowich R, Harrison RW, Hay D, Fremes S, Duncan A, Soltesz EG, Lubner J, Park S, Argenziano M, Murphy E, Marcel R, Kalavrouziotis D, Nagpal D, Bozinovski J, Toller W, Heringlake M, Goodman SG, Levy JH, Harrington RA, Anstrom KJ, Alexander JH, (2017) Levosimendan in Patients with Left Ventricular Dysfunction Undergoing Cardiac Surgery. *N Engl J Med* 376: 2032-2042
22. Landoni G, Lomivorotov VV, Alvaro G, Lobreglio R, Pisano A, Guarracino F, Calabro MG, Grigoryev EV, Likhvantsev VV, Salgado-Filho MF, Bianchi A, Pasyuga VV, Baiocchi M, Pappalardo F, Monaco F, Boboshko VA, Abubakirov MN, Amantea B, Lembo R, Brazzi L, Verniero L, Bertini P, Scandroglio AM, Bove T, Belletti A, Michienzi MG, Shukevich DL, Zabelina TS, Bellomo R, Zangrillo A, (2017) Levosimendan for Hemodynamic Support after Cardiac Surgery. *N Engl J Med* 376: 2021-2031
23. Affronti A, di Bella I, Carino D, Ragni T, (2013) Levosimendan may improve weaning outcomes in venoarterial ECMO patients. *Asaio j* 59: 554-557
24. Tafer N, Megret F, Dolbeau JB, Coiffic A, Calderon J, Mauriat P, Ouattara A, (2013) Intérêt du lévosimendan dans le sevrage difficile des assistances circulatoires temporaires. *Annales francaises d'anesthesie et de reanimation* 32: A211-A212
25. Distelmaier K, Roth C, Schrutka L, Binder C, Steinlechner B, Heinz G, Lang IM, Maurer G, Koinig H, Niessner A, Hulsman M, Speidl W, Goliash G, (2016) Beneficial effects of levosimendan on survival in patients undergoing extracorporeal membrane oxygenation after cardiovascular surgery. *Br J Anaesth* 117: 52-58
26. Ponikowski P, Voors AA, Anker SD, Bueno H, Cleland JG, Coats AJ, Falk V, Gonzalez-Juanatey JR, Harjola VP, Jankowska EA, Jessup M, Linde C, Nihoyannopoulos P, Parissis JT, Pieske B, Riley JP, Rosano GM, Ruilope LM, Ruschitzka F, Rutten FH, van der Meer P, (2016) 2016 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: The Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 37: 2129-2200
27. Altenberger J, Parissis JT, Costard-Jaeckle A, Winter A, Ebner C, Karavidas A, Sihorsch K, Avgeropoulou E, Weber T, Dimopoulos L, Ulmer H, Poelzl G, (2014) Efficacy and safety of the pulsed infusions of levosimendan in outpatients with advanced heart failure (LevoRep) study: a multicentre randomized trial. *Eur J Heart Fail* 16: 898-906
28. Husebye T, Eritsland J, Muller C, Sandvik L, Arnesen H, Seljeflot I, Mangschau A, Bjornerheim R, Andersen GO, (2013) Levosimendan in acute heart failure following primary percutaneous coronary intervention-treated acute ST-elevation myocardial infarction. Results from the LEAF trial: a randomized, placebo-controlled study. *Eur J Heart Fail* 15: 565-572
29. Jamali IN, Kersten JR, Pagel PS, Hettrick DA, Warltier DC, (1997) Intracoronary levosimendan enhances contractile function of stunned myocardium. *Anesth Analg* 85: 23-29
30. Kopustinskiene DM, Pollesello P, Saris NE, (2004) Potassium-specific effects of levosimendan on heart mitochondria. *Biochemical pharmacology* 68: 807-812

31. Sonntag S, Sundberg S, Lehtonen LA, Kleber FX, (2004) The calcium sensitizer levosimendan improves the function of stunned myocardium after percutaneous transluminal coronary angioplasty in acute myocardial ischemia. *J Am Coll Cardiol* 43: 2177-2182
32. Antoniadou C, Tousoulis D, Koumallos N, Marinou K, Stefanadis C, (2007) Levosimendan: beyond its simple inotropic effect in heart failure. *Pharmacology & therapeutics* 114: 184-197
33. Beiras-Fernandez A, Weis FC, Kur F, Kaczmarek I, Schmoeckel M, Weis M, Reichart B, (2008) Primary graft failure and Ca²⁺ sensitizers after heart transplantation. *Transplant Proc* 40: 951-952
34. Fries M, Ince C, Rossaint R, Bleilevens C, Bickenbach J, Rex S, Mik EG, (2008) Levosimendan but not norepinephrine improves microvascular oxygenation during experimental septic shock. *Crit Care Med* 36: 1886-1891
35. Nanas S, Gerovasili V, Dimopoulos S, Pierrakos C, Kourtidou S, Kaldara E, Sarafoglou S, Venetsanakis J, Roussos C, Nanas J, Anastasiou-Nana M, (2008) Inotropic agents improve the peripheral microcirculation of patients with end-stage chronic heart failure. *J Card Fail* 14: 400-406
36. Aissaoui N, El-Banayasy A, Combes A, (2015) How to wean a patient from veno-arterial extracorporeal membrane oxygenation. *Intensive Care Med* 41: 902-905
37. ELSO guidelines, <https://www.else.org/Resources/Guidelines.aspx>, Last accessed August 27th 2017
38. Latouche A, Porcher R, Chevret S, (2004) Sample size formula for proportional hazards modelling of competing risks. *Stat Med* 23: 3263-3274

18 LIST OF ADDENDA

Every addendum and the log of addenda versions are attached, independently of the protocol. Every addendum can be modified (change of addendum version) without modifying the version of the protocol.

18.1 List of Investigators

18.2 Serious Adverse Events report form

18.3 Pregnancy report form

18.4 Questionnaire/scale

- SOFA
- GLASGOW
- SAPS2

18.5 SCP's Levosimendan

- Levosimendan, ZIMINO
<http://agence-prd.ansm.sante.fr/php/ecodex/rcp/R0274395.htm>

**" LEVOSIMENDAN to facilitate weaning
from ECMO in severe cardiogenic shock patients "
LEVOECMO**

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

Version N° 2.0 of 31/05/2023

Project code number: P170914J - EUDRACT No.: 2019-004319-29

Coordinating Investigator:

Alain COMBES

Intensive Care Unit, Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Tel. : + 33 1 42 16 38 18

Email : alain.combes@aphp.fr

Methodologist:

Pr Florence TUBACH

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Email: florence.tubach@aphp.fr

Sponsor:

Assistance publique – Hôpitaux de Paris and by delegation: Clinical Research and Innovation
Delegation (DRCI) - Hôpital Saint-Louis - 1, avenue Claude Vellefaux

DRCI head office project advisor: Carla VANDENABELE

Tel : 01 40 27 57 47

Fax : 01 44 84 17 01

Email : carla.vandenabele@aphp.fr

Entity responsible for monitoring the trial:

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

Head: Pr Florence TUBACH

Project advisor: Jessica PALMYRE

Email: jessica.palmyre@aphp.fr

Tel: 01 42 16 19 35

Fax : 01 42 16 24 40

Agence générale des équipements et produits de santé, (APHP)

AGEPS, Département Essais Cliniques - 7, rue du Fer à Moulin, 75005 Paris

Céline DUPRE : Email : celine.dupre@aphp.fr

Nancy TAMIR-GEDDIS: Email : nancy.tamirgeddis@aphp.fr

Clinical Research and Innovation Direction (DRCI)
Hôpital Saint Louis 75010 PARIS

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

PROTOCOL SIGNATURE PAGE

Research code number: P170914J

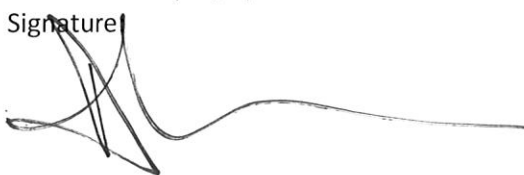
Title: LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients – “LEVOECMO”

Version N° of: 2.0 of 31/05/2023

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:

Pr. Alain COMBES
Intensive Care Unit
Hospital Pitié Salpêtrière
47 - 83 Boulevard de l'Hôpital
75651 Paris Cedex 13

Date: 18/7/2023
Signature: 

Sponsor

Assistance Publique-Hôpitaux de Paris
Clinical Research and Innovation Delegation
Saint Louis
1 avenue Claude Vellefaux
75010 PARIS

Date:/...../.....
Hôpital Signature:

The study was approved by the Ethic committee (CPP) of Ouest III on 04/01/2021 and authorised by the ANSM on 30/11/2020.

TABLE OF CONTENTS

1	SUMMARY	5
2	SCIENTIFIC JUSTIFICATION FOR THE TRIAL	9
2.1	SHORT-TERM MECHANICAL CIRCULATORY SUPPORT FOR REFRACTORY CARDIOGENIC SHOCK	9
2.2	LEVOSIMENDAN, AN ORIGINAL INODILATOR	9
2.3	WHY RECENT LARGE LARGE RANDOMIZED CONTROLLED TRIALS OF LEVOSIMENDAN WERE ALL NEGATIVE?.....	10
2.4	RATIONALE FOR A TRIAL TESTING LEVOSIMENDAN IN VA-ECMO PATIENTS AND JUSTIFICATION FOR THE CHOICE OF SUBJECTS	11
2.5	NAME AND DESCRIPTION OF THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	11
2.6	DESCRIPTION AND JUSTIFICATION OF THE DOSAGE, ROUTE OF ADMINISTRATION, ADMINISTRATION SCHEDULE AND TREATMENT DURATION	11
2.7	SUMMARY OF THE KNOWN AND FORESEEABLE BENEFITS AND RISKS FOR THE STUDY PARTICIPANTS.....	12
3	OBJECTIVES	12
3.1	PRIMARY OBJECTIVE	12
3.2	SECONDARY OBJECTIVES	12
4	DESCRIPTION OF THE TRIAL.....	13
4.1	CONCISE DESCRIPTION OF THE PRIMARY AND SECONDARY ENDPOINTS.....	13
4.2	RESEARCH METHODOLOGY.....	13
5	PROCEDURE FOR THE TRIAL	16
5.1	BASELINE VISIT = J1 (INCLUSION AND RANDOMIZATION)	16
5.2	FOLLOW-UP VISITS	17
5.3	EXPECTED LENGTH OF PARTICIPATION, CHRONOLOGY AND DURATION OF THE STUDY.....	18
5.4	TABLE OR DIAGRAM SUMMARISING THE CHRONOLOGY OF THE STUDY	18
5.5	DISTINCTION BETWEEN STANDARD CARE AND RESEARCH	18
5.6	TERMINATION AND EXIT RULES.....	19
6	SELECTION CRITERIA.....	21
6.1	IDENTIFICATION OF ELIGIBLE PATIENTS: INCLUSION AND EXCLUSION CRITERIA.....	21
6.2	CONSENT AND RANDOMIZATION PROCEDURE.....	22
6.3	RECRUITMENT METHODS	22
7	TREATMENT ADMINISTERED TO STUDY PARTICIPANTS.....	23
7.1	THE INVESTIGATIONAL MEDICINAL PRODUCT(S).....	23
7.2	TRACEABILITY INFORMATION FOR THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	24
7.3	COMMON MANAGEMENT REGARDING OTHER INOTROPIC DRUGS, HEART FAILURE MEDICATIONS AND ANTIAGREGANTS AND ANTICOAGULANTS.....	25
7.4	COMMON MANAGEMENT REGARDING THE ECMO CIRCUIT	25
7.5	AUTHORISED AND PROHIBITED TREATMENTS (MEDICINAL, NON-MEDICINAL, SURGICAL), INCLUDING EMERGENCY MEDICATIONS	25
7.6	METHODS FOR MONITORING COMPLIANCE WITH THE TREATMENT	26
8	EFFICACY ASSESSMENT.....	26
8.1	ANTICIPATED METHODS AND TIMETABLE FOR MEASURING, COLLECTING AND ANALYSING THE EFFICACY DATA	26
9	SPECIFIC COMMITTEES FOR THE TRIAL	26
9.1	STEERING COMMITTEE (SC)	26
10	SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY	27
10.1	RECORDING AND REPORTING ADVERSE EVENTS.....	27
11	DATA MANAGEMENT	33
11.1	DATA COLLECTION PROCEDURE	33
11.2	IDENTIFICATION OF DATA RECORDED DIRECTLY IN THE CRFs WHICH WILL BE CONSIDERED AS SOURCE DATA.....	33
11.3	RIGHT TO ACCESS SOURCE DATA AND DOCUMENTS.....	33
11.4	DATA PROCESSING AND STORAGE OF DOCUMENTS AND DATA	34

11.5	OWNERSHIP OF THE DATA.....	34
12	STATISTICAL ASPECTS	34
12.1	PLANNED STATISTICAL METHODS, INCLUDING THE TIMETABLE FOR ANY PLANNED INTERIM ANALYSES	34
12.2	SAMPLE SIZE CALCULATION	37
12.3	STATE WHETHER SUBJECTS WHO EXIT THE STUDY PREMATURELY WILL BE REPLACED AND IN WHAT PROPORTION.	37
12.4	ANTICIPATED LEVEL OF STATISTICAL SIGNIFICANCE.....	37
12.5	MANAGEMENT OF MODIFICATIONS MADE TO THE ANALYSIS PLAN FOR THE INITIAL STRATEGY.	37
12.6	SELECTION OF POPULATIONS	37
13	QUALITY CONTROL AND ASSURANCE	37
13.1	GENERAL ORGANISATION	37
13.2	QUALITY CONTROL	38
13.3	CASE REPORT FORM.....	38
13.4	MANAGEMENT OF NON-COMPLIANCES.....	38
13.5	AUDITS/INSPECTIONS.....	39
13.6	PRINCIPAL INVESTIGATOR'S DECLARATION OF RESPONSIBILITY	39
14	ETHICAL AND LEGAL CONSIDERATIONS	39
14.1	METHODS FOR INFORMING AND OBTAINING CONSENT FROM THE RESEARCH PARTICIPANTS.....	39
14.2	<i>PROHIBITION OF CONCOMITANT CLINICAL STUDIES PARTICIPATION AND EXCLUSION PERIOD AFTER THE TRIAL, IF APPLICABLE</i>	40
14.3	COMPENSATION FOR SUBJECTS	40
14.4	REGISTRATION ON THE NATIONAL REGISTER OF SUBJECTS PARTICIPATING IN HUMAN RESEARCH TRIALS ON THE PRODUCTS LISTED IN ARTICLE L. 5311-1 OF THE FRENCH PUBLIC HEALTH CODE.....	40
14.5	AUTHORISATION FOR THE RESEARCH LOCATION.....	40
14.6	LEGAL OBLIGATIONS.....	40
15	FUNDING AND INSURANCE	41
15.1	SOURCES OF FUNDING FOR THE TRIAL	42
15.2	INSURANCE	42
16	PUBLICATION	42
16.1	MENTION OF AP-HP AFFILIATION FOR PROJECTS SPONSORED OR MANAGED BY AP-HP	42
16.2	MENTION OF THE AP-HP MANAGER (DRCI) IN THE ACKNOWLEDGEMENTS OF THE TEXT	42
16.3	MENTION OF THE FUNDER IN THE ACKNOWLEDGEMENTS OF THE TEXT	42
17	BIBLIOGRAPHY	43
18	LIST OF ADDENDA	46
18.1	LIST OF INVESTIGATORS	46
18.2	SERIOUS ADVERSE EVENTS REPORT FORM	46
18.3	PREGNANCY REPORT FORM	46
18.4	QUESTIONNAIRE/SCALE.....	46
18.5	SCP'S LEVOSIMENDAN.....	46

1 SUMMARY

Full title	LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients
Acronym	LEVOECMO
Coordinating Investigator	COMBES ALAIN, GROUPE HOSP. PITIE-SALPETRIERE (AP-HP), PARIS, France
Sponsor	Assistance Publique-Hôpitaux de Paris
Scientific justification	In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with refractory cardiogenic shock. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery. In patients with potentially reversible cardiac failure (e.g. myocarditis, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery. Although considered as the ultimate life-saving technology for refractory cardiac failure, veno-arterial ECMO is still associated with severe complications. Specifically, excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. ECMO support also exposes to many complications such as infections, hemorrhage or peripheral vascular embolism. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs. Levosimendan, which acts to sensitize myocardial contractile proteins to calcium, improves cardiac contractility without increasing the intracellular calcium concentration. Unlike traditional inotropes such as dobutamine, levosimendan neither increases myocardial oxygen consumption nor impairs diastolic function or possess proarrhythmic effects. It also influences the opening of ATP-dependent potassium channels, including those in vascular smooth muscle cells, leading to coronary, pulmonary, and peripheral vasodilation and antiinflammatory, antioxidative, antiapoptotic, anti-stunning and cardioprotective effects. Additionally, Levosimendan which has a long lasting action (up to 7–9 d), resulting from the formation of active metabolite, may be used as a single 24h perfusion. In recent preliminary studies, the drug was associated with accelerated weaning from VA-ECMO and even improved survival. Therefore, a multicenter randomized trial with sufficient statistical power is needed in refractory cardiogenic shock patients supported by VA-ECMO to test if the early administration of Levosimendan can facilitate and accelerate VA-ECMO weaning, and ultimately translate in significantly less morbidity, reduced ICU and hospital length of stays and associated costs.
Main objective and primary endpoint	Main objective: To assess the efficacy of early administration of Levosimendan on the time to successful ECMO weaning (defined below) within the 30 days following randomization, in patients with VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Primary endpoint: Time to successful ECMO weaning within the 30 days following randomization.

	- ECMO weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal. - Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization). - Patients still under ECMO 30 days after randomization will be considered as censored. - The number of days alive without ECMO was not an appropriate primary outcome due to the high proportion of patients dying with ECMO and thus ex-aequo (0 days). The analysis will therefore model the risk of successful ECMO ablation in the presence of two competing risks (death and weaning failure).
Secondary objectives and endpoints	Secondary objectives/endpoints: To assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function. - mortality on D30 and D60 - duration of ECMO support between inclusion and D30, and between inclusion and D60; number of ECMO-free days between inclusion and D30/D60 - duration of ICU stay and duration of hospitalization; - major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, rehospitalization for heart failure on days 30 and 60 - time to improvement into hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization - number of days with organ failure, defined by SOFA score, and number of days alive without organ failure between inclusion and D30; - duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30; - duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60; - LV function assessed with echocardiography at Day 30 following randomization - Incidence of adverse drug reactions (atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia). The primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.
Design of the trial	Multicenter, Double Blind Randomized clinical trial
Target Population	Adults patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support and for whom LV recovery is likely.
Inclusion criteria	1. Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h. 2. Obtain informed consent from a close relative or surrogate. According to the specifications of emergency consent, randomization without the close relative or surrogate consent

	could be performed. Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.
Exclusion criteria	1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized. 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) History of torsades de pointes in the 30 days prior to inclusion 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study 21) Patient with a weight over 180 kg 22) Known hypersensitivity to polyvitamin CERNEVIT® 23) In case of hypervitaminosis to any vitamin contained in this formulation, 24) In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria 25) In combination with vitamin A or retinoids
Investigational medicinal product(s)	A continuous infusion of Levosimendan administered over 24h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate limiting side effects.
Comparator treatment	The placebo will be made of polyvitamin CERNEVIT® in order to reproduce the yellow color of levosimendan. The route of administration, administration procedures and duration of treatment will be similar to those of the experimental arm.
Risks added by the trial	Treatment by levosimendan or placebo
Procedure of the trial	<u>Inclusion visit (= Randomisation) Day 1</u> – Informed consent form. – Randomisation – Administration of a continuous infusion of Levosimendan or Placebo – Comorbidities – Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses – Record of AE/SAE

	Day 2 to Day 30 or date of death or date of discharge from the ICU – Record of AE/SAE – Clinical data – ECMO parameters – Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses Day 30, Day 60 – Vital status – ECMO: yes/no – On other mechanical cardiac support device – Heart transplantation – Mechanical ventilation – ICU discharge and date – Hospital discharge and date
Number of subjects included	206 (103 per arm)
Number of sites	National trial involving 14 centers
Duration of the trial	- inclusion period: 24 + 12 months - participation period (treatment+follow-up):2 months - total duration: 26 + 12 months
Number of enrolments expected per site and per month	0.4
Statistical analysis	The analysis will model the risk of successful VA-ECMO ablation in the presence of two competing risks (death and weaning failure).
Trial will have a Data Monitoring Committee	YES
Funding source	PHRC 2017 / French Ministry of Health

2 SCIENTIFIC JUSTIFICATION FOR THE TRIAL

2.1 Short-term mechanical circulatory support for refractory cardiogenic shock

In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with acute cardiogenic shock in the setting of acute myocardial infarction, myocarditis, intoxication with cardiotoxic drugs, end-stage dilated cardiomyopathy or post-cardiotomy [1-6]. Most of these “crash and burn” patients receive the device as salvage therapy after having already developed signs of multiple organ failure. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery [1, 2]. Several considerations must be taken into account before instituting ECMO. First, the device should be inserted before the patient has developed multiple organ failure or myocardial failure has led to refractory cardiac arrest, since these conditions are associated with significantly poorer outcomes [2, 3, 6]. Second, highly unstable patients may benefit from urgent on-site ECMO initiation by a rapid resuscitation team able to operate a portable and quick-to-prime ECMO circuit before transportation to the ECMO referral center [4]. Third, cardiac failure and other organ injuries should be deemed reversible and the patient’s underlying condition should not contraindicate a bridge to a more permanent device or to transplantation. Fourth, management of patients on ECMO for refractory cardiogenic shock is complex and should be conducted in experienced medical-surgical centers [7].

VA-ECMO might be used as a bridge to decision-making if the patient survives the first days to reach the “decision-making” point. In patients with potentially reversible cardiac failure (e.g. myocarditis, drug intoxication, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery [1]. Alternatively, it might be used as a bridge to transplantation or VAD implantation because of absence of LV functional recovery [8]. ECMO can also be simply withdrawn in cases of therapeutic futility (severe brain lesions, end-stage multiple organ failure or absence of myocardial recovery in the context of a definitive contraindication to transplantation or VAD implantation). Long-term survival after VA-ECMO is 70-80% after myocarditis or cardiotoxic drug poisoning, 40-50% after myocardial infarction and 15-25% when the device is used to rescue refractory cardiac arrest [2, 3, 6, 8-10]. Survivors reported a preserved quality of life, despite some limitations in physical activities and social functioning in previous series [2, 3, 6].

Although considered as the ultimate life-saving technology for refractory cardiac failure, venoarterial ECMO is still associated with severe complications. It increases left ventricular (LV) afterload that may delay myocardial recovery. Excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. Prolonged ECMO support also exposes to many complications such as local hemorrhage (10-20%), cannulation site infection and bacteremia (10-15%), limb ischemia (5-10%), ischemic or hemorrhagic stroke (5%) [2, 3, 6, 11, 12]. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs [2, 11-13].

2.2 Levosimendan, an original inodilator

Levosimendan is a calcium sensitizer with inotropic, vasodilatory and cardioprotective properties. It facilitates contact between actin and myosin and prolongs the time of contact between these two proteins, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect that unlike other positive inotropes, does not modify intracellular calcium concentration (which can cause arrhythmia and cell death) or myocardial diastolic relaxation. It also increases the contractile force of the myocardium without modifying its O₂ consumption. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on coronary arteries). The drug was also reported to exert cardioprotective, anti-stunning and anti-ischemic effect through this K⁺-ATP-dependent channel pathway.

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four

to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with prolonged half-life of about three days [16].

2.3 Why recent large randomized controlled trials of levosimendan were all negative?

Levosimendan has been tested in large RCTs against dobutamine or placebo in patients with decompensated heart failure, septic shock and the low cardiac output syndrome after cardiac surgery.

In patients with acute decompensated heart failure, while a first proof of concept trial suggested that levosimendan was associated with improved haemodynamic performance and lower mortality than dobutamine [15], the large (n=1,327) randomized double-blind SURVIVE clinical trial failed to confirm these findings [17]. In this trial, patients received a loading dose of levosimendan (12 µg/kg) followed by an infusion of 0.1 up-to 0.2 µg/kg/min for an additional 23 hours, as tolerated. Although a greater decrease in B-type natriuretic peptide level was observed in the first 5 days of treatment in the levosimendan group, all-cause mortality at 180 days was similar between groups, respectively occurring in 26% in the levosimendan group and 28% in the dobutamine group (HR 0.91; [0.74-1.13]; p=0.40). Marked hypotension, probably related to the levosimendan loading dose, atrial fibrillation, and hypokalemia were more frequent in the levosimendan group, possibly explaining the negative results of this trial.

The Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis (LeoPARDS) trial tested whether the addition of levosimendan to standard care would reduce the severity of organ dysfunction among patients with septic shock [18]. The administration of levosimendan or placebo was started at a rate of 0.1 µg/kg/min, increased to 0.2 µg/kg/min for a further 20 to 22 hours in the absence of rate-limiting side effects. The addition of levosimendan to standard treatment was not associated with less severe organ dysfunction or lower mortality. The mean (±SD) SOFA score over the ICU stay was 6.68±3.96 in the levosimendan group and 6.06±3.89 in the placebo group (P = 0.053) and mortality at 28 days was 34.5% in the levosimendan group and 30.9% in the placebo group (P = 0.43). More patients in the levosimendan group than in the placebo group had supraventricular tachyarrhythmia, tachycardia and hypotension. The negative results of this trial may relate to the inclusion of very few patients with septic shock and confirmed myocardial dysfunction and to the administration of relatively high dose of levosimendan (up to 0.2 µg/kg/min), which induced marked tachycardia and hypotension, the vasodilating properties of drug possibly aggravating sepsis-induced vasoplegia in this setting.

Meta-analyses of small trials [19] suggested that levosimendan may result in a higher rate of survival among patients at risk of developing or presenting the low cardiac output syndrome after cardiac surgery and set the stage for 3 large RCTs testing the drug in this setting. The LICORN [20] (n=336 patients) and the LEVO-CTS [21] (n=882 patients) trials assessed the ability of preoperative levosimendan to prevent postoperative low cardiac output syndrome in patients with pre-operative left ventricular dysfunction. The continuous prophylactic infusion of 0.1 µg/kg/min was started just after anesthetic induction and was maintained over 24 hours in the absence of serious adverse reactions. The occurrence of the composite end-points of prolonged catecholamine infusion, use of left ventricular mechanical assist device, or renal replacement therapy (LICORN) and of death, renal-replacement therapy, perioperative myocardial infarction, or use of a mechanical cardiac assist device (LEVO-CTS) and the rates of adverse events were not different between levosimendan and placebo treated patients in these studies. It should be mentioned that only 22% of the patients developed low cardiac output syndrome in the LEVO-CTS trial and that in both the LICORN and the LEVO-CTS trials, only half of the patients needed catecholamine support for >24 hours. Thus, the preventive use of levosimendan does not seem to confer clinical benefits in this setting. Alternatively, the CHEETAH trial [22] included only patients who developed perioperative or early post-operative cardiovascular dysfunction and low cardiac output syndrome, defined as the need for support with an intraaortic balloon pump or high-dose inotropic support in patients with preoperative left ventricular ejection fraction of less than 25%. Patients were randomly assigned to receive levosimendan at a dose of 0.025 to 0.2 µg/kg/min or placebo, with no loading dose, for up to 48 hours or until discharge from the intensive care unit. The trial was stopped early for futility after 506 patients were enrolled and there

was no significant difference in the primary end-point of 30-day mortality between the levosimendan group and the placebo group (32 patients [12.9%] and 33 patients [12.8%], respectively; absolute risk difference, 0.1 percentage points; 95% confidence interval [CI], -5.7 to 5.9; P = 0.97). Rates of hypotension or cardiac arrhythmias were also not different between the levosimendan and placebo groups. It should however be noted that the mean infusion dose of levosimendan (0.07 µg/kg/min) was lower in this trial than in the 2 other RCTs of cardiac surgery patients.

2.4 Rationale for a trial testing levosimendan in VA-ECMO patients and justification for the choice of subjects

Preliminary retrospective cohort studies suggested a benefit of the early administration of levosimendan in refractory cardiogenic shock patients on VA-ECMO. In a small “before-after” study of medical cardiogenic shock patients, Affronti et al. [23] showed that the drug was associated with a higher rate of VA-ECMO weaning compared to historical controls, respectively 83.3% vs 27.3% (p=0.498). Similarly, Tafer et al [24] reported more ECMO-free days (22 vs 9) and lower 30-day mortality (24% vs 56%, p<0.05) in a cohort of 80 patients of whom 37 had received levosimendan early after ECMO initiation. In a series of 240 patients undergoing VA-ECMO after cardio-vascular surgery, of whom 75% received levosimendan immediately after ECMO initiation [25], the treatment was also associated with higher rates of successful ECMO weaning [HR; 0.41; 95% CI, 0.22–0.80; P=0.008]. Additionally, lower 30-day (adjusted HR 0.52; 95% CI 0.30–0.89; P=0.016), and long-term survival (adjusted HR 0.64; 95% CI 0.42–0.98; P=0.04) were reported, although levosimendan-treated patients had higher baseline EuroSCORE (P<0.001) and SAPS-3 (P=0.02) and lower LVEF (p=0.04).

Indeed, the rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support. Furthermore, it might be possible to administer higher doses of levosimendan to this specific group of patients, since VA-ECMO also provides robust peripheral vascular support that might protect against vasoplegia and hypotension that were suspected to hamper the benefit of the drug in previous trials.

In conclusion, the probability of demonstrating lower mortality with levosimendan in cardiogenic shock patients on ECMO appears unlikely based on the results of recent large RCTs. However, any medical therapy that may be able to hasten myocardial recovery and shorten the duration of VA-ECMO support might translate in significantly less morbidity and ultimately reduced ICU and hospital length of stays and associated costs. Highly relevant pathophysiological and clinical reasons indicate that levosimendan might fulfill these objectives. A multicenter randomized trial with sufficient statistical power is therefore urgently needed to test the early administration of Levosimendan to facilitate and accelerate VA-ECMO weaning in refractory cardiogenic shock patients supported by VA-ECMO.

2.5 Name and description of the investigational medicinal product(s)

- Levosimendan (ZIMINO®)
- CERNEVIT®

See section 7 for more details

2.6 Description and justification of the dosage, route of administration, administration schedule and treatment duration

See section 7 for further information about the administration of the product

2.7 Summary of the known and foreseeable benefits and risks for the study participants

Reducing the duration of ECMO support with levosimendan may be associated with less ECMO-related complications such as hemorrhages, stroke or infections. It may also be accompanied by a reduction in the duration of ICU and hospital stays, therefore reducing the costs associated with the care of these critically ill patients. Better long-term outcomes might also be expected with earlier ECMO weaning in these severe cardiogenic shock patients.

Known risk associated with levosimendan infusion include hypotension and supraventricular and ventricular arrhythmias. However, this side effects may be significantly mitigated in patient on VA-ECMO.

3 OBJECTIVES

3.1 Primary objective

The primary objective will be to assess the efficacy of the early administration of levosimendan on the time to successful ECMO weaning in patients on VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory device or heart transplantation 30 days after ECMO removal [8, 36]. Patients still under ECMO 30 days after randomization will be considered as censored at 30 days.

Patients with unsuccessful ECMO weaning will be considered as weaning failure (competing risk) at the date of the weaning attempt.

3.2 Secondary objectives

Secondary objectives of the trial are to assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function.

- Mortality on D30 and D60
- Duration of ECMO support between inclusion and D30, and between inclusion and D60; and number of ECMO-free days between inclusion and D30/D60
- Duration of ICU stay; of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60.
- Time to improvement in hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization
- Number of days with organ failure, defined by SOFA score, and number of days without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of drug adverse effects (atrial fibrillation, ventricular arrhythmias, ventricular fibrillation, “torsades de pointes”, hypokalemia)
- Primary and secondary endpoints in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4 DESCRIPTION OF THE TRIAL

4.1 Concise description of the primary and secondary endpoints

4.1.1 Primary endpoint

- **Time to successful ECMO weaning within the 30 days following randomization**

- The event of interest is the successful weaning of ECMO (within 30 days after randomization) defined as: the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal [8, 36]. Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization).
- ECMO weaning failure (i.e. unsuccessful weaning), within 30 days after randomization, is a competing event.
- Death within 30 days after randomization, without an attempt to wean, is a competing event.
- No event after 30 days after randomization will be considered and all patients without any of the three events defined above are censored at 30 days post randomization.
- A protocolized management regarding weaning of VA-ECMO will be applied to both groups (see paragraph 5.2.1 for details) [8, 36].
- The number of days alive without ECMO was not an appropriate primary outcome criterion due to the high proportion of patients dying on ECMO and thus ex-aequo (0 days). The analysis will therefore ***model the risk of successful ECMO weaning in the presence of competing risks (death and weaning failure)***.

4.1.2 Secondary endpoints

- Mortality on Day 30 and Day 60
- Total duration of ECMO support between inclusion and D30 and between inclusion and D60;
- Number of ECMO-free days between inclusion and D30/D60.
- Duration of ICU stay and of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60
- Time to improvement in hemodynamic parameters (systolic, diastolic, mean blood pressure, heart rate ...) and time to hemodynamic stabilization
- Days with organ failure, defined by the SOFA score, and days alive without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of adverse drug reactions (including atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia).

The Primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4.2 Research methodology

4.2.1 Design of the trial

Characteristics of the trial are as follows:

- Multicenter clinical trial
- Phase III
- Randomized
- Double blind
- Comparative, Placebo controlled
- Superiority design
- Two treatment arms
 - Early administration of Levosimendan vs. placebo
 - 1:1 ratio

All consecutive eligible patients will be proposed to participate.

Allocation concealment will be guaranteed by use of a centralized randomization process with a web server (see below) and use of the minimization method of randomization. Investigator masking to group assignment after randomization will be guaranteed by use of a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

Because acute cardiogenic shock is a complex and multifactorial process, one key issue in every trial evaluating a new therapeutic strategy is how patients in both groups are managed and how standard care for these patients recommended by current guidelines are implemented in the participating ICUs. In the LEVOECMO trial, common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

4.2.2 Number of participating sites

The study will involve 14 sites in France. Subjects will be recruited in ICU departments.

4.2.3 Identification of the subjects

All consecutive eligible patients will be proposed to participate.

The subjects participating in this study will be identified as follows:

Site number (3 digits) - Sequential selection number for the site (4 digits) - surname initial - first name initial

This reference number is unique and will be used for the entire duration of the trial.

A randomisation number will also be assigned when the participant is randomised. This number will have the following format: XXX.

A treatment number corresponding to the treatment arm will be also assigned to the patient when the participant is allocated to a treatment arm. This number will have the following format: TXXXXX.

A registry of eligible patients not included will be completed, with reason for non-inclusion.

4.2.4 Randomization

After baseline screening, patients will be randomly assigned in a 1:1 ratio to receive levosimendan or placebo. The computer-generated randomization process will be implemented using the minimization method with a probability of assignment of 0.8 after the 20 first included patients (being assigned by balanced random) by a statistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence TUBACH. Stochastic minimization will be according to the primary etiology of cardiogenic shock (acute myocardial infarction, myocarditis, post-heart surgery and other causes) and by center. The assignments of participants will be concealed by use of a centralized 24-hour Internet service (CleanWeb, Télémédecine Technologies, Boulogne, France) and thus without knowledge of the next assignment in the sequence.

4.2.5 Blinding methods and measures put in place to protect the blinding

Investigator masking to group assignment after randomization will be guaranteed by use a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

The study will be double blinded. Treatments will be prepared at ICU by a nurse "off-protocol handler" (specifically trained caregiver) opens the box given by the PUI and prepares the infusion Levosimendan or CERNEVIT®.

Levosimendan and the placebo will be administred as non-identifiable preparation.

Trial participants, care providers, and outcome assessors will be masked to the patient assignment.

4.2.6 Unblinding procedures, if applicable

The investigating doctor may request unblinding for any reason he considers essential, by contacting:

- the DRCI (sponsor) **in non-urgent situations**, on working days and during working hours; the request should be addressed to the DRCI's project advisor, tel.: 01 40 27 57 47
- the poison centre of Fernand Widal Hospital, in the case of an emergency (see emergency situations requiring unblinding), on weekends, bank holidays, when the DRCI is closed and when unblinding cannot be carried out by the DRCI.
Telephone: **+33 (0)1 40 05 48 48**
- the hospital pharmacy

5 PROCEDURE FOR THE TRIAL

5.1 Baseline visit = J1 (inclusion and randomization)

5.1.1 Verification of inclusion and exclusion criteria (see paragraph 6)

5.1.2 Consent (see paragraph 14.1 for details)

Whose consent must be obtained	Who informs the individual and collects their consent	When is the individual informed	When is the individual's consent collected
• Patient's legal representative, family member or a close relative	• Local site investigator	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow

5.1.3 Treatment

- Reconstitution/Preparation of the treatment before administration
- Administration by IV perfusion during 24h
- Administration as soon as possible after randomization; no later than 24 hours after randomization
- Prohibited administration of CERNEVIT® during experimental treatment administration

5.1.4 Clinical data

- Date, time of hospital and ICU admission
- Date, time of ECMO initiation
- Reason for ECMO initiation
- Date of birth
- Sex
- McCabe and Jackson score
- Charlson comorbidity index score
- history of systemic hypertension, myocardial infarction, diabetes mellitus, preexisting cardiovascular diseases (stroke, carotid surgery), severe left ventricle dysfunction (NYHA stage and ejection fraction determined by echocardiography or angiography), chronic renal insufficiency (serum creatinine, chronic dialysis), and hepatic insufficiency, defined as serum bilirubin >100 µmol/L or albuminemia <35 g/L with no other cause found
- immunocompromised: yes/no,
- SAPS II (table to check-off)
- Sequential organ-failure assessment (SOFA) score (table to check-off)
- Hemodynamic parameters (including arterial pressure and heart rate)
- Catecholamine doses

5.1.5 Biological and echocardiographic data

- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers

- Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,)

5.1.6 ECMO parameters

- FiO₂ and membrane ventilation
- Pump turns per minute and flow
- Fibrin/clots on the membrane: yes/no
- Clogged circuit requiring change: yes/no
- Intravascular hemolysis caused by the circuit: yes/no
- Hemorrhage at the cannula insertion site: yes/no
- Infection at the cannula insertion site: yes/no
- Circuit changed: yes/no
- Leg ischemia: yes/no

5.2 Follow-up visits

5.2.1 The following Information will be collected on days 1, 2, 3... 30 or date of death or date of ICU discharge

- Hemodynamic parameters (arterial pressure, heart rate).
- Catecholamine doses
- Mechanical ventilation: yes/no
- Renal replacement therapy: yes/no
- Urine output
- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers
- SOFA score
- ECG
- Echocardiographic parameters (LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus).
- Weanability test [8, 36]: yes/no
 - A protocolized management regarding weaning of VA-ECMO will be applied to both groups.
 - An ECMO-weaning trial will be undertaken every day as soon as the patient will be considered hemodynamically stable, i.e. baseline MBP >60 mm Hg while receiving no or low-dose vasopressors and a pulsatile arterial waveform maintained for at least 24 h.
 - The ECMO flow will be decreased gradually decreased to 66%, and to 33% of its baseline value and then to a minimum of 1–1.5 L/min.
 - If MBP drops significantly and constantly <60 mm Hg during the trial, ECMO flow will be returned to 100% of the initial flow and the trial will be stopped.
 - Patients achieving aortic VTI ≥12 cm, LVEF >20-25% and TDSa ≥6 cm/s at minimal ECMO flow support, will be considered weanable from ECMO [8, 36].
- ECMO parameters until device removal
 - FiO₂ and membrane ventilation
 - Pump turns per minute and flow
 - Fibrin/clots on the membrane: yes/no
 - Clogged circuit requiring change: yes/no
 - Intravascular hemolysis caused by the circuit: yes/no
 - Hemorrhage at the cannula insertion site: yes/no
 - Infection at the cannula insertion site: yes/no
 - Circuit changed: yes/no

- Leg ischemia: yes/no

5.2.2 Outcomes on days 30 and 60

- Living/dead; died: date and hour
- On ECMO: yes/no
- In case of ECMO weaning, need for a second ECMO run yes/no
- On other mechanical cardiac support device: yes/no
- Heart transplantation: yes/no
- Mechanical ventilation: yes/no
- In the ICU: yes/no; discharged from the ICU: date
- Hospitalized; yes/no; discharged from the hospital: date
- Patient's contact information will be collected at ICU discharge to permit 60 day outcome data collection.

5.3 Expected length of participation, chronology and duration of the study.

- Length of enrolment period: 24 months + 12 months
- Duration of participation for each subject: 60 days
- Treatment period: 24 hours
- Follow-up period: 60 days
- Total study duration: (start = trial start date; end = trial end date) = 26 months + 12 months
- When randomization and blinding will take place: immediately after patients are deemed eligible.

5.4 Table or diagram summarising the chronology of the study

Actions	D1	D2 to D30 or date of death or date of ICU discharge	D60
Information	X		
Signature of the consent form	X		
Randomization	X		
Past medical history	X		
Clinical exam	X	X	
Medical procedures (ECG, Echo.)	X	X	
Dispensation of LEVOSIMENDAN or Placebo	X		
Adverse events	X	X	X
Vital status	X	X	X
ECMO status	X	X	X
Pursuit consent		X	X (if not obtained before)

5.5 Distinction between standard care and research

Procedures and treatments carried out as part of the research	Procedures and treatments associated with <u>care</u>	Procedures and treatments added because of <u>the research</u>
Treatments	None	Levosimendan or Placebo at D1
Consultations	D1 to D30 : During patient's hospitalisation's	Follow up by phone :Day 60

Blood samples	Hematology - Biochemistry	None
Imaging, etc.	ECG Echocardiography	None
Act	ECMO	None

In the LEVOECMO trial, only the initial treatment undergoing randomization (levosimendan vs. placebo) will differ between groups.

Common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

5.6 Termination and exit rules

5.6.1 Criteria and procedures for prematurely terminating the study treatment

- Temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the subject's source file and the case report form (CRF). The subject remains enrolled in the study
- Premature termination of treatment, but the subject remains enrolled in the study until the end of the subject's participation: the investigator must document the reason
- Premature termination of treatment and exit from the study. The only reasons for premature exit from the study are: death, lost to follow-up, or patient's consent withdrawal. Premature termination of treatment or adverse reaction should not imply premature exit from the study.

The investigator must:

- Document the reason(s)
- Collect all endpoints at the moment the subject exits from the study, if the subject agrees
- Schedule further follow-up visits, especially in case of a serious adverse event.

Study treatment must be discontinued if they experience any of the following:

- Anaphylactic shock
- Ventricular fibrillation
- "Torsades de pointes"
- Ventricular tachycardia
- Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for more than 60 minutes despite:
 - vascular filling with two 500 mL successive crystalloid boluses AND
 - norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.

Criteria and procedure for premature withdrawals and exits from the study

- Subject may exit the study at any time and for any reason.
- The investigator can temporarily or permanently withdraw a subject from the study for any safety reason or if it is in the subject's best interests
- Subject lost to follow-up: the subject cannot be located. The investigator must make every effort to reconnect with the subject and record his attempts in the source file (at least to determine whether the subject is alive or dead).

If a participant leaves the research prematurely, data relating to the participant can be used unless an objection was recorded when the participant signed the consent form.

If a subject exits the trial prematurely or withdraws consent, any data collected prior to the date of premature exit may still be used.

The case report form must list the various reasons why the subject exited or was withdrawn from the study:

- ☐ Lack of efficacy
- ☐ Adverse reaction
- ☐ Other medical problem
- ☐ Subject's personal reasons
- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up
- ☐ Death

5.6.2 Monitoring subjects after the premature termination of treatment

If a subject exits the trial this will in no way affect the standard care received for his/her condition. In case of severe adverse events, the investigator must notify the sponsor and monitor the subject for 30 days following the premature termination of treatment. If the treatment is stopped prematurely due to a serious adverse event, a serious adverse event report will be sent by mail (vigilance.drc@aphp.fr) to the sponsor. The serious adverse reaction will be monitored until it is resolved.

5.6.3 Full or partial cancellation of the study

AH-HP (the sponsor) or the Competent Authority (ANSM) may prematurely discontinue all or part of the trial, temporarily or permanently, upon the recommendation of the Data Monitoring Committee in the following situations:

- first, if suspected unexpected serious adverse reactions (SUSARs) are observed in one of the treatment arms or if there is a discrepancy in the serious adverse reactions between the two treatment arms, requiring a reassessment of the benefit-risk ratio for the trial.
- similarly, AH-HP, as the sponsor, or the Competent Authority (ANSM) may prematurely cancel the trial due to the discovery of unexpected facts or new information about the product, in light of which the objectives of the study or clinical programme are unlikely to be achieved .
- AP-HP, as the sponsor, reserves the right to permanently suspend enrolment at any time if the enrolment targets have not been met.

If the study is cancelled prematurely, AP-HP will inform the Competent Authority (ANSM) and the Institutional Review Board of its decision within 15 days, together with justification for the decision and any recommendations from the Data Monitoring Committee (if applicable).

6 **SELECTION CRITERIA**

6.1 **Identification of Eligible Patients: Inclusion and Exclusion Criteria**

Adult patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support will be screened for eligibility.

6.1.1 **Inclusion criteria**

1. **Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h.**

(The rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support).

2. **Obtain informed consent from a close relative or surrogate.** According to the specifications of emergency consent, randomization without the close relative or surrogate consent could be performed.

Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

6.1.2 **Exclusion criteria**

1. Age <18
2. Pregnant or lactating women
3. Initiation of ECMO >48 h
4. Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized
5. Irreversible neurological pathology
6. End-stage cardiomyopathy with no hope of LV function recovery
7. Mechanical complication of myocardial infarction
8. Aortic regurgitation > II
9. VA-ECMO for pulmonary embolism
10. VA-ECMO for cardiotoxic drug intoxication
11. VA-ECMO after left-ventricle assist device implantation
12. VA-ECMO in heart transplant patients
13. Patient moribund on the day of randomization, SAPS II >90
14. Liver cirrhosis (Child B or C) and other severe hepatic insufficiency
15. Chronic renal failure requiring hemodialysis
16. Known hypersensitivity to levosimendan
17. History of "torsades de pointes" in the 30 days prior to inclusion
18. History of epilepsy
19. Individuals under guardianship, or permanently legally incompetent adults
20. Participation to another interventional study
21. Patient with a weight over 180 kg

22. Known hypersensitivity to polyvitamin CERNEVIT®
23. In case of hypervitaminosis to any vitamin contained in this formulation,
24. In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria
25. In combination with vitamin A or retinoids

6.2 Consent and randomization procedure

Patients satisfying all of the Inclusion and Exclusion Criteria are classified as 'Eligible'.

To randomize an eligible patient, consent must be obtained **from a close relative or surrogate**.

Should such a person be absent, the patient will be randomized according to the specifications of **emergency enrollment** and the patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

Reasons for not enrolling an eligible patient will be recorded (lack of consent from substitute decision maker; refusal from attending physician; study team not available; study team missed the patient, patient enrolled in another trial).

6.3 Recruitment methods

Patients will be recruited in medical and surgical intensive care units participating to the study.

Number of subjects to be included during the enrolment period, number of subjects to be enrolled per site per month, based on the number of sites and the duration of the enrolment period are provided in the following tables.

Study characteristics	Value
Total number of subjects	206
Number of sites	14
Enrolment period (months)	24 + 12
Number of subjects/site	15,8
Number of subjects/site/month	0.4

7 TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

7.1 The investigational medicinal product(s) : levosimendan and placebo

7.1.1 Levosimendan characteristics

Levosimendan facilitates contact between actin and myosin and prolongs the time of contact between these two molecules, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on the coronary arteries) and a cardio protective anti-ischemic effect [29-35].

Metabolism and Pharmacokinetics

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with a half-life of about three days. The metabolite OR-1896 has been shown to have hemodynamic and pharmacologic properties similar to those of the parent drug in preclinical models [16].

Pharmacodynamics

Levosimendan produces significant, dose-dependent increases in cardiac output, stroke volume and heart rate, and decreases in pulmonary capillary wedge pressure, mean blood pressure, mean pulmonary artery pressure, mean right atrial pressure and total peripheral resistance.

There is no sign of development of tolerance even with a prolonged infusion up to 48 hours. Due to the formation of the active metabolite, the hemodynamic effects are maintained several days after stopping levosimendan infusion. Compared with dobutamine, levosimendan produces a slightly greater increase in cardiac output and a profoundly greater decrease in pulmonary capillary wedge pressure. In contrast to dobutamine, the hemodynamic effects are not attenuated with concomitant beta-blocker use. It has also been shown that at 48 hours after the start of infusion, a 24-hour infusion achieves superior hemodynamic effects over a 48-hour dobutamine infusion in patients with severe acute decompensated heart failure on beta-blockers. decrease in natriuretic peptides. The beneficial effects of levosimendan on hemodynamics and neurohormones are not associated with any significant increase in myocardial energy consumption, as evidenced using dynamic positron emission tomography (PET) in hospitalized patients with NYHA III-IV heart failure.

7.1.2 Description of investigational products

7.1.2.1 Levosimendan

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml containing 5 mL of solution for infusion will be provided free of charge by the Orion Corporation.

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml will be packaged in boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP).

Each boxes contains 3 vials of 5 ml of levosimendan .

7.1.2.2 Placebo (CERNEVIT®)

The placebo will be made of polyvitamin CERNEVIT®.

Vials of CERNEVIT® containing 5 mL of powder for solution for injection will be provided by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP) .

Vials of CERNEVIT® will be packaged in boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical trial Department (CTD) of AGEPS.

Each boxes contains 1 vial of 5 mL of CERNEVIT®.

7.1.2.3 Boxes of levosimendan or placebo

Each box will be assigned a unique treatment number according to the treatment numbers list.

Supply :

The numbered boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo will be sent on investigational sites by CTD of AGEPS.

Storage

Boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo must be kept at +2 to +8 °C in the outside packaging, shielded from the light.

Administration

The products (levosimendan or placebo) will be prepared by non-protocol manipulators (nurses) A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered.

A continuous infusion of Levosimendan or placebo will be administered over 24 h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate-limiting side effects.

7.1.3 Treatment preparation and dispensation

The pharmacy of each center will be responsible dispensing treatment's boxes in the ICU after initiation visit. The products will be packaged and labelled as follows: packs of three vials of levosimendan and a single vial of placebo. Each box will be appropriately labelled and numbered according to the list of treatment numbers provided by the sponsor.

The treatment (levosimendan or placebo) will be prepared by non-protocol manipulators (nurse) in the ICU (according the following table).

A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered.

Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h
40 to 46 kg	1	250 mL	8	11
47 to 89 kg	2	500 mL	16	22
90 to 132 kg	3	1000 mL	32	44
133 to 180 kg	4	1000 mL	32	44

7.2 Traceability information for the investigational medicinal products

Sponsor will provide traceability documents to enable the regulatory traceability of the investigational medicinal products on each investigational site.

7.3 Common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants

Additional inotropic agents may be used at any time in either treatment group as clinically indicated and as recommended by the latest European Society of Cardiology [26]. Dobutamine will be the inotropic agent of choice.

Others heart failure medications (such as diuretics, ACE inhibitors, steroidal antimineralocorticoids), antiagregants and anticoagulants will be prescribed according to the latest European Society of Cardiology [26].

Open-label Levosimendan (ZIMINO) will not be possible in either group during the entire course of the study.

7.4 Common management regarding the ECMO circuit

– Parameter settings for ECMO

The pump flow will be set to 3.5-5 l/min, to provide adequate systemic perfusion. Pump flow might be reduced in case of cessation of LV ejection or major pulmonary edema. Percentage of oxygen contained in the ventilating gaseous air–oxygen mixture will be adjusted to obtain PaO₂ between 65 and 90 mmHg and/or arterial oxygen saturation >90%. The membrane ventilation will be adjusted to maintain PaCO₂ between 40 and 45 mmHg. Extracorporeal circulation will be run at normal body temperature.

– ECMO monitoring (extracorporeal circuit, anticoagulation, possible complications)

The ECMO circuit will be monitored several times daily by the medical and nursing team caring for the patient and at least once every 48 hours by a perfusionist. Circuit and cannula surveillance is intended to verify the correct functioning of the device and early screening for complications (leg ischemia, fibrin deposits or clots on the ECMO membrane, clots in the cannulae or in the pump, bleeding or signs of inflammation or cutaneous infection at the cannula insertion sites, unexpected drop of the ECMO outflow, appearance of clinical or biological signs of intravascular hemolysis). Should any of these complications occur, a medical–surgical consultation will be held to discuss the best therapeutic approach to take.

Anticoagulation will be obtained with non-fractionated heparin to a target aPTT of 55-70 sec or heparinemia (antiXa activity) between 0.4 and 0.6 IU/ml. Should severe bleeding occur that is not immediately controllable by specific treatment, heparin will be discontinued.

Intravascular hemolysis will be sought should unexpected dark urine be excreted or in the case of obvious circuit dysfunction. It is recommended that plasma free hemoglobin be measured every 48 hours and immediately if hemolysis linked to the circuit is suspected.

The membrane and ECMO circuit will be changed in the following situations: massive intravascular hemolysis linked to the device, severe thrombopenia linked to the circuit, clots preventing the pump and/or lines from functioning properly or systemic defibrination.

The hemoglobin threshold for red cells transfusion will be 9-10 g/dl (may be decreased to 7-8 g/dl if the patient is stabilized and does not have residual myocardial ischemia). Platelet transfusion will be discouraged except when severe thrombopenia is associated by bleeding complications or when platelet count will be <20 G/L.

Connecting an extrarenal dialysis circuit to the ECMO circuit will be permitted under strict supervision of perfusionists.

7.5 Authorised and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications

No treatment is prohibited during the study, except open-label Levosimendan and CERNEVIT® (only during experimental treatment administration). Additional inotropic agents (except digoxine) may be used at any time in either treatment group as clinically indicated. Dobutamine will be the inotropic agent of choice.

7.6 Methods for monitoring compliance with the treatment

Compliance to study drug will be assured by the physician in charge of the patient during the 24 hour infusion of the treatment.

8 EFFICACY ASSESSMENT

8.1 Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

The measures of all efficacy parameters rely on the clinical evaluation of the patients from the day of randomization (Day 1) up-to day 60 post-randomization. All information will be recorded by the investigator, nurses or study coordinator.

9 SPECIFIC COMMITTEES FOR THE TRIAL

9.1 Steering Committee (SC)

- Members of the committee: All primary investigators at study sites.
- Missions: The SC will define the overall structure of the study, coordinate information, determine the initial methodology and oversee the trial. Also determine the objective, write the protocol, recommend changes to the protocol during the trial

10 **SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY**

10.1 **Recording and reporting adverse events**

10.1.1 **Definitions**

According to Article R1123-46 of the French Public Health Code:

- **Adverse event**

Any untoward medical occurrence in a trial subject, which does not necessarily have a causal relationship with the clinical trial or with the investigational product.

- **Adverse reaction to an investigational medicinal product**

Any adverse event occurred in a trial subject, which has a causal relationship with the clinical trial or with the investigational medicinal product

- **Serious adverse event or reaction**

Any adverse event or reaction that at any dose of medication, results in death, threatens the life of the research subject, requires hospitalisation or prolongs hospitalisation, causes a severe or long-term disability or handicap, or results in a congenital abnormality or deformity.

- **Unexpected adverse reaction to an investigational medicinal product**

Any adverse reaction to the product, whose nature, severity, frequency or outcome is inconsistent with the safety information described in the Reference Safety Information (summary of product characteristics, or the investigator's brochure if the product is not authorised).

According to Article R.1123-46 of the Code de la Santé Publique and the guidelines for clinical trial sponsors (ANSM):

- **Emerging safety issue**

Any new safety information that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials..

For the clinical trials involving the first administration or use of an investigational medicinal product in healthy volunteers, any serious adverse reaction.

Examples:

- a) any clinically significant increase in the frequency of an expected serious adverse reaction
- b) suspected unexpected serious adverse reactions in patients who have terminated their participation in the clinical trial that are notified by the investigator to the sponsor together with follow-up reports
- c) any new safety issue relating to the conduct of the clinical trial or the development of the investigational medicinal product, that may impact the safety of the trial subjects.

Examples:

- a serious adverse event likely to be related to the interventions and the trial's diagnostic procedures and which may impact the conduct of the clinical trial,
- a significant risk on the trial subjects such as ineffectiveness of the investigational medicinal product in treating a life-threatening illness under investigation,
- significant safety results from a recently completed non-clinical study (such as a carcinogenicity study),
- the premature termination, or temporary suspension, of a trial conducted on the same investigational medicinal product in another country, for safety reasons,
- an unexpected serious adverse reaction associated with a non-experimental medication required for the conduct of the clinical trial, (e.g. challenge agents, rescue treatment)

- d) recommendations from the Data Safety Monitoring Board (DSMB), if applicable, that may affect the safety of the trial subjects

e) any suspected unexpected serious adverse reaction (SUSAR) reported to the sponsor by another sponsor of a trial carried out in a different country but relating to the same medication.

10.1.2 The role of the investigator

The investigator must **assess the seriousness criteria of each adverse event** and record all serious and non-serious adverse events in the case report form eCRF.

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the severity** of the adverse events by using:

- general terms:

- *Mild: tolerated by the patient, does not interfere with daily activities*
- *Moderate: sufficiently uncomfortable to affect daily activities*
- *Serious: preventing daily activities*

- or a severity grading scale for adverse events, attached to the protocol

The investigator must assess the **causal relationship** between the serious adverse events and the investigational medicinal products.

The method used by the investigator is based on the WHO Uppsala Monitoring Centre method and uses the following causality terms:

- Certain
- Probable/likely
- Possible
- Unlikely (not ruled out).

These terms are defined as follows (extracted from the WHO-UMC causality categories, version dated 17/04/2012).

Table : WHO-UMC causality categories (extract)

Causality term	Assessment criteria*
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake ** • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognized pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable / Likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake** • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with reasonable time relationship to drug intake ** • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake ** • that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations

*All points should be reasonably complied with

** Or study procedures

10.1.2.1 Serious adverse events that require a notification without delay by the investigator to the sponsor

According to the article R.1123-49 of the French Public Health Code (CSP), the investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any serious adverse event which occurs during a trial as described in Article L.1121-1(1) CSP, except those which are listed in the protocol (see corresponding section) and, if applicable, in the investigator's brochure as not requiring a notification without delay. These latter should be notified by the investigator to the sponsor

in an appropriate delay taking into consideration the specific features of the trial, the serious adverse events and the modalities specified in the protocol or the investigator's brochure.

A serious adverse event is any untoward medical occurrence that:

- 1- results in death
- 2- is life-threatening
- 3- requires inpatient hospitalisation or prolongation of existing hospitalisation
- 4- results in persistent or significant disability/incapacity
- 5- is a congenital anomaly/birth defect

10.1.2.2 Specific features of the protocol

10.1.2.2.1 Other events that require the investigator to notify the sponsor without delay

– Under VA-ECMO

The mortality rate of patients with refractory cardiogenic shock supported by VA-ECMO is very high (40-60%) [2, 11-13].

VA-ECMO provides both respiratory and cardiac support, therefore, **only deaths considered as directly related the experimental treatment by the investigator** will be notified to the sponsor without delay during the period: under VA-ECMO support .

A CRF extraction of all cause of death under VA-ECMO (including : randomization group/cause of death/ date of start and end of experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

– After weaning ECMO

After ECMO weaning because of cardiac recovery, only the following serious adverse events leading to life threatening or death will be notified without delay:

- Ventricular fibrillation,
- « Torsades de pointes »,
- Ventricular tachycardia (HR $\geq$ 110 bpm),
- Premature ventricular contraction,
- Premature atrial contraction, atrial fibrillation, Hypotension grade 3 and grade 4 and refractory hypotension defined as a mean arterial blood pressure <45 mmHg for > 60 minutes despite vascular filling with two 500 mL successive crystalloid boluses AND norepinephrine ≥ 4 $\mu\text{g/kg/min}$ OR epinephrine ≥ 2 $\mu\text{g/kg/min}$.

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of these adverse events, according to the same modalities and within the same timeline as for serious adverse events (see above).

- **In utero exposure**

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any pregnancy that occurs during the trial, even if not associated with an adverse event.

10.1.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are simply recorded in the case report form. A CRF extraction of these serious adverse events will be done according Data Safety Monitoring Board (DSMB) or AP-HP's Safety department's request. It will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

Special rules for trials with a high mortality rate:

The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13].

Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form.

A CRF extraction of all cause of death under VA-ECMO and after weaning ECMO (including randomization group/cause of death/ date of start/date of end of the experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay.

NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor.

– ***Under VA-ECMO***

All SAEs, which occurred under VA-ECMO, do not require the investigator to notify the sponsor without delay especially SAEs with fatal outcome which are considered by the investigator to be **not related** to the experimental treatment.

NB: Cf Special rules for trials with a high mortality rate.

– ***After weaning ECMO***

- *Normal and natural course of the condition:*
 - Death considered as **exclusively related** to the natural evolution of cardiogenic shock or multiple organ failure and not
 - Prolongation of hospitalization due to surgery
 - Rehospitalization considered as related to cardiogenic shock or for disease follow-up
 - Events likely **to be exclusively** due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion, septic shock, multiorgan failure, hemorrhagic shock, renal failure, ventilator associate pneumonia, acute respiratory distress syndrome ...)

NB: Cf Special rules for trials with a high mortality rate.

- *Special circumstances*
 - Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation)
 - Moderate tachycardia (HR <110 bpm)
 - Premature ventricular contraction except those leading to life threatening or death
 - Premature atrial contraction, atrial fibrillation except those leading to life threatening or death
 - Mild to moderate hypokalemia (3 to 3.5 mmol/L)
- *Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care*

The investigator must report these events to his Centre Régional de Pharmacovigilance (CRPV).

10.1.2.3 Period during which the investigator must send notification of SAEs to the sponsor without delay

The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section:

- starting from the date on which the subject begins treatment with the investigational medicinal product/undergoes the first intervention specific to the study
- throughout the whole follow-up period intended by the trial (60 days after randomization)
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear at long term after exposure to the medication, such as cancers or congenital abnormalities)

10.1.2.4 Procedures and deadlines for notifying the sponsor

The investigator should initially complete a SAE reporting form (contained in the case report form). This report must be signed by the investigator.

The investigator must complete every section of the SAE form so that the sponsor can carry out the appropriate assessment.

The initial report sent to the sponsor must be rapidly followed up by one or more additional written reports describing the course of the event and any complementary information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful for medical assessment of the case (medical reports, laboratory test results, results of additional exams, etc.). These documents must be anonymized. In addition, the investigator must state the study acronym and the number and initials of the study participant on each paper.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the subject has terminated his participation in the trial.

The initial report, the SAE follow-up reports, and all other documents must be sent to AP-HP by e-mail (eig-vigilance.drc@aphp.fr). It is possible to send the SAE to the AP-HP's Safety department by fax No. **+33 (0)1 44 84 17 99** only in case of unsuccessful attempt to send the SAE by e-mail. This is to avoid duplicated reports.

For trials which use e-CRF

- the investigator completes the SAE report form in the e-CRF, then validates, prints and signs the form before sending it by mail;
- In case of failure to connect to the e-CRF, the investigator should complete, sign and send the SAE report form to the safety Department. As soon as the connection is restored, the investigator must complete the SAE report form in the e-CRF.

The investigator must comply with all requests for additional information from the sponsor.

For all questions relating to an adverse event report, the safety Department can be contacted via email at vigilance.drc@aphp.fr.

10.1.3 Role of the sponsor

The sponsor, represented by its safety Department, shall continuously assess the safety of each investigational medicinal product throughout the trial.

10.1.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all reported adverse events,

- the **causal relationship** between these adverse events and investigational medicinal product and any other treatments,
All serious adverse events for which the investigator and/or the sponsor suspect a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.
- the **expectedness assessment** of the serious adverse reactions
Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.
The sponsor, acting through its safety Department, assesses the expectedness of the serious adverse reaction based on the information described below.

- ❖ For serious adverse events likely to be related to the investigational medicinal product(s):
 - refer to the SmPC for Levosimendan (Zimino®) enclosed in appendix N°18.4.
 - refer to the SmPC for Cernevit®

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs), within the regulatory time frame, to the ANSM (French Health Products Safety Agency).

- The sponsor must send the initial report without delay upon receipt of the unexpected serious adverse reaction if it is fatal or life-threatening, or otherwise within 15 days from receipt of any other type of unexpected serious adverse reaction;
- The sponsor must provide all relevant additional information by sending follow-up reports, within 8 calendar days following receipt.

Any suspected unexpected serious adverse reaction must also be declared electronically using the Eudravigilance European adverse drug reactions database managed by the European Medicines Agency (EMA).

The sponsor must notify all the investigators about any information that could adversely affect the safety of the trial subjects.

Special rules for double blind trials

After unblinding, if the administered product is the investigational medicinal product, the case will be immediately reported as suspected serious unexpected adverse reaction (SUSAR); however, if the administered product is the comparator, the unexpected nature of the adverse reaction will be re-assessed according to the reference document of the comparator found in the protocol. In the exceptional case of a clinical study about a disease with high mortality or morbidity, the sponsor may request the ANSM a readjustment of the conditions for unblinding and reporting suspected adverse reaction. These conditions should be thoroughly defined in the study protocol.

10.1.3.2 Analysis and declaration of other safety data

This relates to any new safety data that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials.

The sponsor will inform the competent authority and the Ethics committee without delay upon knowledge any emerging safety issues and, if applicable, describe what urgent safety measures have been taken by the sponsor.

Following the initial declaration of emerging safety issue, the sponsor will declare to ANSM any additional relevant information about the new safety issues in the form of a follow-up report, which must be sent no later than 8 days after becoming aware of the information.

If the suspected unexpected serious adverse reaction meets the definition of an emerging safety issue, the sponsor will report both the SUSAR and the emerging safety issue to the ANSM according to the appropriate modalities and within the regulatory timelines as previously described.

10.1.3.3 Annual safety report

The sponsor must prepare once yearly throughout the trial duration an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- an analysis of safety data concerning trial subjects
- a description of the patients included in the trial (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- cumulative summary tabulation of all the serious adverse events that have occurred since the beginning of the clinical trial,

The report must be transmitted to ANSM and the Ethics Committee no later than 60 days after the anniversary date corresponding to the date of authorization of the clinical trial by ANSM.

10.1.4 Independent Data Safety Monitoring Board (DSMB)

An independent Data Safety Monitoring Board (DSMB) will be set up by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).

The sponsor is responsible for justifying the creation or absence of a DSMB to the Competent Authority (ANSM) and to the Ethics committee.

A DSMB will be set up for this trial. Members will include experts in the field of ECMO and severe cardiogenic shock and a biostatistician/methodologist.

The DSMB must hold its first meeting before the first subject is enrolled. The DSMB's preliminary meeting should take place before the protocol submission to competent health authority (ANSM) and Ethics committee.

The members of the DSMB are:

Pr Alain VUYLSTEKE a.vuylsteke@nhs.net

Pr Christian RICHARD christian.richard@aphp.fr

Dr Caroline ELIE, caroline.elie@aphp.fr

11 DATA MANAGEMENT

11.1 Data collection procedure

During and after the clinical study, all data will be collected by the investigator and others specialised collaborators.

11.2 Identification of data recorded directly in the CRFs which will be considered as source data

None

11.3 Right to access source data and documents

11.3.1.1 Access to data

In accordance with GCP:

- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures.
- the investigators will ensure the persons in charge of monitoring and auditing the clinical trial and of quality control have access to the documents and personal data strictly necessary for these tasks, in

accordance with the statutory and regulatory provisions in force (Articles L.1121-3 and R.5121-13 of the French Public Health Code)

11.3.1.2 Source documents

The source documents are any original document or item that proves the existence or accuracy of a data-point or fact recorded during the trial. Source documents will be kept by the investigator, or by the hospital in the case of hospital medical records, for the statutory period.

11.3.1.3 Data confidentiality

The persons responsible for the quality control of clinical studies (Article L.1121-3 of the French Public Health Code) will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular the identity of the participants and the results obtained.

These persons, as well as the investigators themselves, are bound by professional secrecy (in accordance with the conditions set out in Articles 226-13 and 226-14 of the French Criminal Code).

During and after the clinical study, all data collected about the study participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be anonymised.

Under no circumstances shall the names and addresses of the participants involved be shown.

The sponsor will ensure that each subject has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4 Data processing and storage of documents and data

11.4.1 Identification of the data processing manager and the location(s)

Data management of the study will be the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix. The data will be collected in the eCRFs by clinical study technicians and investigators. The data file will be declared to the Commission Nationale de l'Informatique et des Libertés (CNIL).

The statistical analysis of the data will be performed under the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix.

11.4.2 Data entry

Non-identifying data will be entered electronically via a web browser.

11.5 Ownership of the data

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12 STATISTICAL ASPECTS

12.1 Planned statistical methods, including the timetable for any planned interim analyses

The statistical analysis will be performed by a biostatistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence Tubach.

All comparisons will be done with a two-sided type I error of 5%.

Descriptive analysis will always be overall and by treatment group.

Comparative analysis between groups will be systematically performed:

- Without adjustment;

- With adjustment on baseline prognostic factors whose distribution could be, despite the randomization, unbalanced between treatment groups. These adjustments will potentially require using the appropriate models, whose relevance will be discussed according to the distribution and to the type of variables.

12.1.1 Descriptive methods

Qualitative variables will be described by numbers and percentages.

Quantitative variables will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range.

Censored data will be described by Kaplan-Meier estimation or cumulative incidence function, with rate at 15, 30 and 60 days if appropriate with 95% confidence interval.

12.1.2 Software used

Statistical analyses will be performed with the SAS® or R software with the most recent version at the time of the analysis.

12.1.3 Analysis plan

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.1.4 Description of the inclusions and follow-up

- Number of included participants
- Number of participants included in the analyses and reasons for exclusion if any
- Protocol deviations
- Inclusions curve (evolution of participants frequency between first and last inclusion)

A flow chart according to the CONSORT recommendations will be drawn.

12.1.5 Patients included in the analysis

Analyses will be conducted on the intention-to-treat population (all randomized patients), with the exception of patients who withdraw consent and asked for their data not being analysed.

12.1.6 Baseline characteristics

The following variables will be described:

- Respect of the eligibility criteria;
- Epidemiological characteristics;
- Clinical characteristics;
- Biological characteristics;
- Treatment characteristics.

12.1.7 Main outcome analysis

The main outcome is the time to successful ECMO weaning within the 30 days following randomization in the presence of competing risks.

5 types of follow-up termination can appear within the 30 days following randomization:

- Subject not weaned from ECMO at 30 days: censorship
- Death before D30 after randomization, without ECMO weaning: competitive event
- Subject with successful ECMO weaning: event of interest

- Subjects with unsuccessful ECMO weaning (i.e. death, need for repeat ECMO, need for another mechanical circulatory device or heart transplantation within 30 days after ECMO removal): competitive event
- Lost to follow-up within the 30 days following randomization: censorship.

Time to successful ECMO weaning will be defined as the difference between the date of randomization and the weaning date.

Time to unsuccessful ECMO weaning will be defined as the difference between the date of randomization and the ECMO removal date, in patients for whom repeat ECMO, need for another mechanical circulatory device or heart transplantation or death will occur within 30 days after ECMO removal.

Time to death without ECMO weaning will be defined as the difference between the date of randomization and the date of death.

Time to lost to follow-up will be defined as the difference between the date of randomization and the date of the latest news.

All other subjects will be followed up to 30 days: censorship. The time will be 30 days.

This outcome will be summarized by the cumulative incidence function, with rate at 15 and 30 days with 95% confidence intervals.

The treatments groups will be compared using the Fine and Gray model to account for competing risks. We will also use this model to account for prognostic factors.

Proportional hazard assumption will be studied with Schoenfeld residuals and the loglinearity hypothesis for quantitative covariate will be studied.

With the Fine and Gray model, subdistribution hazard ratio will be reported with 95% confidence interval.

Cumulative incidence function will be drawn from the non-parametric estimation or from the model.

12.1.8 Secondary outcomes analysis

Qualitative outcomes:

- will be described by numbers and percentages;
- will be tested with χ^2 test, corrected χ^2 test or Fisher test, as appropriate.
- To account for prognostic factors or for exploratory analyses we will use a logistic model.

Quantitative outcomes:

- will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range;
- will be tested with the Student t-test, the Student t-test or the Mann-Whitney-Wilcoxon test, as appropriate
- To adjust on prognostic factors, a linear regression model will be used.

Censored outcomes:

- Mortality will be described by survival analysis with Kaplan-Meier estimation and 95% confidence interval. Only survival analysis within 60 days will be performed. In this analysis, time to death will be defined as the difference between the date of randomization and the date of death (whatever ECMO weaning had occurred before or not) or the date of the last news or the end of the follow-up (60 days).

Difference between groups will be tested with the logrank test. To adjust for prognostic factors, a cox model will be used.

12.2 Sample size calculation

We estimated the cumulative incidence of successful ECMO weaning in the control arm at 50% in the presence of competing causes (death or weaning failure) from a 30-day cohort study of similar patients. According to Latouche et al. [38], with HR=1.75 (subdistribution hazard ratio), an alpha risk of 5% and a power of 80%, we need 101 events (= successful ECMO weaning) and 206 subjects in total (103 per group).

12.3 State whether subjects who exit the study prematurely will be replaced and in what proportion.

No replacement is foreseen.

12.4 Anticipated level of statistical significance

All comparisons will be done with a two-sided type I error of 5%.

12.5 Management of modifications made to the analysis plan for the initial strategy.

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.6 Selection of populations

An intention-to-treat (ITT) analysis will be performed as the principal analysis, i.e. all randomized patients will be included in the analysis in the group in which they first were randomized and all their data will be used, regardless of any changes of treatment over the study duration.

13 QUALITY CONTROL AND ASSURANCE

Every research project involving human participants managed by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored interventional research involving human participants. Every clinical study managed by AP-HP is ranked according to the projected risk incurred by the study participants using a classification system specific to AP-HP-sponsored clinical trials.

13.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the trial. The sponsor must have a quality assurance system for monitoring the implementation of the study at the research centers.

For this purpose, the sponsor shall appoint Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study sites, after completing their initial visits.

The purpose of monitoring the study, as defined in the Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1 Strategy for site opening

The strategy for opening the sites established for this study is determined using the appropriate monitoring plan.

13.1.2 Scope of site monitoring

In the case of this C risk study, the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact resulting in a study monitoring level to be implemented: level: **high level**.

13.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practice as well as with the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits by the Clinical Research Associate. During these visits, the following elements will be reviewed:

- written consent
- compliance with the study protocol and its procedures
- quality of the data collected in the case report forms: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

13.3 Case Report Form

All information required by the protocol must be entered in the case report forms. The data must be collected as and when it is obtained, and clearly recorded in these case report forms. Any missing data must be coded.

Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given instructions for using this tool.

Using on-line case report forms means the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, there are consistency checks to ensure the data are verified immediately upon being entered. The investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment. A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the study. The investigator must archive a copy of the authenticated document that was delivered to the sponsor.

13.4 Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study - with the protocol, standard operating procedures, good clinical practice or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

13.5 Audits/inspections

The investigators agree to accept the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the ground of medical secrecy.

An audit can be carried out at any time by individuals appointed by the [sponsor](#). The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the trial agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results and the storage of the data used or produced as part of the study.

13.6 Principal Investigator's declaration of responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal curriculum vitae, signed and dated less than one year, with his/her and RPPS number (Répertoire Partagé des Professionnels de Santé, Collective Database of Health Professionals). The CV must include any previous involvement in clinical research and related training.

Each investigator will commit to comply with legislation and to conduct the trial in line with GCP, in accordance with the Declaration of Helsinki.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCl document) which will be sent to the sponsor's representative.

The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

14 ETHICAL AND LEGAL CONSIDERATIONS

14.1 Methods for informing and obtaining consent from the research participants

In accordance with Article L.1122-1-1 of the French Public Health Code, no research can be carried out on a person without his/her free and informed consent, obtained in writing after the person has been given the information specified in Article L.1122-1 of said Code.

However, if the person is unable to give his or her written consent, consent may be obtained, from a legal representative, family member or a close relative. These persons must have no connection whatsoever to the investigator or the sponsor.

Therefore, the site investigator will obtain consent for inclusion in LEVOECMO (as approved by local IRBs) from a legal representative, family member or a close relative prior to randomization. The investigator will retain the original signed and dated consent form. The research subject relative will keep a signed and dated copy of the consent form.

For LEVOECMO trial, if patient Should a close relative be absent, eligible patients will be randomized according to the specifications of emergency consent (Article L.1122-1-3 CSP) after IRB approval. The patient's legal representative will be informed as soon as possible of the inclusion and asked to give his or her written consent. The patient will be informed and asked to give his/her consent for the continuation of the trial when his/her condition will allow during the subject's study participation. Investigators should to indicate in the medical record the inclusion of a patient unable to consent in emergency situations in the research protocol.

Two formulatess of consent form will be prepared: one for legal representative/family member/close relative of patient when patient is unable to sign consent; one for patient for authorization for pursuit of research.

When eligible patients will not be randomized, the reason for nonrandomization (lack of consent from substitute decision maker; refusal from attending physician; prior or current enrolment in a confounding RCT, patient missed by the research team) will be recorded in the screening log file.

14.2 Prohibition of concomitant clinical studies participation and exclusion period after the trial, if applicable

Whilst participating in this trial, subjects may not take part in any other interventional clinical study (except IMPULMACS, NCT04951999). For non-interventional or observational study, subjects may not take part without first speaking to the doctor in charge of this trial.

14.3 Compensation for subjects

No compensation will be paid to patients included in this trial.

14.4 Registration on the National Register of subjects participating in human research trials on the products listed in Article L. 5311-1 of the French Public Health Code

Not applicable

14.5 Authorisation for the research location

The study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely performed at those units. Therefore, the research location does not require specific authorisation.

14.6 Legal obligations

14.6.1 The sponsor's role

Assistance Publique Hôpitaux de Paris (AP-HP) is the sponsor of this study and by delegation, the DRCI (Clinical Research and Innovation Delegation) carries out the study's missions in accordance with Article L.1121-1 of the French Public Health Code. Assistance Publique - Hôpitaux de Paris (AP-HP) reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.6.2 Request for approval from the CPP (Research Ethics Committee)

Prior to starting the study, AP-HP, as sponsor, must obtain for this interventional study involving human participants concerning a medicinal product for human use, approval from the appropriate CPP (Research Ethics Committee), within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.6.3 Request for approval from the ANSM

Prior to starting the study, AP-HP, as sponsor, must obtains authorisation from the ANSM (French Health Products Safety Agency) for the interventional study involving human participants concerning a medicinal products for human use, within the scope of the ANSM's authority and in accordance with in force legislation and regulatory requirements.

14.6.4 Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended “Informatique et Libertés” law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

- Request for authorisation by the CNIL (French Data Protection Agency)

This research is not governed by the CNIL “Reference Method” (MR-001) because emergency consent and inability to obtain a written consent at inclusion.

The sponsor must obtain the authorisation of the CNIL (French Data Protection Agency) before implementing any data processing involving the data required to conduct the research.

14.6.5 Modifications to the trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, approval from the CPP (Research Ethics Committee) and authorisation from the ANSM within the scope of their respective authorities, before the amendment can be implemented.

The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.6.6 Final study report

The final report for the research involving human participants referred to in Article R1123-67 of the *Code de la Santé Publique* (French Public Health Code) is written and signed by the sponsor and the investigator. A report summary drafted according to the reference plan of the competent authority must be sent to the competent authority within a period of one year following the end of the study, i.e., the end of the participation of the last participant in the study.

14.6.7 Archiving

Specific documents for an interventional study involving human participants concerning a medicinal product for human use will be archived by the investigator and the sponsor for 15 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list) :
 - the successive versions of the protocol (identified by the version number and its date), and any appendices
 - the ANSM authorisations and CPP (Research Ethics Committee) decisions
 - any correspondence
 - the enrolment list or register
 - the appendices specific to the research
 - final study report
- The data collection documents

15 FUNDING AND INSURANCE

15.1 Sources of funding for the trial

Grant from french Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministère de la Santé).

15.2 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

16 PUBLICATION

16.1 Mention of AP-HP affiliation for projects sponsored or managed by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is not important
- However, if the trial is funded by an internal call for tenders at the AP-HP, the first affiliation must be "AP-HP"
- Each of these affiliations must be identified by an address and separated by a semicolon
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: AP-HP, hospital, department, city, postcode, France

16.2 Mention of the AP-HP manager (DRCI) in the acknowledgements of the text

- "The sponsor was *Assistance Publique – Hôpitaux de Paris* (Clinical Research and Development Department)"

16.3 Mention of the funder in the acknowledgements of the text

- "*The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministry of Health)*"

Mention that the study has been registered on the <http://clinicaltrials.gov/> website under registration number NCTXXXXX

17 BIBLIOGRAPHY

1. Werdan K, Gielen S, Ebel H, Hochman JS, (2014) Mechanical circulatory support in cardiogenic shock. *Eur Heart J* 35: 156-167
2. Combes A, Leprince P, Luyt CE, Bonnet N, Trouillet JL, Leger P, Pavie A, Chastre J, (2008) Outcomes and long-term quality-of-life of patients supported by extracorporeal membrane oxygenation for refractory cardiogenic shock. *Crit Care Med* 36: 1404-1411
3. Mirabel M, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Chastre J, Combes A, (2011) Outcomes, long-term quality of life, and psychologic assessment of fulminant myocarditis patients rescued by mechanical circulatory support. *Crit Care Med* 39: 1029-1035
4. Beurtheret S, Mordant P, Paoletti X, Marijon E, Celermajer DS, Leger P, Pavie A, Combes A, Leprince P, (2013) Emergency circulatory support in refractory cardiogenic shock patients in remote institutions: a pilot study (the cardiac-RESCUE program). *Eur Heart J* 34: 112-120
5. Abrams D, Combes A, Brodie D, (2014) Extracorporeal Membrane Oxygenation in Cardiopulmonary Disease in Adults. *J Am Coll Cardiol* 63: 2769-2778
6. Muller G, Flecher E, Lebreton G, Luyt CE, Trouillet JL, Brechot N, Schmidt M, Mastroianni C, Chastre J, Leprince P, Anselmi A, Combes A, (2016) The ENCOURAGE mortality risk score and analysis of long-term outcomes after VA-ECMO for acute myocardial infarction with cardiogenic shock. *Intensive Care Med* 42: 370-378
7. Barbaro RP, Odetola FO, Kidwell KM, Paden ML, Bartlett RH, Davis MM, Annich GM, (2015) Association of hospital-level volume of extracorporeal membrane oxygenation cases and mortality. Analysis of the extracorporeal life support organization registry. *Am J Respir Crit Care Med* 191: 894-901
8. Aissaoui N, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Diebold B, Chastre J, Combes A, (2011) Predictors of successful extracorporeal membrane oxygenation (ECMO) weaning after assistance for refractory cardiogenic shock. *Intensive Care Med* 37: 1738-1745
9. Chen YS, Lin JW, Yu HY, Ko WJ, Jerng JS, Chang WT, Chen WJ, Huang SC, Chi NH, Wang CH, Chen LC, Tsai PR, Wang SS, Hwang JJ, Lin FY, (2008) Cardiopulmonary resuscitation with assisted extracorporeal life-support versus conventional cardiopulmonary resuscitation in adults with in-hospital cardiac arrest: an observational study and propensity analysis. *Lancet* 372: 554-561
10. Sheu JJ, Tsai TH, Lee FY, Fang HY, Sun CK, Leu S, Yang CH, Chen SM, Hang CL, Hsieh YK, Chen CJ, Wu CJ, Yip HK, (2010) Early extracorporeal membrane oxygenator-assisted primary percutaneous coronary intervention improved 30-day clinical outcomes in patients with ST-segment elevation myocardial infarction complicated with profound cardiogenic shock. *Crit Care Med* 38: 1810-1817
11. Schmidt M, Brechot N, Hariri S, Guiguet M, Luyt CE, Makri R, Leprince P, Trouillet JL, Pavie A, Chastre J, Combes A, (2012) Nosocomial infections in adult cardiogenic shock patients supported by venoarterial extracorporeal membrane oxygenation. *Clin Infect Dis* 55: 1633-1641
12. Cheng R, Hachamovitch R, Kittleson M, Patel J, Arabia F, Moriguchi J, Esmailian F, Azarbal B, (2014) Complications of extracorporeal membrane oxygenation for treatment of cardiogenic shock and cardiac arrest: a meta-analysis of 1,866 adult patients. *Ann Thorac Surg* 97: 610-616
13. Bisdas T, Beutel G, Warnecke G, Hoepfer MM, Kuehn C, Haverich A, Teebken OE, (2011) Vascular complications in patients undergoing femoral cannulation for extracorporeal membrane oxygenation support. *Ann Thorac Surg* 92: 626-631
14. Sundberg S, Lilleberg J, Nieminen MS, Lehtonen L, (1995) Hemodynamic and neurohumoral effects of levosimendan, a new calcium sensitizer, at rest and during exercise in healthy men. *Am J Cardiol* 75: 1061-1066
15. Follath F, Cleland JG, Just H, Papp JG, Scholz H, Peuhkurinen K, Harjola VP, Mitrovic V, Abdalla M, Sandell EP, Lehtonen L, (2002) Efficacy and safety of intravenous levosimendan compared with dobutamine in severe low-output heart failure (the LIDO study): a randomised double-blind trial. *Lancet* 360: 196-202
16. Kivikko M, Anttila S, Eha J, Lehtonen L, Pentikainen PJ, (2002) Pharmacokinetics of levosimendan and its metabolites during and after a 24-hour continuous infusion in patients with severe heart failure. *International journal of clinical pharmacology and therapeutics* 40: 465-471

17. Mebazaa A, Nieminen MS, Packer M, Cohen-Solal A, Kleber FX, Pocock SJ, Thakkar R, Padley RJ, Poder P, Kivikko M, (2007) Levosimendan vs dobutamine for patients with acute decompensated heart failure: the SURVIVE Randomized Trial. *Jama* 297: 1883-1891
18. Gordon AC, Perkins GD, Singer M, McAuley DF, Orme RM, Santhakumaran S, Mason AJ, Cross M, Al-Beidh F, Best-Lane J, Brealey D, Nutt CL, McNamee JJ, Reschreiter H, Breen A, Liu KD, Ashby D, (2016) Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis. *N Engl J Med* 375: 1638-1648
19. Gong B, Li Z, Yat Wong PC, (2015) Levosimendan Treatment for Heart Failure: A Systematic Review and Meta-Analysis. *J Cardiothorac Vasc Anesth* 29: 1415-1425
20. Cholley B, Caruba T, Grosjean S, Amour J, Ouattara A, Villacorta J, Miguet B, Guinet P, Levy F, Squara P, Ait Hamou N, Carillon A, Boyer J, Boughenou MF, Rosier S, Robin E, Radutoiu M, Durand M, Guidon C, Desebbe O, Charles-Nelson A, Menasche P, Rozec B, Girard C, Fellahi JL, Pirracchio R, Chatellier G, (2017) Effect of Levosimendan on Low Cardiac Output Syndrome in Patients With Low Ejection Fraction Undergoing Coronary Artery Bypass Grafting With Cardiopulmonary Bypass: The LICORN Randomized Clinical Trial. *Jama* 318: 548-556
21. Mehta RH, Leimberger JD, van Diepen S, Meza J, Wang A, Jankowich R, Harrison RW, Hay D, Fremes S, Duncan A, Soltesz EG, Lubner J, Park S, Argenziano M, Murphy E, Marcel R, Kalavrouziotis D, Nagpal D, Bozinovski J, Toller W, Heringlake M, Goodman SG, Levy JH, Harrington RA, Anstrom KJ, Alexander JH, (2017) Levosimendan in Patients with Left Ventricular Dysfunction Undergoing Cardiac Surgery. *N Engl J Med* 376: 2032-2042
22. Landoni G, Lomivorotov VV, Alvaro G, Lobreglio R, Pisano A, Guarracino F, Calabro MG, Grigoryev EV, Likhvantsev VV, Salgado-Filho MF, Bianchi A, Pasyuga VV, Baiocchi M, Pappalardo F, Monaco F, Boboshko VA, Abubakirov MN, Amantea B, Lembo R, Brazzi L, Verniero L, Bertini P, Scandroglio AM, Bove T, Belletti A, Michienzi MG, Shukevich DL, Zabelina TS, Bellomo R, Zangrillo A, (2017) Levosimendan for Hemodynamic Support after Cardiac Surgery. *N Engl J Med* 376: 2021-2031
23. Affronti A, di Bella I, Carino D, Ragni T, (2013) Levosimendan may improve weaning outcomes in venoarterial ECMO patients. *Asaio j* 59: 554-557
24. Tafer N, Megret F, Dolbeau JB, Coiffic A, Calderon J, Mauriat P, Ouattara A, (2013) Intérêt du lévosimendan dans le sevrage difficile des assistances circulatoires temporaires. *Annales francaises d'anesthesie et de reanimation* 32: A211-A212
25. Distelmaier K, Roth C, Schrutka L, Binder C, Steinlechner B, Heinz G, Lang IM, Maurer G, Koinig H, Niessner A, Hulsman M, Speidl W, Goliash G, (2016) Beneficial effects of levosimendan on survival in patients undergoing extracorporeal membrane oxygenation after cardiovascular surgery. *Br J Anaesth* 117: 52-58
26. Ponikowski P, Voors AA, Anker SD, Bueno H, Cleland JG, Coats AJ, Falk V, Gonzalez-Juanatey JR, Harjola VP, Jankowska EA, Jessup M, Linde C, Nihoyannopoulos P, Parissis JT, Pieske B, Riley JP, Rosano GM, Ruilope LM, Ruschitzka F, Rutten FH, van der Meer P, (2016) 2016 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: The Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 37: 2129-2200
27. Altenberger J, Parissis JT, Costard-Jaeckle A, Winter A, Ebner C, Karavidas A, Sihorsch K, Avgeropoulou E, Weber T, Dimopoulos L, Ulmer H, Poelzl G, (2014) Efficacy and safety of the pulsed infusions of levosimendan in outpatients with advanced heart failure (LevoRep) study: a multicentre randomized trial. *Eur J Heart Fail* 16: 898-906
28. Husebye T, Eritsland J, Muller C, Sandvik L, Arnesen H, Seljeflot I, Mangschau A, Bjornerheim R, Andersen GO, (2013) Levosimendan in acute heart failure following primary percutaneous coronary intervention-treated acute ST-elevation myocardial infarction. Results from the LEAF trial: a randomized, placebo-controlled study. *Eur J Heart Fail* 15: 565-572
29. Jamali IN, Kersten JR, Pagel PS, Hettrick DA, Warltier DC, (1997) Intracoronary levosimendan enhances contractile function of stunned myocardium. *Anesth Analg* 85: 23-29
30. Kopustinskiene DM, Pollesello P, Saris NE, (2004) Potassium-specific effects of levosimendan on heart mitochondria. *Biochemical pharmacology* 68: 807-812

31. Sonntag S, Sundberg S, Lehtonen LA, Kleber FX, (2004) The calcium sensitizer levosimendan improves the function of stunned myocardium after percutaneous transluminal coronary angioplasty in acute myocardial ischemia. *J Am Coll Cardiol* 43: 2177-2182
32. Antoniadou C, Tousoulis D, Koumallos N, Marinou K, Stefanadis C, (2007) Levosimendan: beyond its simple inotropic effect in heart failure. *Pharmacology & therapeutics* 114: 184-197
33. Beiras-Fernandez A, Weis FC, Kur F, Kaczmarek I, Schmoeckel M, Weis M, Reichart B, (2008) Primary graft failure and Ca²⁺ sensitizers after heart transplantation. *Transplant Proc* 40: 951-952
34. Fries M, Ince C, Rossaint R, Bleilevens C, Bickenbach J, Rex S, Mik EG, (2008) Levosimendan but not norepinephrine improves microvascular oxygenation during experimental septic shock. *Crit Care Med* 36: 1886-1891
35. Nanas S, Gerovasili V, Dimopoulos S, Pierrakos C, Kourtidou S, Kaldara E, Sarafoglou S, Venetsanakis J, Roussos C, Nanas J, Anastasiou-Nana M, (2008) Inotropic agents improve the peripheral microcirculation of patients with end-stage chronic heart failure. *J Card Fail* 14: 400-406
36. Aissaoui N, El-Banayasy A, Combes A, (2015) How to wean a patient from veno-arterial extracorporeal membrane oxygenation. *Intensive Care Med* 41: 902-905
37. ELSO guidelines, <https://www.else.org/Resources/Guidelines.aspx>, Last accessed August 27th 2017
38. Latouche A, Porcher R, Chevret S, (2004) Sample size formula for proportional hazards modelling of competing risks. *Stat Med* 23: 3263-3274

18 LIST OF ADDENDA

Every addendum and the log of addenda versions are attached, independently of the protocol. Every addendum can be modified (change of addendum version) without modifying the version of the protocol.

18.1 List of Investigators

18.2 Serious Adverse Events report form

18.3 Pregnancy report form

18.4 Questionnaire/scale

- SOFA
- GLASGOW
- SAPS2

18.5 SmpC Levosimendan

- Levosimendan, ZIMINO®
<http://agence-prd.ansm.sante.fr/php/ecodex/rcp/R0274395.htm>

18.6 SmPC CERNEVIT®

<https://base-donnees-publique.medicaments.gouv.fr/extraire.php?specid=61001031>

**" LEVOSIMENDAN to facilitate weaning
from ECMO in severe cardiogenic shock patients "
LEVOECMO**

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

Version N° 3.0 du 26/06/2024

Project code number: P170914J - EUDRACT No.: 2019-004319-29

Coordinating Investigator:

Alain COMBES

Intensive Care Unit, Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Tel. : + 33 1 42 16 38 18

Email : alain.combes@aphp.fr

Methodologist:

Pr Florence TUBACH

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

47 - 83 Boulevard de l'Hôpital – 75651 Paris Cedex 13

Email: florence.tubach@aphp.fr

Sponsor:

Assistance publique – Hôpitaux de Paris and by delegation: Clinical Research and Innovation
Delegation (DRCI) - Hôpital Saint-Louis - 1, avenue Claude Vellefaux

DRCI head office project advisor: Zahia BEN ABDESSELAM

Tel : 01 40 27 56 97

Fax : 01 44 84 17 01

Email : zahia.ben-abdesslam@aphp.fr

Entity responsible for monitoring the trial:

Unité de Recherche Clinique (URC) - Hospital Pitié Salpêtrière

Head: Pr Florence TUBACH

Project advisor: Jessica PALMYRE

Email: jessica.palmyre@aphp.fr

Tel: 01 42 16 19 35

Fax : 01 42 16 24 40

Agence générale des équipements et produits de santé, (APHP)

AGEPS, Département Essais Cliniques - 7, rue du Fer à Moulin, 75005 Paris

Céline DUPRE : Email : celine.dupre@aphp.fr

Nancy TAMIR-GEDDIS: Email : nancy.tamirgeddis@aphp.fr

Clinical Research and Innovation Direction (DRCI)
Hôpital Saint Louis 75010 PARIS

**INTERVENTIONAL RESEARCH PROTOCOL
RELATING TO A MEDICINAL PRODUCT FOR HUMAN USE**

PROTOCOL SIGNATURE PAGE

Research code number: P170914J

Title: LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients – "LEVOECMO"

Version N° of: 3.0 of 26/06/2024

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:
Pr. Alain COMBES
Intensive Care Unit
Hospital Pitié Salpêtrière
47 - 83 Boulevard de l'Hôpital
75651 Paris Cedex 13

Date: 26/06/2024
Signature:

GH Pitié Salpêtrière Charles Foix
Médecine Intensive Réanimation Pr COMBES
Professeur COMBES Alain
Institut de Cardiologie
Pôle Coeur et Métabolisme
Tél : 01 42 18 33 18

Sponsor

Assistance Publique-Hôpitaux de Paris
Clinical Research and Innovation Delegation Hôpital
Saint Louis
1 avenue Claude Vellefaux
75010 PARIS

Date: 02/07/2024
Signature:


Yannick VACHER
Responsable centre - Hôpital
HCL

The study was approved by the Ethic committee (CPP) of Dapst III on 01/01/2021 and authorised by the ANSM on 30/11/2020

"LEVOECMO" protocol, version 3.0 of 26/06/2024

2/46

This document is the property of DRCI / AP-HP. All reproduction is strictly prohibited.
Version date: April 2018

TABLE OF CONTENTS

1	SUMMARY.....	5
2	SCIENTIFIC JUSTIFICATION FOR THE TRIAL.....	9
2.1	SHORT-TERM MECHANICAL CIRCULATORY SUPPORT FOR REFRACTORY CARDIOGENIC SHOCK	9
2.2	LEVOSIMENDAN, AN ORIGINAL INODILATOR	9
2.3	WHY RECENT LARGE LARGE RANDOMIZED CONTROLLED TRIALS OF LEVOSIMENDAN WERE ALL NEGATIVE?.....	10
2.4	RATIONALE FOR A TRIAL TESTING LEVOSIMENDAN IN VA-ECMO PATIENTS AND JUSTIFICATION FOR THE CHOICE OF SUBJECTS	11
2.5	NAME AND DESCRIPTION OF THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	11
2.6	DESCRIPTION AND JUSTIFICATION OF THE DOSAGE, ROUTE OF ADMINISTRATION, ADMINISTRATION SCHEDULE AND TREATMENT DURATION	11
2.7	SUMMARY OF THE KNOWN AND FORESEEABLE BENEFITS AND RISKS FOR THE STUDY PARTICIPANTS.....	12
3	OBJECTIVES	12
3.1	PRIMARY OBJECTIVE	12
3.2	SECONDARY OBJECTIVES	12
4	DESCRIPTION OF THE TRIAL.....	13
4.1	CONCISE DESCRIPTION OF THE PRIMARY AND SECONDARY ENDPOINTS.....	13
4.2	RESEARCH METHODOLOGY.....	13
5	PROCEDURE FOR THE TRIAL	16
5.1	BASELINE VISIT = J1 (INCLUSION AND RANDOMIZATION)	16
5.2	FOLLOW-UP VISITS	17
5.3	EXPECTED LENGTH OF PARTICIPATION, CHRONOLOGY AND DURATION OF THE STUDY.....	18
5.4	TABLE OR DIAGRAM SUMMARISING THE CHRONOLOGY OF THE STUDY	18
5.5	DISTINCTION BETWEEN STANDARD CARE AND RESEARCH	18
5.6	TERMINATION AND EXIT RULES.....	19
6	SELECTION CRITERIA.....	21
6.1	IDENTIFICATION OF ELIGIBLE PATIENTS: INCLUSION AND EXCLUSION CRITERIA.....	21
6.2	CONSENT AND RANDOMIZATION PROCEDURE.....	22
6.3	RECRUITMENT METHODS	22
7	TREATMENT ADMINISTERED TO STUDY PARTICIPANTS.....	23
7.1	THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	23
7.2	TRACEABILITY INFORMATION FOR THE INVESTIGATIONAL MEDICINAL PRODUCT(S)	24
7.3	COMMON MANAGEMENT REGARDING OTHER INOTROPIC DRUGS, HEART FAILURE MEDICATIONS AND ANTIAGREGANTS AND ANTICOAGULANTS.....	25
7.4	COMMON MANAGEMENT REGARDING THE ECMO CIRCUIT	25
7.5	AUTHORISED AND PROHIBITED TREATMENTS (MEDICINAL, NON-MEDICINAL, SURGICAL), INCLUDING EMERGENCY MEDICATIONS	25
7.6	METHODS FOR MONITORING COMPLIANCE WITH THE TREATMENT	26
8	EFFICACY ASSESSMENT.....	26
8.1	ANTICIPATED METHODS AND TIMETABLE FOR MEASURING, COLLECTING AND ANALYSING THE EFFICACY DATA	26
9	SPECIFIC COMMITTEES FOR THE TRIAL	26
9.1	STEERING COMMITTEE (SC)	26
10	SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY	27
10.1	RECORDING AND REPORTING ADVERSE EVENTS.....	27
11	DATA MANAGEMENT	33
11.1	DATA COLLECTION PROCEDURE	33
11.2	IDENTIFICATION OF DATA RECORDED DIRECTLY IN THE CRFs WHICH WILL BE CONSIDERED AS SOURCE DATA.....	33
11.3	RIGHT TO ACCESS SOURCE DATA AND DOCUMENTS.....	33

11.4	DATA PROCESSING AND STORAGE OF DOCUMENTS AND DATA	34
11.5	OWNERSHIP OF THE DATA.....	34
12	STATISTICAL ASPECTS	34
12.1	PLANNED STATISTICAL METHODS, INCLUDING THE TIMETABLE FOR ANY PLANNED INTERIM ANALYSES	34
12.2	SAMPLE SIZE CALCULATION	37
12.3	STATE WHETHER SUBJECTS WHO EXIT THE STUDY PREMATURELY WILL BE REPLACED AND IN WHAT PROPORTION.	37
12.4	ANTICIPATED LEVEL OF STATISTICAL SIGNIFICANCE.....	37
12.5	MANAGEMENT OF MODIFICATIONS MADE TO THE ANALYSIS PLAN FOR THE INITIAL STRATEGY.	37
12.6	SELECTION OF POPULATIONS	37
13	QUALITY CONTROL AND ASSURANCE	37
13.1	GENERAL ORGANISATION	37
13.2	QUALITY CONTROL	38
13.3	CASE REPORT FORM.....	38
13.4	MANAGEMENT OF NON-COMPLIANCES.....	38
13.5	AUDITS/INSPECTIONS.....	39
13.6	PRINCIPAL INVESTIGATOR'S DECLARATION OF RESPONSIBILITY	39
14	ETHICAL AND LEGAL CONSIDERATIONS	39
14.1	METHODS FOR INFORMING AND OBTAINING CONSENT FROM THE RESEARCH PARTICIPANTS.....	39
14.2	<i>PROHIBITION OF CONCOMITANT CLINICAL STUDIES PARTICIPATION AND EXCLUSION PERIOD AFTER THE TRIAL, IF APPLICABLE</i>	40
14.3	COMPENSATION FOR SUBJECTS	40
14.4	REGISTRATION ON THE NATIONAL REGISTER OF SUBJECTS PARTICIPATING IN HUMAN RESEARCH TRIALS ON THE PRODUCTS LISTED IN ARTICLE L. 5311-1 OF THE FRENCH PUBLIC HEALTH CODE.....	40
14.5	AUTHORISATION FOR THE RESEARCH LOCATION.....	40
14.6	LEGAL OBLIGATIONS.....	40
15	FUNDING AND INSURANCE	41
15.1	SOURCES OF FUNDING FOR THE TRIAL	42
15.2	INSURANCE	42
16	PUBLICATION	42
16.1	MENTION OF AP-HP AFFILIATION FOR PROJECTS SPONSORED OR MANAGED BY AP-HP	42
16.2	MENTION OF THE AP-HP MANAGER (DRCI) IN THE ACKNOWLEDGEMENTS OF THE TEXT.....	42
16.3	MENTION OF THE FUNDER IN THE ACKNOWLEDGEMENTS OF THE TEXT	42
17	BIBLIOGRAPHY	43
18	LIST OF ADDENDA	46
18.1	LIST OF INVESTIGATORS	46
18.2	SERIOUS ADVERSE EVENTS REPORT FORM	46
18.3	PREGNANCY REPORT FORM	46
18.4	QUESTIONNAIRE/SCALE.....	46
18.5	SCP'S LEVOSIMENDAN.....	46

1 SUMMARY

Full title	LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients
Acronym	LEVOECMO
Coordinating Investigator	COMBES ALAIN, GROUPE HOSP. PITIE-SALPETRIERE (AP-HP), PARIS, France
Sponsor	Assistance Publique-Hôpitaux de Paris
Scientific justification	In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with refractory cardiogenic shock. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery. In patients with potentially reversible cardiac failure (e.g. myocarditis, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery. Although considered as the ultimate life-saving technology for refractory cardiac failure, veno-arterial ECMO is still associated with severe complications. Specifically, excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. ECMO support also exposes to many complications such as infections, hemorrhage or peripheral vascular embolism. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs. Levosimendan, which acts to sensitize myocardial contractile proteins to calcium, improves cardiac contractility without increasing the intracellular calcium concentration. Unlike traditional inotropes such as dobutamine, levosimendan neither increases myocardial oxygen consumption nor impairs diastolic function or possess proarrhythmic effects. It also influences the opening of ATP-dependent potassium channels, including those in vascular smooth muscle cells, leading to coronary, pulmonary, and peripheral vasodilation and antiinflammatory, antioxidative, antiapoptotic, anti-stunning and cardioprotective effects. Additionally, Levosimendan which has a long lasting action (up to 7–9 d), resulting from the formation of active metabolite, may be used as a single 24h perfusion. In recent preliminary studies, the drug was associated with accelerated weaning from VA-ECMO and even improved survival. Therefore, a multicenter randomized trial with sufficient statistical power is needed in refractory cardiogenic shock patients supported by VA-ECMO to test if the early administration of Levosimendan can facilitate and accelerate VA-ECMO weaning, and ultimately translate in significantly less morbidity, reduced ICU and hospital length of stays and associated costs.
Main objective and primary endpoint	Main objective: To assess the efficacy of early administration of Levosimendan on the time to successful ECMO weaning (defined below) within the 30 days following randomization, in patients with VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Primary endpoint: Time to successful ECMO weaning within the 30 days following randomization.

	- ECMO weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal. - Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization). - Patients still under ECMO 30 days after randomization will be considered as censored. - The number of days alive without ECMO was not an appropriate primary outcome due to the high proportion of patients dying with ECMO and thus ex-aequo (0 days). The analysis will therefore model the risk of successful ECMO ablation in the presence of two competing risks (death and weaning failure).
Secondary objectives and endpoints	Secondary objectives/endpoints: To assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function. - mortality on D30 and D60 - duration of ECMO support between inclusion and D30, and between inclusion and D60; number of ECMO-free days between inclusion and D30/D60 - duration of ICU stay and duration of hospitalization; - major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, rehospitalization for heart failure on days 30 and 60 - time to improvement into hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization - number of days with organ failure, defined by SOFA score, and number of days alive without organ failure between inclusion and D30; - duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30; - duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60; - LV function assessed with echocardiography at Day 30 following randomization - Incidence of adverse drug reactions (atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia). The primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.
Design of the trial	Multicenter, Double Blind Randomized clinical trial
Target Population	Adults patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support and for whom LV recovery is likely.
Inclusion criteria	1. Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h. 2. Obtain informed consent from a close relative or surrogate. According to the specifications of emergency consent, randomization without the close relative or surrogate consent

	could be performed. Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.
Exclusion criteria	1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized. 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) History of torsades de pointes in the 30 days prior to inclusion 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study 21) Patient with a weight over 180 kg 22) Known hypersensitivity to polyvitamin CERNEVIT® 23) In case of hypervitaminosis to any vitamin contained in this formulation, 24) In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria 25) In combination with vitamin A or retinoids
Investigational medicinal product(s)	A continuous infusion of Levosimendan administered over 24h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate limiting side effects.
Comparator treatment	The placebo will be made of polyvitamin CERNEVIT® in order to reproduce the yellow color of levosimendan. The route of administration, administration procedures and duration of treatment will be similar to those of the experimental arm.
Risks added by the trial	Treatment by levosimendan or placebo
Procedure of the trial	<u>Inclusion visit (= Randomisation) Day 1</u> – Informed consent form. – Randomisation – Administration of a continuous infusion of Levosimendan or Placebo – Comorbidities – Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses – Record of AE/SAE

	Day 2 to Day 30 or date of death or date of discharge from the ICU – Record of AE/SAE – Clinical data – ECMO parameters – Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers – Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,) – SOFA score – Catecholamine doses Day 30, Day 60 – Vital status – ECMO: yes/no – On other mechanical cardiac support device – Heart transplantation – Mechanical ventilation – ICU discharge and date – Hospital discharge and date
Number of subjects included	206 (103 per arm)
Number of sites	National trial involving 14 centers
Duration of the trial	- inclusion period: 36 + 3 months - participation period (treatment+follow-up):2 months - total duration: 38 + 3 months
Number of enrolments expected per site and per month	0.4
Statistical analysis	The analysis will model the risk of successful VA-ECMO ablation in the presence of two competing risks (death and weaning failure).
Trial will have a Data Monitoring Committee	YES
Funding source	PHRC 2017 / French Ministry of Health

2 SCIENTIFIC JUSTIFICATION FOR THE TRIAL

2.1 Short-term mechanical circulatory support for refractory cardiogenic shock

In the last decade, venoarterial extracorporeal membrane oxygenation (VA-ECMO) has become the first-line therapy in patients with acute cardiogenic shock in the setting of acute myocardial infarction, myocarditis, intoxication with cardiotoxic drugs, end-stage dilated cardiomyopathy or post-cardiotomy [1-6]. Most of these “crash and burn” patients receive the device as salvage therapy after having already developed signs of multiple organ failure. VA-ECMO provides both respiratory and cardiac support, is easy to insert, even at the bedside, provides stable flow rates, and is associated with less organ failure after implantation compared to large biventricular assist-devices that require open-heart surgery [1, 2]. Several considerations must be taken into account before instituting ECMO. First, the device should be inserted before the patient has developed multiple organ failure or myocardial failure has led to refractory cardiac arrest, since these conditions are associated with significantly poorer outcomes [2, 3, 6]. Second, highly unstable patients may benefit from urgent on-site ECMO initiation by a rapid resuscitation team able to operate a portable and quick-to-prime ECMO circuit before transportation to the ECMO referral center [4]. Third, cardiac failure and other organ injuries should be deemed reversible and the patient’s underlying condition should not contraindicate a bridge to a more permanent device or to transplantation. Fourth, management of patients on ECMO for refractory cardiogenic shock is complex and should be conducted in experienced medical-surgical centers [7].

VA-ECMO might be used as a bridge to decision-making if the patient survives the first days to reach the “decision-making” point. In patients with potentially reversible cardiac failure (e.g. myocarditis, drug intoxication, myocardial stunning post-myocardial infarction, post-cardiotomy or post-cardiac arrest), VA-ECMO might be weaned after a few days of support and used as a bridge to recovery [1]. Alternatively, it might be used as a bridge to transplantation or VAD implantation because of absence of LV functional recovery [8]. ECMO can also be simply withdrawn in cases of therapeutic futility (severe brain lesions, end-stage multiple organ failure or absence of myocardial recovery in the context of a definitive contraindication to transplantation or VAD implantation). Long-term survival after VA-ECMO is 70-80% after myocarditis or cardiotoxic drug poisoning, 40-50% after myocardial infarction and 15-25% when the device is used to rescue refractory cardiac arrest [2, 3, 6, 8-10]. Survivors reported a preserved quality of life, despite some limitations in physical activities and social functioning in previous series [2, 3, 6].

Although considered as the ultimate life-saving technology for refractory cardiac failure, venoarterial ECMO is still associated with severe complications. It increases left ventricular (LV) afterload that may delay myocardial recovery. Excessive LV afterload and lack of LV unloading under VA-ECMO might induce LV stasis with thrombus formation, pulmonary edema, myocardial ischemia caused by ventricular distension and ultimately increase mortality. Prolonged ECMO support also exposes to many complications such as local hemorrhage (10-20%), cannulation site infection and bacteremia (10-15%), limb ischemia (5-10%), ischemic or hemorrhagic stroke (5%) [2, 3, 6, 11, 12]. These complications are more frequent with prolonged support and are responsible for significant morbidity and mortality, prolonged ICU and hospital stays and higher costs [2, 11-13].

2.2 Levosimendan, an original inodilator

Levosimendan is a calcium sensitizer with inotropic, vasodilatory and cardioprotective properties. It facilitates contact between actin and myosin and prolongs the time of contact between these two proteins, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect that unlike other positive inotropes, does not modify intracellular calcium concentration (which can cause arrhythmia and cell death) or myocardial diastolic relaxation. It also increases the contractile force of the myocardium without modifying its O₂ consumption. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on coronary arteries). The drug was also reported to exert cardioprotective, anti-stunning and anti-ischemic effect through this K⁺-ATP-dependent channel pathway.

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four

to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with prolonged half-life of about three days [16].

2.3 Why recent large randomized controlled trials of levosimendan were all negative?

Levosimendan has been tested in large RCTs against dobutamine or placebo in patients with decompensated heart failure, septic shock and the low cardiac output syndrome after cardiac surgery.

In patients with acute decompensated heart failure, while a first proof of concept trial suggested that levosimendan was associated with improved haemodynamic performance and lower mortality than dobutamine [15], the large (n=1,327) randomized double-blind SURVIVE clinical trial failed to confirm these findings [17]. In this trial, patients received a loading dose of levosimendan (12 µg/kg) followed by an infusion of 0.1 up-to 0.2 µg/kg/min for an additional 23 hours, as tolerated. Although a greater decrease in B-type natriuretic peptide level was observed in the first 5 days of treatment in the levosimendan group, all-cause mortality at 180 days was similar between groups, respectively occurring in 26% in the levosimendan group and 28% in the dobutamine group (HR 0.91; [0.74-1.13]; p=0.40). Marked hypotension, probably related to the levosimendan loading dose, atrial fibrillation, and hypokalemia were more frequent in the levosimendan group, possibly explaining the negative results of this trial.

The Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis (LeoPARDS) trial tested whether the addition of levosimendan to standard care would reduce the severity of organ dysfunction among patients with septic shock [18]. The administration of levosimendan or placebo was started at a rate of 0.1 µg/kg/min, increased to 0.2 µg/kg/min for a further 20 to 22 hours in the absence of rate-limiting side effects. The addition of levosimendan to standard treatment was not associated with less severe organ dysfunction or lower mortality. The mean (±SD) SOFA score over the ICU stay was 6.68±3.96 in the levosimendan group and 6.06±3.89 in the placebo group (P = 0.053) and mortality at 28 days was 34.5% in the levosimendan group and 30.9% in the placebo group (P = 0.43). More patients in the levosimendan group than in the placebo group had supraventricular tachyarrhythmia, tachycardia and hypotension. The negative results of this trial may relate to the inclusion of very few patients with septic shock and confirmed myocardial dysfunction and to the administration of relatively high dose of levosimendan (up to 0.2 µg/kg/min), which induced marked tachycardia and hypotension, the vasodilating properties of drug possibly aggravating sepsis-induced vasoplegia in this setting.

Meta-analyses of small trials [19] suggested that levosimendan may result in a higher rate of survival among patients at risk of developing or presenting the low cardiac output syndrome after cardiac surgery and set the stage for 3 large RCTs testing the drug in this setting. The LICORN [20] (n=336 patients) and the LEVO-CTS [21] (n=882 patients) trials assessed the ability of preoperative levosimendan to prevent postoperative low cardiac output syndrome in patients with pre-operative left ventricular dysfunction. The continuous prophylactic infusion of 0.1 µg/kg/min was started just after anesthetic induction and was maintained over 24 hours in the absence of serious adverse reactions. The occurrence of the composite end-points of prolonged catecholamine infusion, use of left ventricular mechanical assist device, or renal replacement therapy (LICORN) and of death, renal-replacement therapy, perioperative myocardial infarction, or use of a mechanical cardiac assist device (LEVO-CTS) and the rates of adverse events were not different between levosimendan and placebo treated patients in these studies. It should be mentioned that only 22% of the patients developed low cardiac output syndrome in the LEVO-CTS trial and that in both the LICORN and the LEVO-CTS trials, only half of the patients needed catecholamine support for >24 hours. Thus, the preventive use of levosimendan does not seem to confer clinical benefits in this setting. Alternatively, the CHEETAH trial [22] included only patients who developed perioperative or early post-operative cardiovascular dysfunction and low cardiac output syndrome, defined as the need for support with an intraaortic balloon pump or high-dose inotropic support in patients with preoperative left ventricular ejection fraction of less than 25%. Patients were randomly assigned to receive levosimendan at a dose of 0.025 to 0.2 µg/kg/min or placebo, with no loading dose, for up to 48 hours or until discharge from the intensive care unit. The trial was stopped early for futility after 506 patients were enrolled and there

was no significant difference in the primary end-point of 30-day mortality between the levosimendan group and the placebo group (32 patients [12.9%] and 33 patients [12.8%], respectively; absolute risk difference, 0.1 percentage points; 95% confidence interval [CI], -5.7 to 5.9; $P = 0.97$). Rates of hypotension or cardiac arrhythmias were also not different between the levosimendan and placebo groups. It should however be noted that the mean infusion dose of levosimendan ($0.07 \mu\text{g/kg/min}$) was lower in this trial than in the 2 other RCTs of cardiac surgery patients.

2.4 Rationale for a trial testing levosimendan in VA-ECMO patients and justification for the choice of subjects

Preliminary retrospective cohort studies suggested a benefit of the early administration of levosimendan in refractory cardiogenic shock patients on VA-ECMO. In a small “before-after” study of medical cardiogenic shock patients, Affronti et al. [23] showed that the drug was associated with a higher rate of VA-ECMO weaning compared to historical controls, respectively 83.3% vs 27.3% ($p=0.498$). Similarly, Tafer et al [24] reported more ECMO-free days (22 vs 9) and lower 30-day mortality (24% vs 56%, $p<0.05$) in a cohort of 80 patients of whom 37 had received levosimendan early after ECMO initiation. In a series of 240 patients undergoing VA-ECMO after cardio-vascular surgery, of whom 75% received levosimendan immediately after ECMO initiation [25], the treatment was also associated with higher rates of successful ECMO weaning [HR; 0.41; 95% CI, 0.22–0.80; $P=0.008$]. Additionally, lower 30-day (adjusted HR 0.52; 95% CI 0.30–0.89; $P=0.016$), and long-term survival (adjusted HR 0.64; 95% CI 0.42–0.98; $P=0.04$) were reported, although levosimendan-treated patients had higher baseline EuroSCORE ($P<0.001$) and SAPS-3 ($P=0.02$) and lower LVEF ($p=0.04$).

Indeed, the rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support. Furthermore, it might be possible to administer higher doses of levosimendan to this specific group of patients, since VA-ECMO also provides robust peripheral vascular support that might protect against vasoplegia and hypotension that were suspected to hamper the benefit of the drug in previous trials.

In conclusion, the probability of demonstrating lower mortality with levosimendan in cardiogenic shock patients on ECMO appears unlikely based on the results of recent large RCTs. However, any medical therapy that may be able to hasten myocardial recovery and shorten the duration of VA-ECMO support might translate in significantly less morbidity and ultimately reduced ICU and hospital length of stays and associated costs. Highly relevant pathophysiological and clinical reasons indicate that levosimendan might fulfill these objectives. A multicenter randomized trial with sufficient statistical power is therefore urgently needed to test the early administration of Levosimendan to facilitate and accelerate VA-ECMO weaning in refractory cardiogenic shock patients supported by VA-ECMO.

2.5 Name and description of the investigational medicinal product(s)

- Levosimendan (ZIMINO®)
- CERNEVIT®

See section 7 for more details

2.6 Description and justification of the dosage, route of administration, administration schedule and treatment duration

See section 7 for further information about the administration of the product

2.7 Summary of the known and foreseeable benefits and risks for the study participants

Reducing the duration of ECMO support with levosimendan may be associated with less ECMO-related complications such as hemorrhages, stroke or infections. It may also be accompanied by a reduction in the duration of ICU and hospital stays, therefore reducing the costs associated with the care of these critically ill patients. Better long-term outcomes might also be expected with earlier ECMO weaning in these severe cardiogenic shock patients.

Known risk associated with levosimendan infusion include hypotension and supraventricular and ventricular arrhythmias. However, this side effects may be significantly mitigated in patient on VA-ECMO.

3 OBJECTIVES

3.1 Primary objective

The primary objective will be to assess the efficacy of the early administration of levosimendan on the time to successful ECMO weaning in patients on VA-ECMO support for acute cardiogenic shock refractory to conventional therapy. Weaning will be considered successful only if the patient survives without ECMO, other mechanical circulatory device or heart transplantation 30 days after ECMO removal [8, 36]. Patients still under ECMO 30 days after randomization will be considered as censored at 30 days.

Patients with unsuccessful ECMO weaning will be considered as weaning failure (competing risk) at the date of the weaning attempt.

3.2 Secondary objectives

Secondary objectives of the trial are to assess the efficacy of early administration of Levosimendan on mortality, durations of ECMO support, mechanical ventilation, ICU and hospital stay, the occurrence of major adverse cardiac events, the evolution of hemodynamic parameters and organ function.

- Mortality on D30 and D60
- Duration of ECMO support between inclusion and D30, and between inclusion and D60; and number of ECMO-free days between inclusion and D30/D60
- Duration of ICU stay; of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60.
- Time to improvement in hemodynamics (systolic, diastolic, mean blood pressure, heart rate...) and time to hemodynamic stabilization
- Number of days with organ failure, defined by SOFA score, and number of days without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and number of days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and number of days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of drug adverse effects (atrial fibrillation, ventricular arrhythmias, ventricular fibrillation, “torsades de pointes”, hypokalemia)
- Primary and secondary endpoints in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4 DESCRIPTION OF THE TRIAL

4.1 Concise description of the primary and secondary endpoints

4.1.1 Primary endpoint

- Time to successful ECMO weaning within the 30 days following randomization

- The event of interest is the successful weaning of ECMO (within 30 days after randomization) defined as: the patient survives without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal [8, 36]. Thus all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization).
- ECMO weaning failure (i.e. unsuccessful weaning), within 30 days after randomization, is a competing event.
- Death within 30 days after randomization, without an attempt to wean, is a competing event.
- No event after 30 days after randomization will be considered and all patients without any of the three events defined above are censored at 30 days post randomization.
- A protocolized management regarding weaning of VA-ECMO will be applied to both groups (see paragraph 5.2.1 for details) [8, 36].
- The number of days alive without ECMO was not an appropriate primary outcome criterion due to the high proportion of patients dying on ECMO and thus ex-aequo (0 days). The analysis will therefore **model the risk of successful ECMO weaning in the presence of competing risks (death and weaning failure)**.

4.1.2 Secondary endpoints

- Mortality on Day 30 and Day 60
- Total duration of ECMO support between inclusion and D30 and between inclusion and D60;
- Number of ECMO-free days between inclusion and D30/D60.
- Duration of ICU stay and of hospitalization;
- Major adverse cardiovascular events defined as death, cardiac transplant, escalation to permanent left ventricular assist device, stroke, dialysis, re-hospitalization for heart failure on days 30 and 60
- Time to improvement in hemodynamic parameters (systolic, diastolic, mean blood pressure, heart rate ...) and time to hemodynamic stabilization
- Days with organ failure, defined by the SOFA score, and days alive without organ failure between inclusion and D30;
- Duration of hemodynamic support with catecholamines and days alive without hemodynamic support between inclusion and D30;
- Duration of mechanical ventilation and days alive without mechanical ventilation between inclusion and D30 and D60;
- LV function assessed with echocardiography at Day 30 following randomization
- Incidence of adverse drug reactions (including atrial fibrillation and other supraventricular arrhythmias, ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes, hypokalemia).

The Primary and secondary endpoints will be also assessed in predefined subgroups: acute myocardial infarction, myocarditis, post-cardiac surgery and post-cardiac arrest patients.

4.2 Research methodology

4.2.1 Design of the trial

Characteristics of the trial are as follows:

- Multicenter clinical trial
- Phase III
- Randomized
- Double blind
- Comparative, Placebo controlled
- Superiority design
- Two treatment arms
 - Early administration of Levosimendan vs. placebo
 - 1:1 ratio

All consecutive eligible patients will be proposed to participate.

Allocation concealment will be guaranteed by use of a centralized randomization process with a web server (see below) and use of the minimization method of randomization. Investigator masking to group assignment after randomization will be guaranteed by use of a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

Because acute cardiogenic shock is a complex and multifactorial process, one key issue in every trial evaluating a new therapeutic strategy is how patients in both groups are managed and how standard care for these patients recommended by current guidelines are implemented in the participating ICUs. In the LEVOECMO trial, common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

4.2.2 Number of participating sites

The study will involve 14 sites in France. Subjects will be recruited in ICU departments.

4.2.3 Identification of the subjects

All consecutive eligible patients will be proposed to participate.

The subjects participating in this study will be identified as follows:

Site number (3 digits) - Sequential selection number for the site (4 digits) - surname initial - first name initial

This reference number is unique and will be used for the entire duration of the trial.

A randomisation number will also be assigned when the participant is randomised. This number will have the following format: XXX.

A treatment number corresponding to the treatment arm will be also assigned to the patient when the participant is allocated to a treatment arm. This number will have the following format: TXXXXX.

A registry of eligible patients not included will be completed, with reason for non-inclusion.

4.2.4 Randomization

After baseline screening, patients will be randomly assigned in a 1:1 ratio to receive levosimendan or placebo. The computer-generated randomization process will be implemented using the minimization method with a probability of assignment of 0.8 after the 20 first included patients (being assigned by balanced random) by a statistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence TUBACH. Stochastic minimization will be according to the primary etiology of cardiogenic shock (acute myocardial infarction, myocarditis, post-heart surgery and other causes) and by center. The assignments of participants will be concealed by use of a centralized 24-hour Internet service (CleanWeb, Télémédecine Technologies, Boulogne, France) and thus without knowledge of the next assignment in the sequence.

4.2.5 Blinding methods and measures put in place to protect the blinding

Investigator masking to group assignment after randomization will be guaranteed by use a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo.

The study will be double blinded. Treatments will be prepared at ICU by a nurse "off-protocol handler" (specifically trained caregiver) opens the box given by the PUI and prepares the infusion Levosimendan or CERNEVIT®.

Levosimendan and the placebo will be administred as non-identifiable preparation.

Trial participants, care providers, and outcome assessors will be masked to the patient assignment.

4.2.6 Unblinding procedures, if applicable

The investigating doctor may request unblinding for any reason he considers essential, by contacting:

- the DRCI (sponsor) **in non-urgent situations**, on working days and during working hours; the request should be addressed to the DRCI's project advisor, tel.: 01 40 27 57 47
- the poison centre of Fernand Widal Hospital, in the case of an emergency (see emergency situations requiring unblinding), on weekends, bank holidays, when the DRCI is closed and when unblinding cannot be carried out by the DRCI.
Telephone: **+33 (0)1 40 05 48 48**
- the hospital pharmacy

5 PROCEDURE FOR THE TRIAL

5.1 Baseline visit = J1 (inclusion and randomization)

5.1.1 Verification of inclusion and exclusion criteria (see paragraph 6)

5.1.2 Consent (see paragraph 14.1 for details)

Whose consent must be obtained	Who informs the individual and collects their consent	When is the individual informed	When is the individual's consent collected
• Patient's legal representative, family member or a close relative	• Local site investigator	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow	• For patient's legal representative, family member or close relative : Before randomization OR as soon as possible after randomization • When his/her condition will allow

5.1.3 Treatment

- Reconstitution/Preparation of the treatment before administration
- Administration by IV perfusion during 24h
- Administration as soon as possible after randomization; no later than 24 hours after randomization
- Prohibited administration of CERNEVIT® during experimental treatment administration

5.1.4 Clinical data

- Date, time of hospital and ICU admission
- Date, time of ECMO initiation
- Reason for ECMO initiation
- Date of birth
- Sex
- McCabe and Jackson score
- Charlson comorbidity index score
- history of systemic hypertension, myocardial infarction, diabetes mellitus, preexisting cardiovascular diseases (stroke, carotid surgery), severe left ventricle dysfunction (NYHA stage and ejection fraction determined by echocardiography or angiography), chronic renal insufficiency (serum creatinine, chronic dialysis), and hepatic insufficiency, defined as serum bilirubin >100 µmol/L or albuminemia <35 g/L with no other cause found
- immunocompromised: yes/no,
- SAPS II (table to check-off)
- Sequential organ-failure assessment (SOFA) score (table to check-off)
- Hemodynamic parameters (including arterial pressure and heart rate)
- Catecholamine doses

5.1.5 Biological and echocardiographic data

- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers

- Echocardiographic parameters (LV/LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus,)

5.1.6 ECMO parameters

- FiO₂ and membrane ventilation
- Pump turns per minute and flow
- Fibrin/clots on the membrane: yes/no
- Clogged circuit requiring change: yes/no
- Intravascular hemolysis caused by the circuit: yes/no
- Hemorrhage at the cannula insertion site: yes/no
- Infection at the cannula insertion site: yes/no
- Circuit changed: yes/no
- Leg ischemia: yes/no

5.2 Follow-up visits

5.2.1 The following Information will be collected on days 1, 2, 3... 30 or date of death or date of ICU discharge

- Hemodynamic parameters (arterial pressure, heart rate).
- Catecholamine doses
- Mechanical ventilation: yes/no
- Renal replacement therapy: yes/no
- Urine output
- Ionogram (including potassium), serum lactate, Blood Gases, WBC, Platelets, renal and hepatic markers
- SOFA score
- ECG
- Echocardiographic parameters (LVEF, LV outflow VTI, Tissue Doppler systolic velocity the lateral mitral annulus).
- Weanability test [8, 36]: yes/no
 - A protocolized management regarding weaning of VA-ECMO will be applied to both groups.
 - An ECMO-weaning trial will be undertaken every day as soon as the patient will be considered hemodynamically stable, i.e. baseline MBP >60 mm Hg while receiving no or low-dose vasopressors and a pulsatile arterial waveform maintained for at least 24 h.
 - The ECMO flow will be decreased gradually decreased to 66%, and to 33% of its baseline value and then to a minimum of 1–1.5 L/min.
 - If MBP drops significantly and constantly <60 mm Hg during the trial, ECMO flow will be returned to 100% of the initial flow and the trial will be stopped.
 - Patients achieving aortic VTI ≥12 cm, LVEF >20-25% and TDSa ≥6 cm/s at minimal ECMO flow support, will be considered weanable from ECMO [8, 36].
- ECMO parameters until device removal
 - FiO₂ and membrane ventilation
 - Pump turns per minute and flow
 - Fibrin/clots on the membrane: yes/no
 - Clogged circuit requiring change: yes/no
 - Intravascular hemolysis caused by the circuit: yes/no
 - Hemorrhage at the cannula insertion site: yes/no
 - Infection at the cannula insertion site: yes/no
 - Circuit changed: yes/no

- Leg ischemia: yes/no

5.2.2 Outcomes on days 30 and 60

- Living/dead; died: date and hour
- On ECMO: yes/no
- In case of ECMO weaning, need for a second ECMO run yes/no
- On other mechanical cardiac support device: yes/no
- Heart transplantation: yes/no
- Mechanical ventilation: yes/no
- In the ICU: yes/no; discharged from the ICU: date
- Hospitalized; yes/no; discharged from the hospital: date
- Patient's contact information will be collected at ICU discharge to permit 60 day outcome data collection.

5.3 Expected length of participation, chronology and duration of the study.

- Length of enrolment period: 36 months + 3 months
- Duration of participation for each subject: 60 days
- Treatment period: 24 hours
- Follow-up period: 60 days
- Total study duration: (start = trial start date; end = trial end date) = 38 months + 3 months
- When randomization and blinding will take place: immediately after patients are deemed eligible.

5.4 Table or diagram summarising the chronology of the study

Actions	D1	D2 to D30 or date of death or date of ICU discharge	D60
Information	X		
Signature of the consent form	X		
Randomization	X		
Past medical history	X		
Clinical exam	X	X	
Medical procedures (ECG, Echo.)	X	X	
Dispensation of LEVOSIMENDAN or Placebo	X		
Adverse events	X	X	X
Vital status	X	X	X
ECMO status	X	X	X
Pursuit consent		X	X (if not obtained before)

5.5 Distinction between standard care and research

Procedures and treatments carried out as part of the research	Procedures and treatments associated with <u>care</u>	Procedures and treatments added because of <u>the research</u>
Treatments	None	Levosimendan or Placebo at D1
Consultations	D1 to D30 : During patient's hospitalisation's	Follow up by phone :Day 60

Blood samples	Hematology - Biochemistry	None
Imaging, etc.	ECG Echocardiography	None
Act	ECMO	None

In the LEVOECMO trial, only the initial treatment undergoing randomization (levosimendan vs. placebo) will differ between groups.

Common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants will be conducted according to the latest European Society of Cardiology [26]. The ECMO circuit will be managed according to recommendations of the ELSO organization [37]. And see paragraph 7.4 for details.

5.6 Termination and exit rules

5.6.1 Criteria and procedures for prematurely terminating the study treatment

- Temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the subject's source file and the case report form (CRF). The subject remains enrolled in the study
- Premature termination of treatment, but the subject remains enrolled in the study until the end of the subject's participation: the investigator must document the reason
- Premature termination of treatment and exit from the study. The only reasons for premature exit from the study are: death, lost to follow-up, or patient's consent withdrawal. Premature termination of treatment or adverse reaction should not imply premature exit from the study.

The investigator must:

- Document the reason(s)
- Collect all endpoints at the moment the subject exits from the study, if the subject agrees
- Schedule further follow-up visits, especially in case of a serious adverse event.

Study treatment must be discontinued if they experience any of the following:

- Anaphylactic shock
- Ventricular fibrillation
- "Torsades de pointes"
- Ventricular tachycardia
- Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for more than 60 minutes despite:
 - vascular filling with two 500 mL successive crystalloid boluses AND
 - norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.

5.6.2 Criteria and procedure for premature withdrawals and exits from the study

- Subject may exit the study at any time and for any reason.
- The investigator can temporarily or permanently withdraw a subject from the study for any safety reason or if it is in the subject's best interests
- Subject lost to follow-up: the subject cannot be located. The investigator must make every effort to reconnect with the subject and record his attempts in the source file (at least to determine whether the subject is alive or dead).

If a participant leaves the research prematurely, data relating to the participant can be used unless an objection was recorded when the participant signed the consent form.

If a subject exits the trial prematurely or withdraws consent, any data collected prior to the date of premature exit may still be used.

The case report form must list the various reasons why the subject exited or was withdrawn from the study:

- ☐ Lack of efficacy
- ☐ Adverse reaction
- ☐ Other medical problem
- ☐ Subject's personal reasons
- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up
- ☐ Death

5.6.3 Monitoring subjects after the premature termination of treatment

If a subject exits the trial this will in no way affect the standard care received for his/her condition. In case of severe adverse events, the investigator must notify the sponsor and monitor the subject for 30 days following the premature termination of treatment. If the treatment is stopped prematurely due to a serious adverse event, a serious adverse event report will be sent by mail (vigilance.drc@aphp.fr) to the sponsor. The serious adverse reaction will be monitored until it is resolved.

5.6.4 Full or partial cancellation of the study

AH-HP (the sponsor) or the Competent Authority (ANSM) may prematurely discontinue all or part of the trial, temporarily or permanently, upon the recommendation of the Data Monitoring Committee in the following situations:

- first, if suspected unexpected serious adverse reactions (SUSARs) are observed in one of the treatment arms or if there is a discrepancy in the serious adverse reactions between the two treatment arms, requiring a reassessment of the benefit-risk ratio for the trial.
- similarly, AH-HP, as the sponsor, or the Competent Authority (ANSM) may prematurely cancel the trial due to the discovery of unexpected facts or new information about the product, in light of which the objectives of the study or clinical programme are unlikely to be achieved .
- AP-HP, as the sponsor, reserves the right to permanently suspend enrolment at any time if the enrolment targets have not been met.

If the study is cancelled prematurely, AP-HP will inform the Competent Authority (ANSM) and the Institutional Review Board of its decision within 15 days, together with justification for the decision and any recommendations from the Data Monitoring Committee (if applicable).

6 **SELECTION CRITERIA**

6.1 **Identification of Eligible Patients: Inclusion and Exclusion Criteria**

Adult patients with acute cardiogenic shock refractory to conventional therapy placed on VA-ECMO support will be screened for eligibility.

6.1.1 **Inclusion criteria**

1. **Acute cardiogenic shock patient refractory to conventional therapy placed on VA-ECMO support in the preceding 48h.**

(The rationale for the early use of levosimendan after VA-ECMO initiation appears strong in patients with refractory cardiogenic shock related to conditions such as acute myocardial infarction, myocarditis, post-cardiac surgery or post-cardiac arrest. Myocardial injuries in these situations share many common pathophysiological features, including ischemia, inflammation and increased oxidative stress leading to extensive myocardial stunning and dysfunction [26-28]. Besides its inotropic properties that might quickly improve myocardial contractility, levosimendan might also exert beneficial antiinflammatory, antioxidative, antiapoptotic, cardioprotective and anti-stunning effects [29-35] that might accelerate cardiac recovery allowing earlier weaning from the support).

2. **Obtain informed consent from a close relative or surrogate.** According to the specifications of emergency consent, randomization without the close relative or surrogate consent could be performed.

Close relative/surrogate/family consent will be asked as soon as possible. The patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

6.1.2 **Exclusion criteria**

1. Age <18
2. Pregnant or lactating women
3. Initiation of ECMO >48 h
4. Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized
5. Irreversible neurological pathology
6. End-stage cardiomyopathy with no hope of LV function recovery
7. Mechanical complication of myocardial infarction
8. Aortic regurgitation > II
9. VA-ECMO for pulmonary embolism
10. VA-ECMO for cardiotoxic drug intoxication
11. VA-ECMO after left-ventricle assist device implantation
12. VA-ECMO in heart transplant patients
13. Patient moribund on the day of randomization, SAPS II >90
14. Liver cirrhosis (Child B or C) and other severe hepatic insufficiency
15. Chronic renal failure requiring hemodialysis
16. Known hypersensitivity to levosimendan
17. History of "torsades de pointes" in the 30 days prior to inclusion
18. History of epilepsy
19. Individuals under guardianship, or permanently legally incompetent adults
20. Participation to another interventional study
21. Patient with a weight over 180 kg

22. Known hypersensitivity to polyvitamin CERNEVIT®
23. In case of hypervitaminosis to any vitamin contained in this formulation,
24. In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria
25. In combination with vitamin A or retinoids

6.2 Consent and randomization procedure

Patients satisfying all of the Inclusion and Exclusion Criteria are classified as 'Eligible'.

To randomize an eligible patient, consent must be obtained **from a close relative or surrogate**.

Should such a person be absent, the patient will be randomized according to the specifications of **emergency enrollment** and the patient will be asked to give his/her consent for the continuation of the trial when his/her condition will allow.

Reasons for not enrolling an eligible patient will be recorded (lack of consent from substitute decision maker; refusal from attending physician; study team not available; study team missed the patient, patient enrolled in another trial).

6.3 Recruitment methods

Patients will be recruited in medical and surgical intensive care units participating to the study.

Number of subjects to be included during the enrolment period, number of subjects to be enrolled per site per month, based on the number of sites and the duration of the enrolment period are provided in the following tables.

Study characteristics	Value
Total number of subjects	206
Number of sites	14
Enrolment period (months)	36 + 3
Number of subjects/site	15,8
Number of subjects/site/month	0.4

7 TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

7.1 The investigational medicinal product(s) : levosimendan and placebo

7.1.1 Levosimendan characteristics

Levosimendan facilitates contact between actin and myosin and prolongs the time of contact between these two molecules, without modifying intracellular calcium concentration or myocardial oxygen consumption [14, 15]. This results in a positive inotropic effect. Through its agonistic effect on K⁺-ATP-dependent channels, levosimendan has a vasodilator effect (particularly on the coronary arteries) and a cardio protective anti-ischemic effect [29-35].

Metabolism and Pharmacokinetics

The pharmacokinetic characteristics of levosimendan are: a biological half-life of 60 minutes, a plasma clearance of 300 mL/min and a plasma protein binding rate of 97%. Equilibrium is reached four to eight hours after injection. Following its intravenous injection, levosimendan is rapidly metabolized by the liver to generate an active metabolite called OR-1986, which has pharmacological properties identical to those of levosimendan, but with a half-life of about three days. The metabolite OR-1896 has been shown to have hemodynamic and pharmacologic properties similar to those of the parent drug in preclinical models [16].

Pharmacodynamics

Levosimendan produces significant, dose-dependent increases in cardiac output, stroke volume and heart rate, and decreases in pulmonary capillary wedge pressure, mean blood pressure, mean pulmonary artery pressure, mean right atrial pressure and total peripheral resistance.

There is no sign of development of tolerance even with a prolonged infusion up to 48 hours. Due to the formation of the active metabolite, the hemodynamic effects are maintained several days after stopping levosimendan infusion. Compared with dobutamine, levosimendan produces a slightly greater increase in cardiac output and a profoundly greater decrease in pulmonary capillary wedge pressure. In contrast to dobutamine, the hemodynamic effects are not attenuated with concomitant beta-blocker use. It has also been shown that at 48 hours after the start of infusion, a 24-hour infusion achieves superior hemodynamic effects over a 48-hour dobutamine infusion in patients with severe acute decompensated heart failure on beta-blockers. decrease in natriuretic peptides. The beneficial effects of levosimendan on hemodynamics and neurohormones are not associated with any significant increase in myocardial energy consumption, as evidenced using dynamic positron emission tomography (PET) in hospitalized patients with NYHA III-IV heart failure.

7.1.2 Description of investigational products

7.1.2.1 Levosimendan

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml containing 5 mL of solution for infusion will be provided free of charge by the Orion Corporation.

Vials of Levosimendan (ZIMINO®) 2,5 mg/ml will be packaged in boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP).

Each boxes contains 3 vials of 5 ml of levosimendan .

7.1.2.2 Placebo (CERNEVIT®)

The placebo will be made of polyvitamin CERNEVIT®.

Vials of CERNEVIT® containing 5 mL of powder for solution for injection will be provided by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP) .

Vials of CERNEVIT® will be packaged in boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical trial Department (CTD) of AGEPS.

Each boxes contains 1 vial of 5 mL of CERNEVIT®.

7.1.2.3 Boxes of levosimendan or placebo

Each box will be assigned a unique treatment number according to the treatment numbers list.

Supply :

The numbered boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo will be sent on investigational sites by CTD of AGEPS.

Storage

Boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo must be kept at +2 to +8 °C in the outside packaging, shielded from the light.

Administration

The products (levosimendan or placebo) will be prepared by non-protocol manipulators (nurses) A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered.

A continuous infusion of Levosimendan or placebo will be administered over 24 h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate-limiting side effects.

7.1.3 Treatment preparation and dispensation

The pharmacy of each center will be responsible dispensing treatment's boxes in the ICU after initiation visit. The products will be packaged and labelled as follows: packs of three vials of levosimendan and a single vial of placebo. Each box will be appropriately labelled and numbered according to the list of treatment numbers provided by the sponsor.

The treatment (levosimendan or placebo) will be prepared by non-protocol manipulators (nurse) in the ICU (according the following table).

A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered.

Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h
40 to 46 kg	1	250 mL	8	11
47 to 89 kg	2	500 mL	16	22
90 to 132 kg	3	1000 mL	32	44
133 to 180 kg	4	1000 mL	32	44

7.2 Traceability information for the investigational medicinal products

Sponsor will provide traceability documents to enable the regulatory traceability of the investigational medicinal products on each investigational site.

7.3 Common management regarding other inotropic drugs, heart failure medications and antiagregants and anticoagulants

Additional inotropic agents may be used at any time in either treatment group as clinically indicated and as recommended by the latest European Society of Cardiology [26]. Dobutamine will be the inotropic agent of choice.

Others heart failure medications (such as diuretics, ACE inhibitors, steroidal antimineralocorticoids), antiagregants and anticoagulants will be prescribed according to the latest European Society of Cardiology [26].

Open-label Levosimendan (ZIMINO) will not be possible in either group during the entire course of the study.

7.4 Common management regarding the ECMO circuit

– Parameter settings for ECMO

The pump flow will be set to 3.5-5 l/min, to provide adequate systemic perfusion. Pump flow might be reduced in case of cessation of LV ejection or major pulmonary edema. Percentage of oxygen contained in the ventilating gaseous air–oxygen mixture will be adjusted to obtain PaO₂ between 65 and 90 mmHg and/or arterial oxygen saturation >90%. The membrane ventilation will be adjusted to maintain PaCO₂ between 40 and 45 mmHg. Extracorporeal circulation will be run at normal body temperature.

– ECMO monitoring (extracorporeal circuit, anticoagulation, possible complications)

The ECMO circuit will be monitored several times daily by the medical and nursing team caring for the patient and at least once every 48 hours by a perfusionist. Circuit and cannula surveillance is intended to verify the correct functioning of the device and early screening for complications (leg ischemia, fibrin deposits or clots on the ECMO membrane, clots in the cannulae or in the pump, bleeding or signs of inflammation or cutaneous infection at the cannula insertion sites, unexpected drop of the ECMO outflow, appearance of clinical or biological signs of intravascular hemolysis). Should any of these complications occur, a medical–surgical consultation will be held to discuss the best therapeutic approach to take.

Anticoagulation will be obtained with non-fractionated heparin to a target aPTT of 55-70 sec or heparinemia (antiXa activity) between 0.4 and 0.6 IU/ml. Should severe bleeding occur that is not immediately controllable by specific treatment, heparin will be discontinued.

Intravascular hemolysis will be sought should unexpected dark urine be excreted or in the case of obvious circuit dysfunction. It is recommended that plasma free hemoglobin be measured every 48 hours and immediately if hemolysis linked to the circuit is suspected.

The membrane and ECMO circuit will be changed in the following situations: massive intravascular hemolysis linked to the device, severe thrombopenia linked to the circuit, clots preventing the pump and/or lines from functioning properly or systemic defibrination.

The hemoglobin threshold for red cells transfusion will be 9-10 g/dl (may be decreased to 7-8 g/dl if the patient is stabilized and does not have residual myocardial ischemia). Platelet transfusion will be discouraged except when severe thrombopenia is associated by bleeding complications or when platelet count will be <20 G/L.

Connecting an extrarenal dialysis circuit to the ECMO circuit will be permitted under strict supervision of perfusionnists.

7.5 Authorised and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications

No treatment is prohibited during the study, except open-label Levosimendan and CERNEVIT® (only during experimental treatment administration). Additional inotropic agents (except digoxine) may be used at any time in either treatment group as clinically indicated. Dobutamine will be the inotropic agent of choice.

7.6 Methods for monitoring compliance with the treatment

Compliance to study drug will be assured by the physician in charge of the patient during the 24 hour infusion of the treatment.

8 EFFICACY ASSESSMENT

8.1 Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

The measures of all efficacy parameters rely on the clinical evaluation of the patients from the day of randomization (Day 1) up-to day 60 post-randomization. All information will be recorded by the investigator, nurses or study coordinator.

9 SPECIFIC COMMITTEES FOR THE TRIAL

9.1 Steering Committee (SC)

- Members of the committee: All primary investigators at study sites.
- Missions: The SC will define the overall structure of the study, coordinate information, determine the initial methodology and oversee the trial. Also determine the objective, write the protocol, recommend changes to the protocol during the trial

10 SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY

10.1 Recording and reporting adverse events

10.1.1 Definitions

According to Article R1123-46 of the French Public Health Code:

- **Adverse event**

Any untoward medical occurrence in a trial subject, which does not necessarily have a causal relationship with the clinical trial or with the investigational product.

- **Adverse reaction to an investigational medicinal product**

Any adverse event occurred in a trial subject, which has a causal relationship with the clinical trial or with the investigational medicinal product

- **Serious adverse event or reaction**

Any adverse event or reaction that at any dose of medication, results in death, threatens the life of the research subject, requires hospitalisation or prolongs hospitalisation, causes a severe or long-term disability or handicap, or results in a congenital abnormality or deformity.

- **Unexpected adverse reaction to an investigational medicinal product**

Any adverse reaction to the product, whose nature, severity, frequency or outcome is inconsistent with the safety information described in the Reference Safety Information (summary of product characteristics, or the investigator's brochure if the product is not authorised).

According to Article R.1123-46 of the Code de la Santé Publique and the guidelines for clinical trial sponsors (ANSM):

- **Emerging safety issue**

Any new safety information that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials..

For the clinical trials involving the first administration or use of an investigational medicinal product in healthy volunteers, any serious adverse reaction.

Examples:

- a) any clinically significant increase in the frequency of an expected serious adverse reaction
- b) suspected unexpected serious adverse reactions in patients who have terminated their participation in the clinical trial that are notified by the investigator to the sponsor together with follow-up reports
- c) any new safety issue relating to the conduct of the clinical trial or the development of the investigational medicinal product, that may impact the safety of the trial subjects.

Examples:

- a serious adverse event likely to be related to the interventions and the trial's diagnostic procedures and which may impact the conduct of the clinical trial,
 - a significant risk on the trial subjects such as ineffectiveness of the investigational medicinal product in treating a life-threatening illness under investigation,
 - significant safety results from a recently completed non-clinical study (such as a carcinogenicity study),
 - the premature termination, or temporary suspension, of a trial conducted on the same investigational medicinal product in another country, for safety reasons,
 - an unexpected serious adverse reaction associated with a non-experimental medication required for the conduct of the clinical trial, (e.g. challenge agents, rescue treatment)
- d) recommendations from the Data Safety Monitoring Board (DSMB), if applicable, that may affect the safety of the trial subjects

e) any suspected unexpected serious adverse reaction (SUSAR) reported to the sponsor by another sponsor of a trial carried out in a different country but relating to the same medication.

10.1.2 The role of the investigator

The investigator must **assess the seriousness criteria of each adverse event** and record all serious and non-serious adverse events in the case report form eCRF.

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the severity** of the adverse events by using:

- general terms:

- *Mild: tolerated by the patient, does not interfere with daily activities*
- *Moderate: sufficiently uncomfortable to affect daily activities*
- *Serious: preventing daily activities*

- or a severity grading scale for adverse events, attached to the protocol

The investigator must assess the **causal relationship** between the serious adverse events and the investigational medicinal products.

The method used by the investigator is based on the WHO Uppsala Monitoring Centre method and uses the following causality terms:

- Certain
- Probable/likely
- Possible
- Unlikely (not ruled out).

These terms are defined as follows (extracted from the WHO-UMC causality categories, version dated 17/04/2012).

Table : WHO-UMC causality categories (extract)

Causality term	Assessment criteria*
Certain	· Event or laboratory test abnormality, with plausible time relationship to drug intake ** · Cannot be explained by disease or other drugs · Response to withdrawal plausible (pharmacologically, pathologically) · Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognized pharmacological phenomenon) · Rechallenge satisfactory, if necessary
Probable / Likely	· Event or laboratory test abnormality, with reasonable time relationship to drug intake** · Unlikely to be attributed to disease or other drugs · Response to withdrawal clinically reasonable · Rechallenge not required
Possible	· Event or laboratory test abnormality, with reasonable time relationship to drug intake ** · Could also be explained by disease or other drugs · Information on drug withdrawal may be lacking or unclear
Unlikely	· Event or laboratory test abnormality, with a time to drug intake ** · that makes a relationship improbable (but not impossible) · Disease or other drugs provide plausible explanations

*All points should be reasonably complied with

** Or study procedures

10.1.2.1 Serious adverse events that require a notification without delay by the investigator to the sponsor

According to the article R.1123-49 of the French Public Health Code (CSP), the investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any serious adverse event which occurs during a trial as described in Article L.1121-1(1) CSP, except those which are listed in the protocol (see corresponding section) and, if applicable, in the investigator's brochure as not requiring a notification without delay. These latter should be notified by the investigator to the sponsor

in an appropriate delay taking into consideration the specific features of the trial, the serious adverse events and the modalities specified in the protocol or the investigator's brochure.

A serious adverse event is any untoward medical occurrence that:

- 1- results in death
- 2- is life-threatening
- 3- requires inpatient hospitalisation or prolongation of existing hospitalisation
- 4- results in persistent or significant disability/incapacity
- 5- is a congenital anomaly/birth defect

10.1.2.2 Specific features of the protocol

10.1.2.2.1 Other events that require the investigator to notify the sponsor without delay

– Under VA-ECMO

The mortality rate of patients with refractory cardiogenic shock supported by VA-ECMO is very high (40-60%) [2, 11-13].

VA-ECMO provides both respiratory and cardiac support, therefore, **only deaths considered as directly related the experimental treatment by the investigator** will be notified to the sponsor without delay during the period: under VA-ECMO support .

A CRF extraction of all cause of death under VA-ECMO (including : randomization group/cause of death/ date of start and end of experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

– After weaning ECMO

After ECMO weaning because of cardiac recovery, only the following serious adverse events leading to life threatening or death will be notified without delay:

- Ventricular fibrillation,
- « Torsades de pointes »,
- Ventricular tachycardia (HR $\geq$ 110 bpm),
- Premature ventricular contraction,
- Premature atrial contraction, atrial fibrillation, Hypotension grade 3 and grade 4 and refractory hypotension defined as a mean arterial blood pressure <45 mmHg for > 60 minutes despite vascular filling with two 500 mL successive crystalloid boluses AND norepinephrine ≥ 4 $\mu\text{g/kg/min}$ OR epinephrine ≥ 2 $\mu\text{g/kg/min}$.

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of these adverse events, according to the same modalities and within the same timeline as for serious adverse events (see above).

- **In utero exposure**

The investigator must notify the sponsor **without delay on the day when the investigator becomes aware** of any pregnancy that occurs during the trial, even if not associated with an adverse event.

10.1.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are simply recorded in the case report form. A CRF extraction of these serious adverse events will be done according Data Safety Monitoring Board (DSMB) or AP-HP's Safety department's request. It will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

Special rules for trials with a high mortality rate:

The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13].

Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form.

A CRF extraction of all cause of death under VA-ECMO and after weaning ECMO (including randomization group/cause of death/ date of start/date of end of the experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay.

NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor.

– ***Under VA-ECMO***

All SAEs, which occurred under VA-ECMO, do not require the investigator to notify the sponsor without delay especially SAEs with fatal outcome which are considered by the investigator to be **not related** to the experimental treatment.

NB: Cf Special rules for trials with a high mortality rate.

– ***After weaning ECMO***

- *Normal and natural course of the condition:*
 - Death considered as **exclusively related** to the natural evolution of cardiogenic shock or multiple organ failure
 - Prolongation of hospitalization due to surgery
 - Rehospitalization considered as related to cardiogenic shock or for disease follow-up
 - Events likely **to be exclusively** due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion, septic shock, multiorgan failure, hemorrhagic shock, renal failure, ventilator associate pneumonia, acute respiratory distress syndrome ...)

NB: Cf Special rules for trials with a high mortality rate.

- *Special circumstances*
 - Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation)
 - Moderate tachycardia (HR <110 bpm)
 - Premature ventricular contraction except those leading to life threatening or death
 - Premature atrial contraction, atrial fibrillation except those leading to life threatening or death
 - Mild to moderate hypokalemia (3 to 3.5 mmol/L)
- *Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care*

The investigator must report these events to his Centre Régional de Pharmacovigilance (CRPV).

10.1.2.3 Period during which the investigator must send notification of SAEs to the sponsor without delay

The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section:

- starting from the date on which the subject begins treatment with the investigational medicinal product/undergoes the first intervention specific to the study
- throughout the whole follow-up period intended by the trial (60 days after randomization)
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear at long term after exposure to the medication, such as cancers or congenital abnormalities)

10.1.2.4 Procedures and deadlines for notifying the sponsor

The investigator should initially complete a SAE reporting form (contained in the case report form). This report must be signed by the investigator.

The investigator must complete every section of the SAE form so that the sponsor can carry out the appropriate assessment.

The initial report sent to the sponsor must be rapidly followed up by one or more additional written reports describing the course of the event and any complementary information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful for medical assessment of the case (medical reports, laboratory test results, results of additional exams, etc.). These documents must be anonymized. In addition, the investigator must state the study acronym and the number and initials of the study participant on each paper.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the subject has terminated his participation in the trial.

The initial report, the SAE follow-up reports, and all other documents must be sent to AP-HP by e-mail (eig-vigilance.drc@aphp.fr). It is possible to send the SAE to the AP-HP's Safety department by fax No. **+33 (0)1 44 84 17 99** only in case of unsuccessful attempt to send the SAE by e-mail. This is to avoid duplicated reports.

For trials which use e-CRF

- the investigator completes the SAE report form in the e-CRF, then validates, prints and signs the form before sending it by mail;
- In case of failure to connect to the e-CRF, the investigator should complete, sign and send the SAE report form to the safety Department. As soon as the connection is restored, the investigator must complete the SAE report form in the e-CRF.

The investigator must comply with all requests for additional information from the sponsor.

For all questions relating to an adverse event report, the safety Department can be contacted via email at vigilance.drc@aphp.fr.

10.1.3 Role of the sponsor

The sponsor, represented by its safety Department, shall continuously assess the safety of each investigational medicinal product throughout the trial.

10.1.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all reported adverse events,

- the **causal relationship** between these adverse events and investigational medicinal product and any other treatments,
All serious adverse events for which the investigator and/or the sponsor suspect a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.
- the **expectedness assessment** of the serious adverse reactions
Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.
The sponsor, acting through its safety Department, assesses the expectedness of the serious adverse reaction based on the information described below.

- ❖ For serious adverse events likely to be related to the investigational medicinal product(s):
 - refer to the SmPC for Levosimendan (Zimino®) enclosed in appendix N°18.4.
 - refer to the SmPC for Cernevit®

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs), within the regulatory time frame, to the ANSM (French Health Products Safety Agency).

- The sponsor must send the initial report without delay upon receipt of the unexpected serious adverse reaction if it is fatal or life-threatening, or otherwise within 15 days from receipt of any other type of unexpected serious adverse reaction;
- The sponsor must provide all relevant additional information by sending follow-up reports, within 8 calendar days following receipt.

Any suspected unexpected serious adverse reaction must also be declared electronically using the Eudravigilance European adverse drug reactions database managed by the European Medicines Agency (EMA).

The sponsor must notify all the investigators about any information that could adversely affect the safety of the trial subjects.

Special rules for double blind trials

After unblinding, if the administered product is the investigational medicinal product, the case will be immediately reported as suspected serious unexpected adverse reaction (SUSAR); however, if the administered product is the comparator, the unexpected nature of the adverse reaction will be re-assessed according to the reference document of the comparator found in the protocol. In the exceptional case of a clinical study about a disease with high mortality or morbidity, the sponsor may request the ANSM a readjustment of the conditions for unblinding and reporting suspected adverse reaction. These conditions should be thoroughly defined in the study protocol.

10.1.3.2 Analysis and declaration of other safety data

This relates to any new safety data that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials.

The sponsor will inform the competent authority and the Ethics committee without delay upon knowledge any emerging safety issues and, if applicable, describe what urgent safety measures have been taken by the sponsor.

Following the initial declaration of emerging safety issue, the sponsor will declare to ANSM any additional relevant information about the new safety issues in the form of a follow-up report, which must be sent no later than 8 days after becoming aware of the information.

If the suspected unexpected serious adverse reaction meets the definition of an emerging safety issue, the sponsor will report both the SUSAR and the emerging safety issue to the ANSM according to the appropriate modalities and within the regulatory timelines as previously described.

10.1.3.3 Annual safety report

The sponsor must prepare once yearly throughout the trial duration an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- an analysis of safety data concerning trial subjects
- a description of the patients included in the trial (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- cumulative summary tabulation of all the serious adverse events that have occurred since the beginning of the clinical trial,

The report must be transmitted to ANSM and the Ethics Committee no later than 60 days after the anniversary date corresponding to the date of authorization of the clinical trial by ANSM.

10.1.4 Independent Data Safety Monitoring Board (DSMB)

An independent Data Safety Monitoring Board (DSMB) will be set up by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).

The sponsor is responsible for justifying the creation or absence of a DSMB to the Competent Authority (ANSM) and to the Ethics committee.

A DSMB will be set up for this trial. Members will include experts in the field of ECMO and severe cardiogenic shock and a biostatistician/methodologist.

The DSMB must hold its first meeting before the first subject is enrolled. The DSMB's preliminary meeting should take place before the protocol submission to competent health authority (ANSM) and Ethics committee.

The members of the DSMB are:

Pr Alain VUYLSTEKE a.vuylsteke@nhs.net

Pr Christian RICHARD christian.richard@aphp.fr

Dr Caroline ELIE, caroline.elie@aphp.fr

11 DATA MANAGEMENT

11.1 Data collection procedure

During and after the clinical study, all data will be collected by the investigator and others specialised collaborators.

11.2 Identification of data recorded directly in the CRFs which will be considered as source data

None

11.3 Right to access source data and documents

11.3.1.1 Access to data

In accordance with GCP:

- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures.
- the investigators will ensure the persons in charge of monitoring and auditing the clinical trial and of quality control have access to the documents and personal data strictly necessary for these tasks, in

accordance with the statutory and regulatory provisions in force (Articles L.1121-3 and R.5121-13 of the French Public Health Code)

11.3.1.2 Source documents

The source documents are any original document or item that proves the existence or accuracy of a data-point or fact recorded during the trial. Source documents will be kept by the investigator, or by the hospital in the case of hospital medical records, for the statutory period.

11.3.1.3 Data confidentiality

The persons responsible for the quality control of clinical studies (Article L.1121-3 of the French Public Health Code) will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular the identity of the participants and the results obtained.

These persons, as well as the investigators themselves, are bound by professional secrecy (in accordance with the conditions set out in Articles 226-13 and 226-14 of the French Criminal Code).

During and after the clinical study, all data collected about the study participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be anonymised.

Under no circumstances shall the names and addresses of the participants involved be shown.

The sponsor will ensure that each subject has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4 Data processing and storage of documents and data

11.4.1 Identification of the data processing manager and the location(s)

Data management of the study will be the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix. The data will be collected in the eCRFs by clinical study technicians and investigators. The data file will be declared to the Commission Nationale de l'Informatique et des Libertés (CNIL).

The statistical analysis of the data will be performed under the responsibility of the Clinical Research Unit of La Pitié-Salpêtrière hospital/Charles Foix.

11.4.2 Data entry

Non-identifying data will be entered electronically via a web browser.

11.5 Ownership of the data

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12 STATISTICAL ASPECTS

12.1 Planned statistical methods, including the timetable for any planned interim analyses

The statistical analysis will be performed by a biostatistician of the Clinical Research Unit of Pitié Salpêtrière Hospital, under the responsibility of Pr Florence Tubach.

All comparisons will be done with a two-sided type I error of 5%.

Descriptive analysis will always be overall and by treatment group.

Comparative analysis between groups will be systematically performed:

- Without adjustment;

- With adjustment on baseline prognostic factors whose distribution could be, despite the randomization, unbalanced between treatment groups. These adjustments will potentially require using the appropriate models, whose relevance will be discussed according to the distribution and to the type of variables.

12.1.1 Descriptive methods

Qualitative variables will be described by numbers and percentages.

Quantitative variables will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range.

Censored data will be described by Kaplan-Meier estimation or cumulative incidence function, with rate at 15, 30 and 60 days if appropriate with 95% confidence interval.

12.1.2 Software used

Statistical analyses will be performed with the SAS® or R software with the most recent version at the time of the analysis.

12.1.3 Analysis plan

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.1.4 Description of the inclusions and follow-up

- Number of included participants
- Number of participants included in the analyses and reasons for exclusion if any
- Protocol deviations
- Inclusions curve (evolution of participants frequency between first and last inclusion)

A flow chart according to the CONSORT recommendations will be drawn.

12.1.5 Patients included in the analysis

Analyses will be conducted on the intention-to-treat population (all randomized patients), with the exception of patients who withdraw consent and asked for their data not being analysed.

12.1.6 Baseline characteristics

The following variables will be described:

- Respect of the eligibility criteria;
- Epidemiological characteristics;
- Clinical characteristics;
- Biological characteristics;
- Treatment characteristics.

12.1.7 Main outcome analysis

The main outcome is the time to successful ECMO weaning within the 30 days following randomization in the presence of competing risks.

5 types of follow-up termination can appear within the 30 days following randomization:

- Subject not weaned from ECMO at 30 days: censorship
- Death before D30 after randomization, without ECMO weaning: competitive event
- Subject with successful ECMO weaning: event of interest

- Subjects with unsuccessful ECMO weaning (i.e. death, need for repeat ECMO, need for another mechanical circulatory device or heart transplantation within 30 days after ECMO removal): competitive event
- Lost to follow-up within the 30 days following randomization: censorship.

Time to successful ECMO weaning will be defined as the difference between the date of randomization and the weaning date.

Time to unsuccessful ECMO weaning will be defined as the difference between the date of randomization and the ECMO removal date, in patients for whom repeat ECMO, need for another mechanical circulatory device or heart transplantation or death will occur within 30 days after ECMO removal.

Time to death without ECMO weaning will be defined as the difference between the date of randomization and the date of death.

Time to lost to follow-up will be defined as the difference between the date of randomization and the date of the latest news.

All other subjects will be followed up to 30 days: censorship. The time will be 30 days.

This outcome will be summarized by the cumulative incidence function, with rate at 15 and 30 days with 95% confidence intervals.

The treatments groups will be compared using the Fine and Gray model to account for competing risks. We will also use this model to account for prognostic factors.

Proportional hazard assumption will be studied with Schoenfeld residuals and the loglinearity hypothesis for quantitative covariate will be studied.

With the Fine and Gray model, subdistribution hazard ratio will be reported with 95% confidence interval.

Cumulative incidence function will be drawn from the non-parametric estimation or from the model.

12.1.8 Secondary outcomes analysis

Qualitative outcomes:

- will be described by numbers and percentages;
- will be tested with χ^2 test, corrected χ^2 test or Fisher test, as appropriate.
- To account for prognostic factors or for exploratory analyses we will use a logistic model.

Quantitative outcomes:

- will be described by numbers, mean, standard deviation, 95% confidence interval of the mean, median, range, and interquartile range;
- will be tested with the Student t-test, the Student t-test or the Mann-Whitney-Wilcoxon test, as appropriate
- To adjust on prognostic factors, a linear regression model will be used.

Censored outcomes:

- Mortality will be described by survival analysis with Kaplan-Meier estimation and 95% confidence interval. Only survival analysis within 60 days will be performed. In this analysis, time to death will be defined as the difference between the date of randomization and the date of death (whatever ECMO weaning had occurred before or not) or the date of the last news or the end of the follow-up (60 days).

Difference between groups will be tested with the logrank test. To adjust for prognostic factors, a cox model will be used.

12.2 Sample size calculation

We estimated the cumulative incidence of successful ECMO weaning in the control arm at 50% in the presence of competing causes (death or weaning failure) from a 30-day cohort study of similar patients. According to Latouche et al. [38], with HR=1.75 (subdistribution hazard ratio), an alpha risk of 5% and a power of 80%, we need 101 events (= successful ECMO weaning) and 206 subjects in total (103 per group).

12.3 State whether subjects who exit the study prematurely will be replaced and in what proportion.

No replacement is foreseen.

12.4 Anticipated level of statistical significance

All comparisons will be done with a two-sided type I error of 5%.

12.5 Management of modifications made to the analysis plan for the initial strategy.

A detailed analysis plan will be defined and validated by the study scientific committee. Later modifications must be made before unblinding of the database and systematically be validated by the scientific committee.

12.6 Selection of populations

An intention-to-treat (ITT) analysis will be performed as the principal analysis, i.e. all randomized patients will be included in the analysis in the group in which they first were randomized and all their data will be used, regardless of any changes of treatment over the study duration.

13 QUALITY CONTROL AND ASSURANCE

Every research project involving human participants managed by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored interventional research involving human participants. Every clinical study managed by AP-HP is ranked according to the projected risk incurred by the study participants using a classification system specific to AP-HP-sponsored clinical trials.

13.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the trial. The sponsor must have a quality assurance system for monitoring the implementation of the study at the research centers.

For this purpose, the sponsor shall appoint Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study sites, after completing their initial visits.

The purpose of monitoring the study, as defined in the Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1 Strategy for site opening

The strategy for opening the sites established for this study is determined using the appropriate monitoring plan.

13.1.2 Scope of site monitoring

In the case of this C risk study, the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact resulting in a study monitoring level to be implemented: level: **high level**.

13.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practice as well as with the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits by the Clinical Research Associate. During these visits, the following elements will be reviewed:

- written consent
- compliance with the study protocol and its procedures
- quality of the data collected in the case report forms: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

13.3 Case Report Form

All information required by the protocol must be entered in the case report forms. The data must be collected as and when it is obtained, and clearly recorded in these case report forms. Any missing data must be coded.

Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given instructions for using this tool.

Using on-line case report forms means the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, there are consistency checks to ensure the data are verified immediately upon being entered. The investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment. A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the study. The investigator must archive a copy of the authenticated document that was delivered to the sponsor.

13.4 Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study - with the protocol, standard operating procedures, good clinical practice or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

13.5 Audits/inspections

The investigators agree to accept the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the ground of medical secrecy.

An audit can be carried out at any time by individuals appointed by the [sponsor](#). The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the trial agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results and the storage of the data used or produced as part of the study.

13.6 Principal Investigator's declaration of responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal curriculum vitae, signed and dated less than one year, with his/her and RPPS number (Répertoire Partagé des Professionnels de Santé, Collective Database of Health Professionals). The CV must include any previous involvement in clinical research and related training.

Each investigator will commit to comply with legislation and to conduct the trial in line with GCP, in accordance with the Declaration of Helsinki.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.

The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

14 ETHICAL AND LEGAL CONSIDERATIONS

14.1 Methods for informing and obtaining consent from the research participants

In accordance with Article L.1122-1-1 of the French Public Health Code, no research can be carried out on a person without his/her free and informed consent, obtained in writing after the person has been given the information specified in Article L.1122-1 of said Code.

However, if the person is unable to give his or her written consent, consent may be obtained, from a legal representative, family member or a close relative. These persons must have no connection whatsoever to the investigator or the sponsor.

Therefore, the site investigator will obtain consent for inclusion in LEVOECMO (as approved by local IRBs) from a legal representative, family member or a close relative prior to randomization. The investigator will retain the original signed and dated consent form. The research subject relative will keep a signed and dated copy of the consent form.

For LEVOECMO trial, if patient Should a close relative be absent, eligible patients will be randomized according to the specifications of emergency consent (Article L.1122-1-3 CSP) after IRB approval. The patient's legal representative will be informed as soon as possible of the inclusion and asked to give his or her written consent. The patient will be informed and asked to give his/her consent for the continuation of the trial when his/her condition will allow during the subject's study participation. Investigators should to indicate in the medical record the inclusion of a patient unable to consent in emergency situations in the research protocol.

Two formulæ of consent form will be prepared: one for legal representative/family member/close relative of patient when patient is unable to sign consent; one for patient for authorization for pursuit of research.

When eligible patients will not be randomized, the reason for nonrandomization (lack of consent from substitute decision maker; refusal from attending physician; prior or current enrolment in a confounding RCT, patient missed by the research team) will be recorded in the screening log file.

14.2 Prohibition of concomitant clinical studies participation and exclusion period after the trial, if applicable

Whilst participating in this trial, subjects may not take part in any other interventional clinical study (except IMPULMACS, NCT04951999). For non-interventional or observational study, subjects may not take part without first speaking to the doctor in charge of this trial.

14.3 Compensation for subjects

No compensation will be paid to patients included in this trial.

14.4 Registration on the National Register of subjects participating in human research trials on the products listed in Article L. 5311-1 of the French Public Health Code

Not applicable

14.5 Authorisation for the research location

The study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely performed at those units. Therefore, the research location does not require specific authorisation.

14.6 Legal obligations

14.6.1 The sponsor's role

Assistance Publique Hôpitaux de Paris (AP-HP) is the sponsor of this study and by delegation, the DRCI (Clinical Research and Innovation Delegation) carries out the study's missions in accordance with Article L.1121-1 of the French Public Health Code. Assistance Publique - Hôpitaux de Paris (AP-HP) reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.6.2 Request for approval from the CPP (Research Ethics Committee)

Prior to starting the study, AP-HP, as sponsor, must obtain for this interventional study involving human participants concerning a medicinal product for human use, approval from the appropriate CPP (Research Ethics Committee), within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.6.3 Request for approval from the ANSM

Prior to starting the study, AP-HP, as sponsor, must obtains authorisation from the ANSM (French Health Products Safety Agency) for the interventional study involving human participants concerning a medicinal products for human use, within the scope of the ANSM's authority and in accordance with in force legislation and regulatory requirements.

14.6.4 Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended “Informatique et Libertés” law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

- Request for authorisation by the CNIL (French Data Protection Agency)

This research is not governed by the CNIL “Reference Method” (MR-001) because emergency consent and inability to obtain a written consent at inclusion.

The sponsor must obtain the authorisation of the CNIL (French Data Protection Agency) before implementing any data processing involving the data required to conduct the research.

14.6.5 Modifications to the trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, approval from the CPP (Research Ethics Committee) and authorisation from the ANSM within the scope of their respective authorities, before the amendment can be implemented.

The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.6.6 Final study report

The final report for the research involving human participants referred to in Article R1123-67 of the *Code de la Santé Publique* (French Public Health Code) is written and signed by the sponsor and the investigator. A report summary drafted according to the reference plan of the competent authority must be sent to the competent authority within a period of one year following the end of the study, i.e., the end of the participation of the last participant in the study.

14.6.7 Archiving

Specific documents for an interventional study involving human participants concerning a medicinal product for human use will be archived by the investigator and the sponsor for 15 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list) :
 - the successive versions of the protocol (identified by the version number and its date), and any appendices
 - the ANSM authorisations and CPP (Research Ethics Committee) decisions
 - any correspondence
 - the enrolment list or register
 - the appendices specific to the research
 - final study report
- The data collection documents

15 FUNDING AND INSURANCE

15.1 Sources of funding for the trial

Grant from french Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministère de la Santé).

15.2 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

16 PUBLICATION

16.1 Mention of AP-HP affiliation for projects sponsored or managed by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is not important
- However, if the trial is funded by an internal call for tenders at the AP-HP, the first affiliation must be "AP-HP"
- Each of these affiliations must be identified by an address and separated by a semicolon
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: AP-HP, hospital, department, city, postcode, France

16.2 Mention of the AP-HP manager (DRCI) in the acknowledgements of the text

- "The sponsor was *Assistance Publique – Hôpitaux de Paris* (Clinical Research and Development Department)"

16.3 Mention of the funder in the acknowledgements of the text

- "*The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2017 (Ministry of Health)*"

Mention that the study has been registered on the <http://clinicaltrials.gov/> website under registration number NCTXXXXX

17 BIBLIOGRAPHY

1. Werdan K, Gielen S, Ebelt H, Hochman JS, (2014) Mechanical circulatory support in cardiogenic shock. *Eur Heart J* 35: 156-167
2. Combes A, Leprince P, Luyt CE, Bonnet N, Trouillet JL, Leger P, Pavie A, Chastre J, (2008) Outcomes and long-term quality-of-life of patients supported by extracorporeal membrane oxygenation for refractory cardiogenic shock. *Crit Care Med* 36: 1404-1411
3. Mirabel M, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Chastre J, Combes A, (2011) Outcomes, long-term quality of life, and psychologic assessment of fulminant myocarditis patients rescued by mechanical circulatory support. *Crit Care Med* 39: 1029-1035
4. Beurtheret S, Mordant P, Paoletti X, Marijon E, Celermajer DS, Leger P, Pavie A, Combes A, Leprince P, (2013) Emergency circulatory support in refractory cardiogenic shock patients in remote institutions: a pilot study (the cardiac-RESCUE program). *Eur Heart J* 34: 112-120
5. Abrams D, Combes A, Brodie D, (2014) Extracorporeal Membrane Oxygenation in Cardiopulmonary Disease in Adults. *J Am Coll Cardiol* 63: 2769-2778
6. Muller G, Flecher E, Lebreton G, Luyt CE, Trouillet JL, Brechot N, Schmidt M, Mastroianni C, Chastre J, Leprince P, Anselmi A, Combes A, (2016) The ENCOURAGE mortality risk score and analysis of long-term outcomes after VA-ECMO for acute myocardial infarction with cardiogenic shock. *Intensive Care Med* 42: 370-378
7. Barbaro RP, Odetola FO, Kidwell KM, Paden ML, Bartlett RH, Davis MM, Annich GM, (2015) Association of hospital-level volume of extracorporeal membrane oxygenation cases and mortality. Analysis of the extracorporeal life support organization registry. *Am J Respir Crit Care Med* 191: 894-901
8. Aissaoui N, Luyt CE, Leprince P, Trouillet JL, Leger P, Pavie A, Diebold B, Chastre J, Combes A, (2011) Predictors of successful extracorporeal membrane oxygenation (ECMO) weaning after assistance for refractory cardiogenic shock. *Intensive Care Med* 37: 1738-1745
9. Chen YS, Lin JW, Yu HY, Ko WJ, Jerng JS, Chang WT, Chen WJ, Huang SC, Chi NH, Wang CH, Chen LC, Tsai PR, Wang SS, Hwang JJ, Lin FY, (2008) Cardiopulmonary resuscitation with assisted extracorporeal life-support versus conventional cardiopulmonary resuscitation in adults with in-hospital cardiac arrest: an observational study and propensity analysis. *Lancet* 372: 554-561
10. Sheu JJ, Tsai TH, Lee FY, Fang HY, Sun CK, Leu S, Yang CH, Chen SM, Hang CL, Hsieh YK, Chen CJ, Wu CJ, Yip HK, (2010) Early extracorporeal membrane oxygenator-assisted primary percutaneous coronary intervention improved 30-day clinical outcomes in patients with ST-segment elevation myocardial infarction complicated with profound cardiogenic shock. *Crit Care Med* 38: 1810-1817
11. Schmidt M, Brechot N, Hariri S, Guiguet M, Luyt CE, Makri R, Leprince P, Trouillet JL, Pavie A, Chastre J, Combes A, (2012) Nosocomial infections in adult cardiogenic shock patients supported by venoarterial extracorporeal membrane oxygenation. *Clin Infect Dis* 55: 1633-1641
12. Cheng R, Hachamovitch R, Kittleson M, Patel J, Arabia F, Moriguchi J, Esmailian F, Azarbal B, (2014) Complications of extracorporeal membrane oxygenation for treatment of cardiogenic shock and cardiac arrest: a meta-analysis of 1,866 adult patients. *Ann Thorac Surg* 97: 610-616
13. Bisdas T, Beutel G, Warnecke G, Hoepfer MM, Kuehn C, Haverich A, Teebken OE, (2011) Vascular complications in patients undergoing femoral cannulation for extracorporeal membrane oxygenation support. *Ann Thorac Surg* 92: 626-631
14. Sundberg S, Lilleberg J, Nieminen MS, Lehtonen L, (1995) Hemodynamic and neurohumoral effects of levosimendan, a new calcium sensitizer, at rest and during exercise in healthy men. *Am J Cardiol* 75: 1061-1066
15. Follath F, Cleland JG, Just H, Papp JG, Scholz H, Peuhkurinen K, Harjola VP, Mitrovic V, Abdalla M, Sandell EP, Lehtonen L, (2002) Efficacy and safety of intravenous levosimendan compared with dobutamine in severe low-output heart failure (the LIDO study): a randomised double-blind trial. *Lancet* 360: 196-202
16. Kivikko M, Antila S, Eha J, Lehtonen L, Pentikainen PJ, (2002) Pharmacokinetics of levosimendan and its metabolites during and after a 24-hour continuous infusion in patients with severe heart failure. *International journal of clinical pharmacology and therapeutics* 40: 465-471

17. Mebazaa A, Nieminen MS, Packer M, Cohen-Solal A, Kleber FX, Pocock SJ, Thakkar R, Padley RJ, Poder P, Kivikko M, (2007) Levosimendan vs dobutamine for patients with acute decompensated heart failure: the SURVIVE Randomized Trial. *Jama* 297: 1883-1891
18. Gordon AC, Perkins GD, Singer M, McAuley DF, Orme RM, Santhakumaran S, Mason AJ, Cross M, Al-Beidh F, Best-Lane J, Brealey D, Nutt CL, McNamee JJ, Reschreiter H, Breen A, Liu KD, Ashby D, (2016) Levosimendan for the Prevention of Acute Organ Dysfunction in Sepsis. *N Engl J Med* 375: 1638-1648
19. Gong B, Li Z, Yat Wong PC, (2015) Levosimendan Treatment for Heart Failure: A Systematic Review and Meta-Analysis. *J Cardiothorac Vasc Anesth* 29: 1415-1425
20. Cholley B, Caruba T, Grosjean S, Amour J, Ouattara A, Villacorta J, Miguet B, Guinet P, Levy F, Squara P, Ait Hamou N, Carillon A, Boyer J, Boughenou MF, Rosier S, Robin E, Radutoiu M, Durand M, Guidon C, Desebbe O, Charles-Nelson A, Menasche P, Rozec B, Girard C, Fellahi JL, Pirracchio R, Chatellier G, (2017) Effect of Levosimendan on Low Cardiac Output Syndrome in Patients With Low Ejection Fraction Undergoing Coronary Artery Bypass Grafting With Cardiopulmonary Bypass: The LICORN Randomized Clinical Trial. *Jama* 318: 548-556
21. Mehta RH, Leimberger JD, van Diepen S, Meza J, Wang A, Jankowich R, Harrison RW, Hay D, Fremes S, Duncan A, Soltesz EG, Lubner J, Park S, Argenziano M, Murphy E, Marcel R, Kalavrouziotis D, Nagpal D, Bozinovski J, Toller W, Heringlake M, Goodman SG, Levy JH, Harrington RA, Anstrom KJ, Alexander JH, (2017) Levosimendan in Patients with Left Ventricular Dysfunction Undergoing Cardiac Surgery. *N Engl J Med* 376: 2032-2042
22. Landoni G, Lomivorotov VV, Alvaro G, Lobreglio R, Pisano A, Guarracino F, Calabro MG, Grigoryev EV, Likhvantsev VV, Salgado-Filho MF, Bianchi A, Pasyuga VV, Baiocchi M, Pappalardo F, Monaco F, Boboshko VA, Abubakirov MN, Amantea B, Lembo R, Brazzi L, Verniero L, Bertini P, Scandroglio AM, Bove T, Belletti A, Michienzi MG, Shukevich DL, Zabelina TS, Bellomo R, Zangrillo A, (2017) Levosimendan for Hemodynamic Support after Cardiac Surgery. *N Engl J Med* 376: 2021-2031
23. Affronti A, di Bella I, Carino D, Ragni T, (2013) Levosimendan may improve weaning outcomes in venoarterial ECMO patients. *Asaio j* 59: 554-557
24. Tafer N, Megret F, Dolbeau JB, Coiffic A, Calderon J, Mauriat P, Ouattara A, (2013) Intérêt du lévosimendan dans le sevrage difficile des assistances circulatoires temporaires. *Annales francaises d'anesthesie et de reanimation* 32: A211-A212
25. Distelmaier K, Roth C, Schrutka L, Binder C, Steinlechner B, Heinz G, Lang IM, Maurer G, Koinig H, Niessner A, Hulsman M, Speidl W, Goliassch G, (2016) Beneficial effects of levosimendan on survival in patients undergoing extracorporeal membrane oxygenation after cardiovascular surgery. *Br J Anaesth* 117: 52-58
26. Ponikowski P, Voors AA, Anker SD, Bueno H, Cleland JG, Coats AJ, Falk V, Gonzalez-Juanatey JR, Harjola VP, Jankowska EA, Jessup M, Linde C, Nihoyannopoulos P, Parissis JT, Pieske B, Riley JP, Rosano GM, Ruilope LM, Ruschitzka F, Rutten FH, van der Meer P, (2016) 2016 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: The Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 37: 2129-2200
27. Altenberger J, Parissis JT, Costard-Jaeckle A, Winter A, Ebner C, Karavidas A, Sihorsch K, Avgeropoulou E, Weber T, Dimopoulos L, Ulmer H, Poelzl G, (2014) Efficacy and safety of the pulsed infusions of levosimendan in outpatients with advanced heart failure (LevoRep) study: a multicentre randomized trial. *Eur J Heart Fail* 16: 898-906
28. Husebye T, Eritsland J, Muller C, Sandvik L, Arnesen H, Seljeflot I, Mangschau A, Bjornerheim R, Andersen GO, (2013) Levosimendan in acute heart failure following primary percutaneous coronary intervention-treated acute ST-elevation myocardial infarction. Results from the LEAF trial: a randomized, placebo-controlled study. *Eur J Heart Fail* 15: 565-572
29. Jamali IN, Kersten JR, Pagel PS, Hettrick DA, Warltier DC, (1997) Intracoronary levosimendan enhances contractile function of stunned myocardium. *Anesth Analg* 85: 23-29
30. Kopustinskiene DM, Pollesello P, Saris NE, (2004) Potassium-specific effects of levosimendan on heart mitochondria. *Biochemical pharmacology* 68: 807-812

31. Sonntag S, Sundberg S, Lehtonen LA, Kleber FX, (2004) The calcium sensitizer levosimendan improves the function of stunned myocardium after percutaneous transluminal coronary angioplasty in acute myocardial ischemia. *J Am Coll Cardiol* 43: 2177-2182
32. Antoniadou C, Tousoulis D, Koumallos N, Marinou K, Stefanadis C, (2007) Levosimendan: beyond its simple inotropic effect in heart failure. *Pharmacology & therapeutics* 114: 184-197
33. Beiras-Fernandez A, Weis FC, Kur F, Kaczmarek I, Schmoeckel M, Weis M, Reichart B, (2008) Primary graft failure and Ca²⁺ sensitizers after heart transplantation. *Transplant Proc* 40: 951-952
34. Fries M, Ince C, Rossaint R, Bleilevens C, Bickenbach J, Rex S, Mik EG, (2008) Levosimendan but not norepinephrine improves microvascular oxygenation during experimental septic shock. *Crit Care Med* 36: 1886-1891
35. Nanas S, Gerovasili V, Dimopoulos S, Pierrakos C, Kourtidou S, Kaldara E, Sarafoglou S, Venetsanakis J, Roussos C, Nanas J, Anastasiou-Nana M, (2008) Inotropic agents improve the peripheral microcirculation of patients with end-stage chronic heart failure. *J Card Fail* 14: 400-406
36. Aissaoui N, El-Banayosy A, Combes A, (2015) How to wean a patient from veno-arterial extracorporeal membrane oxygenation. *Intensive Care Med* 41: 902-905
37. ELSO guidelines, <https://www.else.org/Resources/Guidelines.aspx>, Last accessed August 27th 2017
38. Latouche A, Porcher R, Chevret S, (2004) Sample size formula for proportional hazards modelling of competing risks. *Stat Med* 23: 3263-3274

18 LIST OF ADDENDA

Every addendum and the log of addenda versions are attached, independently of the protocol. Every addendum can be modified (change of addendum version) without modifying the version of the protocol.

18.1 List of Investigators

18.2 Serious Adverse Events report form

18.3 Pregnancy report form

18.4 Questionnaire/scale

- SOFA
- GLASGOW
- SAPS2

18.5 SmpC Levosimendan

- Levosimendan, ZIMINO®
<http://agence-prd.ansm.sante.fr/php/ecodex/rcp/R0274395.htm>

18.6 SmPC CERNEVIT®

<https://base-donnees-publique.medicaments.gouv.fr/extraire.php?specid=61001031>

LEVOECMO Randomized Clinical Trial Protocol

Summary of changes

Version n°1.1 ; 12/04/2020

Version N°2.0 ; 05/31/2023

Pages	Protocole Version n°1.1 du 14/12/2020	Pages	Protocole Version n°2.0 du 31/05/2023
P1 à 47	Entête : Version 1.1 of 04/12/2020	P1 à 47	Entête : Version 2.0 of 31/05/2023
P1	Version 1.1 of 04/12/2020	P1	Version 2.0 of 31/05/2023
P7	Exclusion criteria : 1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes before ECMO 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) Prior history of torsades de pointes 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study	P7	Exclusion criteria : 1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized. 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) Prior to History of torsades de pointes in the 30 days prior to inclusion 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study 21) Patient with a weight over 180 kg 22) Known hypersensitivity to polyvitamin CERNEVIT® 23) In case of hypervitaminosis to any vitamin contained in this formulation, 24) In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria 25) In combination with vitamin A or retinoids
P7	Comparator treatment : The placebo will be made of riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used as a dietary supplement, especially	P7	Comparator treatment : The placebo will be made of polyvitamin CERNEVIT® riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used as a

	in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing levosimendan (Gordon, NEJM 2016)		dietary supplement, especially in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing levosimendan (Gordon, NEJM 2016)
P8	Number of site : National trial involving 13 centers	P8	Number of site : National trial involving 13-14 centers
P8	Duration of the trial : - inclusion period: 24 months - participation period (treatment+follow-up):2 months - total duration: 26 months	P8	Duration of the trial : - inclusion period: 24 + 12 months - participation period (treatment+follow-up):2 months - total duration: 26 + 12 months
P8	Number of enrolments expected per site and per month : 0,7	P8	Number of enrolments expected per site and per month : 0,4
P11	Name and description of the investigational medicinal product(s) • Levosimendan (ZIMINO®) • Placebo of Levosimendan	P11	Name and description of the investigational medicinal product(s) • Levosimendan (ZIMINO®) • Placebo of Levosimendan CERNEVIT®
P15	Blinding methods and measures put in place to protect the blinding : Investigator masking to group assignment after randomization will be guaranteed by use a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo. The study will be double blinded. Treatments will be prepared at ICU by a nurse Levosimendan and the placebo will be administred as non-identifiable preparation. The placebo will be made of riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used daily as a dietary supplement in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing Levosimendan [18]. The appearance (color and presentation) of placebo of levosimendan will be strictly identical to levosimendan to maintain double blind. Trial participants, care providers, and outcome assessors will be masked to the patient assignment.	P15	Blinding methods and measures put in place to protect the blinding : Investigator masking to group assignment after randomization will be guaranteed by use a tested and validated e-CRF and by other blinding methods like patient identification numbers, boxes numbered which contain levosimendan or placebo. The study will be double blinded. Treatments will be prepared at ICU by a nurse "off-protocol handler" (specifically trained caregiver) opens the box given by the PUI and prepares the infusion Levosimendan or CERNEVIT® Levosimendan and the placebo will be administred as non-identifiable preparation. The placebo will be made of riboflavin in order to reproduce the yellow color of levosimendan. Riboflavin, also known as vitamin B2, is a vitamin that is required by the body for cellular respiration. It is used daily as a dietary supplement in ICU patients. Riboflavin has already been used as a control placebo in previous trials testing Levosimendan [18]. The appearance (color and presentation) of placebo of levosimendan will be strictly identical to levosimendan to maintain double blind. Trial participants, care providers, and outcome assessors will be masked to the patient assignment.
P16	Treatment : - Reconstitution/Preparation of the treatment before administration - Administration by IV perfusion during 24h - Administration as soon as possible after randomization; no later than 24 hours after randomization	P16	Treatment : - Reconstitution/Preparation of the treatment before administration - Administration by IV perfusion during 24h - Administration as soon as possible after randomization; no later than 24 hours after randomization - Prohibited administration of CERNEVIT® during experimental treatment administration
P18	Expected length of participation, chronology and duration of the study :	P18	Expected length of participation, chronology and duration of the study :

	- Length of enrolment period: 24 months - Duration of participation for each subject: 60 days - Treatment period: 24 hours - Follow-up period: 60 days - Total study duration: (start = trial start date; end = trial end date) = 26 months		- Length of enrolment period: 24 months + 12 months - Duration of participation for each subject: 60 days - Treatment period: 24 hours - Follow-up period: 60 days - Total study duration: (start = trial start date; end = trial end date) = 26 months + 12 months
P21 à 22	Exclusion criteria : 1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes before ECMO 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) Prior history of torsades de pointes 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study	P21 à 22	Exclusion criteria : 1) Age <18 2) Pregnant or lactating women 3) Initiation of VA-ECMO >48 h 4) Resuscitation >30 minutes in the 48 hours before ECMO (cumulative low-flow time). If a low-flow episode occurs before the 48 hours window prior to ECMO, patients must fully recover consciousness to be randomized. 5) Irreversible neurological pathology 6) End-stage cardiomyopathy with no hope of LV function recovery 7) Mechanical complication of myocardial infarction 8) Aortic regurgitation > II 9) VA-ECMO for pulmonary embolism 10) VA-ECMO for cardiotoxic drug intoxication 11) ECMO after left-ventricle assist device implantation 12) VA-ECMO in heart transplant patients 13) Patient moribund on the day of randomization, SAPS II >90 14) Liver cirrhosis (Child B or C) and other severe hepatic insufficiency 15) Chronic renal failure requiring hemodialysis 16) Known hypersensitivity to levosimendan 17) Prior History of torsades de pointes in the 30 days prior to inclusion 18) History of epilepsy 19) Individuals under guardianship, or permanently legally incompetent adults 20) Participation to another interventional study 21) Patient with a weight over 180 kg 22) Known hypersensitivity to polyvitamin CERNEVIT® 23) In case of hypervitaminosis to any vitamin contained in this formulation, 24) In case of severe hypercalcemia, hypercalciuria, treatment, pathology and/or disorders leading to severe hypercalcemia and/or hypercalciuria 25) In combination with vitamin A or retinoids
P22	Study characteristics :	P22	Study characteristics :

		Study characteristics	Value				Study characteristics	Value	
		Total number of subjects	206				Total number of subjects	206	
		Number of sites	13				Number of sites	13 14	
		Enrolment period (months)	24				Enrolment period (months)	24 + 12	
		Number of subjects/site	15,8				Number of subjects/site	15,8	
		Number of subjects/site/month	0.66				Number of subjects/site/month	0.66 0.4	
P23 à 24	7.1.2	Description of investigational products Levosimendan			P23 à 24	7.1.2	Description of investigational products - Levosimendan		
		Presentation Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, containing 5 mL of solution for infusion will be provided free of charge by the Orion Corporation. Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, will be packaged in double-blinded boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP). Each boxes contains 3 vials of 5 ml of levosimendan or placebo. Each box will be assigned a unique treatment number according to the treatment numbers list. Supply : The numbered boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, will be sent on investigational sites by CTD of AGEPS. Storage Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, must be kept at +2 to +8 °C in the outside packaging, shielded from the light. A continuous infusion of Levosimendan will be administered over 24 h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be					Presentation 7.1.2.1 Levosimendan Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, containing 5 mL of solution for infusion will be provided free of charge by the Orion Corporation. Vials of Levosimendan (ZIMINO®) 2,5 mg/ml and its placebo, will be packaged in double-blinded boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP). Each boxes contains 3 vials of 5 ml of levosimendan or placebo. 7.1.2.2 Placebo (CERNEVIT®) The placebo will be made of polyvitamin CERNEVIT®. Vials of CERNEVIT® containing 5 mL of powder for solution for injection will be provided by the Clinical Trial Department (CTD) of AGEPS (Pharmaceutical establishment of AP-HP) . Vials of CERNEVIT® will be packaged in boxes specifically labelled according to the regulatory requirements of clinical trials by the Clinical trial Department (CTD) of AGEPS. Each boxes contains 1 vial of 5 mL of CERNEVIT®. 7.1.2.3 Boxes of levosimendan or placebo		

	increased to 0.20 µg/kg/min after 2 hours in the absence of rate-limiting side effects.		Each box will be assigned a unique treatment number according to the treatment numbers list. Supply : The numbered boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo and its placebo, will be sent on investigational sites by CTD of AGEPS. Storage Boxes of Levosimendan (ZIMINO®) 2,5 mg/ml and placebo and its placebo, must be kept at +2 to +8 °C in the outside packaging, shielded from the light. Administration The products (levosimendan or placebo) will be prepared by non-protocol manipulators (nurses) A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered. A continuous infusion of Levosimendan or placebo will be administered over 24 h, with no initial bolus. The starting infusion rate will be 0.15 µg/kg/min and will be increased to 0.20 µg/kg/min after 2 hours in the absence of rate-limiting side effects.
P24 à 25	7.1.3 Treatment preparation and dispensation The pharmacy of each center will be responsible dispensing treatment's boxes in the ICU after initiation visit. The products will be packaged and labelled as packs of three vials of levosimendan or placebo labelled and numbered according to the list of treatment numbers provided by the sponsor. The treatment will be prepared by a nurse in the ICU (according the following table).	P24 à 25	7.1.3 Treatment preparation and dispensation The pharmacy of each center will be responsible dispensing treatment's boxes in the ICU after initiation visit. The products will be packaged and labelled as follows: packs of three vials of levosimendan and a single vial of placebo. Each box will be appropriately as packs of three vials of levosimendan or placebo labelled and numbered according to the list of treatment numbers provided by the sponsor. The treatment (levosimendan or placebo) will be prepared by non-protocol manipulators a (nurse) in the ICU (according the following table). A pool of manipulators will have been previously identified and trained in services. Thus, participants, physicians and nurses in charge for the patients will remain blinded of the treatment administered.

P25	<table><tr><th>Weight</th><th>Number of flasks</th><th>Volume of Glucose</th><th>Pump flow during H1-H2 in mL/h</th><th>Pump flow during H3-H24 in mL/h</th></tr><tr><td>40 to 46 kg</td><td>1</td><td>250 mL</td><td>8</td><td>11</td></tr><tr><td>47 to 89 kg</td><td>2</td><td>500 mL</td><td>16</td><td>22</td></tr><tr><td>90 to 132 kg</td><td>3</td><td>1000 mL</td><td>32</td><td>44</td></tr></table>	Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h	40 to 46 kg	1	250 mL	8	11	47 to 89 kg	2	500 mL	16	22	90 to 132 kg	3	1000 mL	32	44
Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h																	
40 to 46 kg	1	250 mL	8	11																	
47 to 89 kg	2	500 mL	16	22																	
90 to 132 kg	3	1000 mL	32	44																	
P26	7.5 Authorised and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications No treatment is prohibited during the study, except open-label Levosimendan (ZIMINO). Additional inotropic agents (except digoxine) may be used at any time in either treatment group as clinically indicated. Dobutamine will be the inotropic agent of choice.																				
P29	10.1.2.2.1 Other events that require the investigator to notify the sponsor without delay • Adverse events judged as being directly responsible for patient death: Under VA-ECMO The mortality rate of patients with refractory cardiogenic shock supported by VA-ECMO is very high (40-60%) [2, 11-13]. Therefore, only deaths considered as directly related the experimental treatment will be notified to the sponsor without delay in the LEVOECMO trial. Specifically, ventricular rhythms disturbances (ventricular fibrillation, « Torsades de pointes », ventricular tachycardia) do not incur a risk of immediate death in patients under VA-ECMO support and should not be notified without delay when occurring in patients under VA-ECMO. After Weaning ECMO																				

P25	<table><tr><th>Weight</th><th>Number of flasks</th><th>Volume of Glucose</th><th>Pump flow during H1-H2 in mL/h</th><th>Pump flow during H3-H24 in mL/h</th></tr><tr><td>40 to 46 kg</td><td>1</td><td>250 mL</td><td>8</td><td>11</td></tr><tr><td>47 to 89 kg</td><td>2</td><td>500 mL</td><td>16</td><td>22</td></tr><tr><td>90 to 132 kg</td><td>3</td><td>1000 mL</td><td>32</td><td>44</td></tr><tr><td>133 to 180 kg</td><td>4</td><td>1000 mL</td><td>32</td><td>44</td></tr></table>	Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h	40 to 46 kg	1	250 mL	8	11	47 to 89 kg	2	500 mL	16	22	90 to 132 kg	3	1000 mL	32	44	133 to 180 kg	4	1000 mL	32	44
Weight	Number of flasks	Volume of Glucose	Pump flow during H1-H2 in mL/h	Pump flow during H3-H24 in mL/h																						
40 to 46 kg	1	250 mL	8	11																						
47 to 89 kg	2	500 mL	16	22																						
90 to 132 kg	3	1000 mL	32	44																						
133 to 180 kg	4	1000 mL	32	44																						
P26	7.5 Authorised and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications No treatment is prohibited during the study, except open-label Levosimendan (ZIMINO) and CERNEVIT® (only during experimental treatment administration). Additional inotropic agents (except digoxine) may be used at any time in either treatment group as clinically indicated. Dobutamine will be the inotropic agent of choice.																									
P29	10.1.2.2.1 Other events that require the investigator to notify the sponsor without delay • Adverse events judged as being directly responsible for patient death: Under VA-ECMO The mortality rate of patients with refractory cardiogenic shock supported by VA-ECMO is very high (40-60%) [2, 11-13]. VA-ECMO provides both respiratory and cardiac support, therefore, only deaths considered as directly related the experimental treatment by the investigator will be notified to the sponsor without delay during the period: under VA-ECMO support in the LEVOECMO trial. A CRF extraction of all cause of death under VA-ECMO (including : randomization group/cause of death/ date of start and end of experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.																									

	After ECMO weaning because of cardiac recovery, the following adverse events will also be notified without delay: - Ventricular fibrillation, - « Torsades de pointes », - Ventricular tachycardia, - Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for > 60 minutes despite vascular filling with two 500 mL successive crystalloid boluses AND norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.		Specifically, ventricular rhythms disturbances (ventricular fibrillation, « Torsades de pointes », ventricular tachycardia) do not incur a risk of immediate death in patients under VA-ECMO support and should not be notified without delay when occurring in patients under VA-ECMO. After Weaning ECMO After ECMO weaning because of cardiac recovery, only the following serious adverse events leading to life threatening or death will also be notified without delay: - Ventricular fibrillation, - « Torsades de pointes », - Ventricular tachycardia (HR ≥ 110 bpm), - Premature ventricular contraction, - Premature atrial contraction, atrial fibrillation, - Hypotension grade 3 and grade 4 and Refractory hypotension defined as a mean arterial blood pressure <45 mmHg for > 60 minutes despite vascular filling with two 500 mL successive crystalloid boluses AND norepinephrine $\geq 4 \mu\text{g/kg/min}$ OR epinephrine $\geq 2 \mu\text{g/kg/min}$.
P30 à 31	These serious adverse events are simply recorded in the case report form. A CRF extraction of these serious adverse events will be done according Data Safety Monitoring Board (DSMB) or AP-HP's Safety department's request. • Normal and natural course of the condition: -Death considered as related to the natural evolution of cardiogenic shock or multiple organ failure and not to the infusion of the study drug -Prolongation of hospitalization due to surgery -Rehospitalization considered as not related to cardiogenic shock or for disease follow-up -Events likely to be exclusively due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion ...) • Normal and natural course of the condition: -Death considered as exclusively related to the natural evolution of cardiogenic shock or multiple organ failure and not to the infusion of the study drug -Prolongation of hospitalization due to surgery	P30 à 31	These serious adverse events are simply recorded in the case report form. A CRF extraction of these serious adverse events will be done according Data Safety Monitoring Board (DSMB) or AP-HP's Safety department's request. It will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr. Special rules for trials with a high mortality rate: The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13]. Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form. A CRF extraction of all cause of death under VA-ECMO and after weaning ECMO (including randomization group/cause of death/ date of start/date of end of the experimental treatment/ investigator's assessment) will be performed every 3 months by clinical research unit and transmit to Safety Department at expertisecsi.drc@aphp.fr.

-Rehospitalization considered as not related to cardiogenic shock or for disease follow-up -Events likely to be exclusively due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion, septic shock, multiorgan failure, hemorrhagic shock, renal failure, ventilator associate pneumonia, acute respiratory distress syndrome ...) Special rules for trials with a high mortality rate: The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13]. Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form. A CRF extraction of all deaths whatever the cause will be realized every months by clinical research unit. These data will be sent to the DSMB members and the AP-HP's safety department's . If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay. NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor. • Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care The investigator must report these events to his Centre Régional de Pharmacovigilance (CRPV). • Adverse events specifically due to the treatment with Levosimendan - Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation) - Moderate tachycardia (HR <110 bpm) -Premature ventricular contraction - Premature atrial contraction, atrial fibrillation	If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay. NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor. Under VA-ECMO All SAEs, which occurred under VA-ECMO, do not require the investigator to notify the sponsor without delay especially SAEs with fatal outcome which are considered by the investigator to be not related to the experimental treatment. NB: Cf Special rules for trials with a high mortality rate. After weaning ECMO • Normal and natural course of the condition: -Death considered as exclusively related to the natural evolution of cardiogenic shock or multiple organ failure and not to the infusion of the study drug -Prolongation of hospitalization due to surgery -Rehospitalization considered as not related to cardiogenic shock or for disease follow-up -Events likely to be exclusively due to ICU care occurring during the hospitalization period and requiring a prolongation of hospitalization (e.g. pleural effusion, septic shock, multiorgan failure, hemorrhagic shock, renal failure, ventilator associate pneumonia, acute respiratory distress syndrome ...) NB: Cf Special rules for trials with a high mortality rate. • Special circumstances -Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation) -Moderate tachycardia (HR <110 bpm) -Premature ventricular contraction except those leading to life threatening or death
--	---

- Nausea - Vomiting - Diarrhea – Constipation - Dizziness - Headache - Insomnia - Mild to moderate hypokalemia (3 to 3.5 mmol/L)	-Premature atrial contraction, atrial fibrillation except those leading to life threatening or death -Mild to moderate hypokalemia (3 to 3.5 mmol/L) • Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care The investigator must report these events to his Centre Régional de Pharmacovigilance (CRPV). Special rules for trials with a high mortality rate: The mortality rate of patients with refractory cardiogenic shock and ECMO VA is 58 % [2, 11-13]. Early administration of continuous infusion of Levosimendan (investigational medicinal product) will facilitate and accelerate VA-ECMO weaning in this patient population supported by VA-ECMO; therefore, deaths considered as related to cardiogenic shock do not need to be notified to the sponsor without delay but will be recorded in the case report form. A CRF extraction of all deaths whatever the cause will be realized every months by clinical research unit. These data will be sent to the DSMB members and the AP-HP's safety department's. If there is any imbalance between the randomization groups or the mortality rate is higher than expected affecting the safety of trial subjects and which requires the sponsor to take urgent safety measures, the ANSM will be informed about the emerging safety issue without delay. NB: any deaths likely to be related to the administration of Levosimendan will be notified by the investigator to the sponsor. -Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care The investigator must report these events to his Centre Régional de Pharmacovigilance (CRPV). • Adverse events specifically due to the treatment with Levosimendan -Hypotension (grades 1 and 2, <30% decrease in mean arterial pressure after treatment initiation) -Moderate tachycardia (HR <110 bpm) -Premature ventricular contraction
--	--

			Premature atrial contraction, atrial fibrillation Nausea Vomiting Diarrhea – Constipation – Dizziness Headache Insomnia Mild to moderate hypokalemia (3 to 3.5 mmol/L)
P31	10.1.2.3 Period during which the investigator must send notification of SAEs to the sponsor without delay The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section: - starting from the date on which the subject signs the consent form/begins treatment with the investigational medicinal product/undergoes the first intervention specific to the study - throughout the whole follow-up period intended by the trial	P31	10.1.2.3 Period during which the investigator must send notification of SAEs to the sponsor without delay The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section: - starting from the date on which the subject signs the consent form/begins treatment with the investigational medicinal product/undergoes the first intervention specific to the study - throughout the whole follow-up period intended by the trial (60 days after randomization)
P32	For serious adverse events likely to be related to the investigational medicinal product(s): - refer to the SmPC for Levosimendan (Zimino®) enclosed in appendix N°18.4.	P32	For serious adverse events likely to be related to the investigational medicinal product(s): - refer to the SmPC for Levosimendan (Zimino®) enclosed in appendix N°18.4. - refer to the SmPC for Cernevit®
P40	14.1 Methods for informing and obtaining consent from the research participants : However, if the person is unable to give his or her written consent, consent may be obtained, in descending order of priority, from a legal representative, family member or a close relative. These persons must have no connection whatsoever to the investigator or the sponsor [...] Should a close relative be absent, eligible patients will be randomized according to the specifications of emergency consent (Article L.1122-1-3 CSP) after IRB approval. The patient's legal representative will be informed as soon as possible of the inclusion and asked to give his or her written consent. The patient will be informed and asked to give his/her consent for the continuation of the trial when his/her condition will allow during the subject's study participation. Investigators should to indicate in the medical record the inclusion of a patient unable to consent in emergency situations in the research protocol.	P40	14.1 Methods for informing and obtaining consent from the research participants : However, if the person is unable to give his or her written consent, consent may be obtained, in descending order of priority, from a legal representative, family member or a close relative. These persons must have no connection whatsoever to the investigator or the sponsor [...] For LEVOECMO trial, if patient Should a close relative be absent, eligible patients will be randomized according to the specifications of emergency consent (Article L.1122-1-3 CSP) after IRB approval. The patient's legal representative will be informed as soon as possible of the inclusion and asked to give his or her written consent. The patient will be informed and asked to give his/her consent for the continuation of the trial when his/her condition will allow during the subject's study participation. Investigators should to indicate in the medical record the inclusion of a patient unable to consent in emergency situations in the research protocol.

P40	14.2 Prohibition of concomitant clinical studies participation and exclusion period after the trial, if applicable Whilst participating in this trial, subjects may not take part in any other interventional clinical study. For non-interventional or observational study, subjects may not take part without first speaking to the doctor in charge of this trial.	P40	14.2 Prohibition of concomitant clinical studies participation and exclusion period after the trial, if applicable Whilst participating in this trial, subjects may not take part in any other interventional clinical study [except IMPULMACS, NCT04951999] . For non-interventional or observational study, subjects may not take part without first speaking to the doctor in charge of this trial.
P47	18.5 SCP's Levosimendan	P47	18.5 SmpCCP's Levosimendan 18.6 SmPC CERNEVIT® https://base-donnees-publique.medicaments.gouv.fr/extrait.php?specid=6100103

Version n°2.0 ; 05/31/2023

Version N°3.0 ; 06/26/2024

Pages	Protocole Version n°2.0 du 31/05/2023			Pages	Protocole Version n°3.0 du 26/06/2024			
P1 à 46	Entête : Version 2.0 of 31/05/2023			P1 à 46	Entête : Version 3.0 of 26/06/2024			
P1	Version 2.0 of 31/05/2023			P1	Version 3.0 of 26/06/2024			
P8	Duration of the trial - inclusion period: 24 + 12 months - participation period (treatment+follow-up): 2 months - total duration: 26 + 12 months			P8	Duration of the trial - inclusion period: 24 + 12-36 + 3 months - participation period (treatment+follow-up): 2 months - total duration: 26 + 12-38 + 3 months			
P18	5.3 Expected length of participation, chronology and duration of the study. - Length of enrolment period: 24 months + 12 months [...] - Total study duration: (start = trial start date; end = trial end date) = 26 months + 12 months			P18	5.3 Expected length of participation, chronology and duration of the study. - Length of enrolment period: 24 36 months + 12-3 months [...] - Total study duration: (start = trial start date; end = trial end date) = 26 38 months + 12 3 months			
P22		Study characteristics	Value		P22		Study characteristics	Value
		Total number of subjects	206				Total number of subjects	206
		Number of sites	14				Number of sites	14
		Enrolment period (months)	24 + 12				Enrolment period (months)	24 + 12 36 + 3
		Number of subjects/site	15,8				Number of subjects/site	15,8
		Number of subjects/site/month	0.4				Number of subjects/site/month	0.4